

# Compal Confidential

## C560 LA-A061P Schematics Document

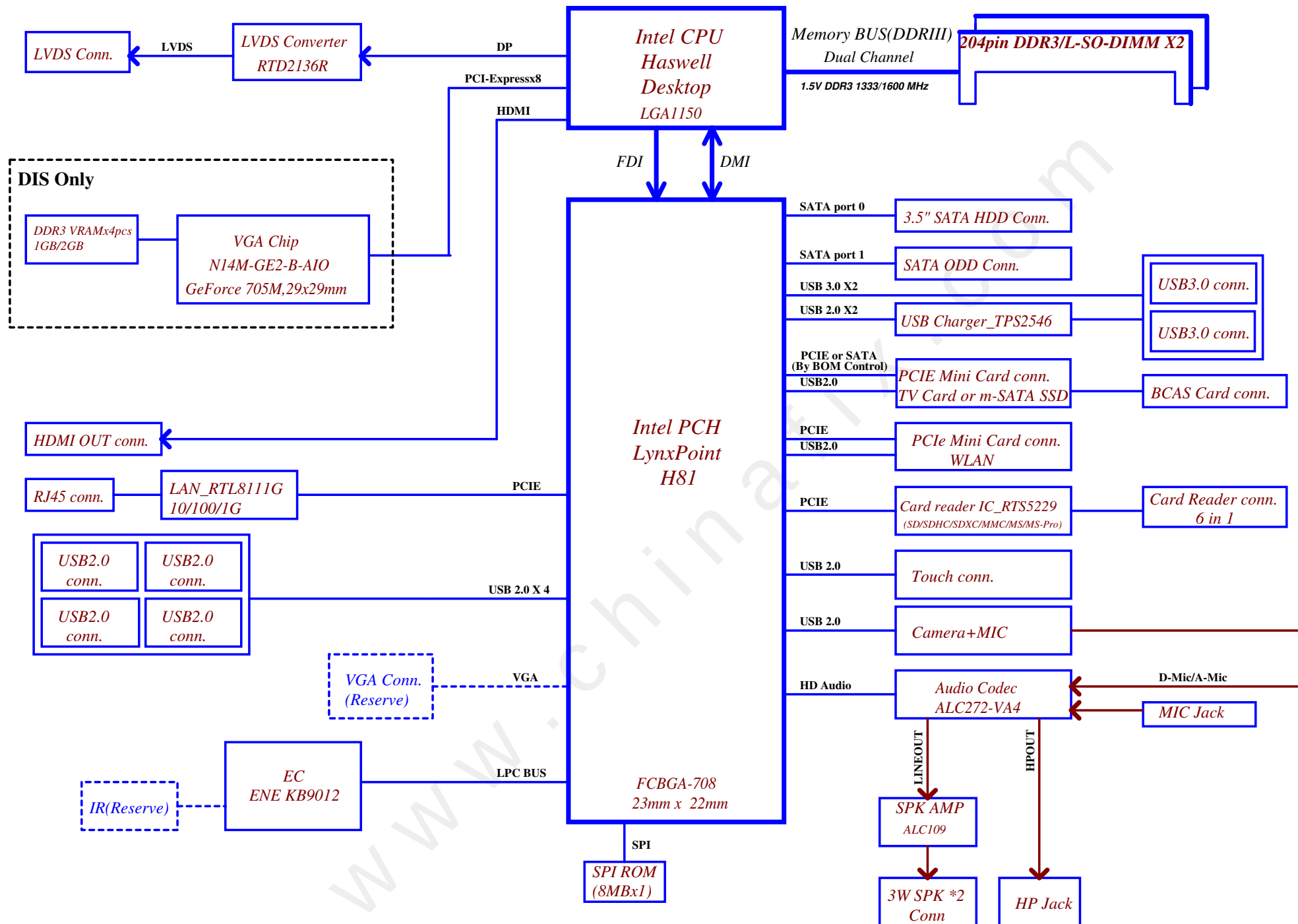
INTEL Haswell CPU with DDRIII + PCH Lynx-Point

AIO M/B

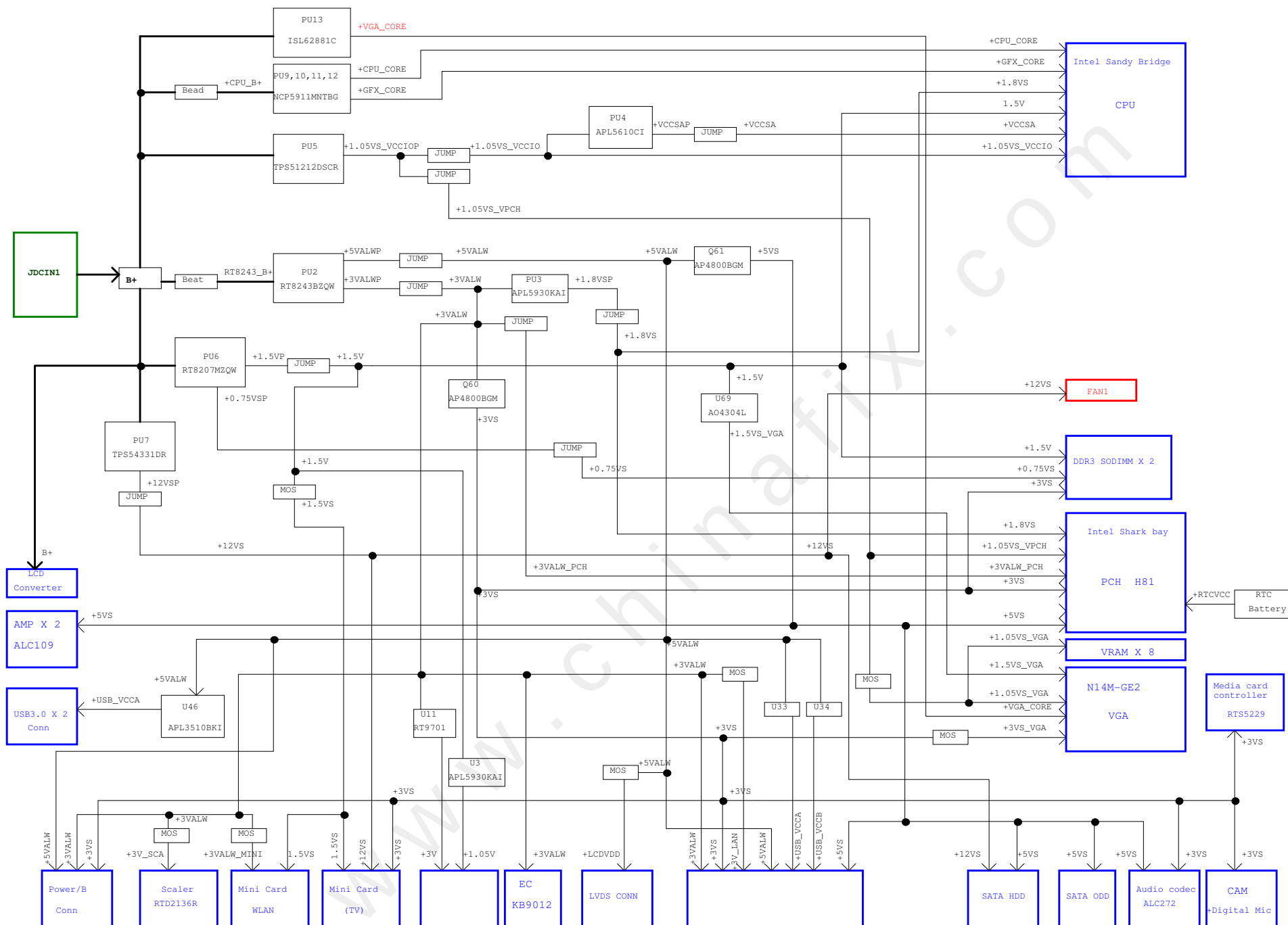
September 24, 2013

REV:1.0

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| Date                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Tuesday, September 24, 2013 | Sheet           | 1                        | of 59      |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |                             |  |                 |  |                          |  |               |  |                    |  |  |  |         |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------|--|-----------------|--|--------------------------|--|---------------|--|--------------------|--|--|--|---------|--|
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |                             |  |                 |  |                          |  | Part Number   |  | Document Number    |  |  |  | Rev 0.3 |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |                             |  |                 |  |                          |  | Custom        |  | ZEA00 LA-A061P M/B |  |  |  |         |  |
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| STATE                 | SIGNAL | SLP_S3# | SLP_S4# | SLP_S5# | +VALW <sup>*1</sup> | +VS <sup>*2</sup> | +1.5V | +0.75VS | +RTCVCC |
|-----------------------|--------|---------|---------|---------|---------------------|-------------------|-------|---------|---------|
| S0 (Full ON)          |        | HIGH    | HIGH    | HIGH    | ON                  | ON                | ON    | ON      | ON      |
| S1 (Power On Suspend) |        | HIGH    | HIGH    | HIGH    | ON                  | ON                | ON    | ON      | ON      |
| S3 (Suspend to RAM)   |        | LOW     | HIGH    | HIGH    | ON                  | OFF               | ON    | OFF     | ON      |
| S4 (Suspend to Disk)  |        | LOW     | LOW     | HIGH    | ON                  | OFF               | OFF   | OFF     | ON      |
| S5 (Soft OFF)         |        | LOW     | LOW     | LOW     | ON                  | OFF               | OFF   | OFF     | ON      |

Note:

\*1: +VALW power reail include +3VALW, +5VALW, B+, +VSB, +3VALW\_PCH

\*2: +VS power reail include +3VS, +5VS, +12VS, +1.05VS\_VPCH, +1.5VS\_VGA, +VGA\_CORE, +CPU\_CORE

| USB Port Table |         |      |                             |
|----------------|---------|------|-----------------------------|
| USB 2.0        | USB 1.1 | Port | Device                      |
| RMH1           | UHCI0   | 0    | Co-lay w/USB30 PORT0        |
|                |         | 1    | Co-lay w/USB30 PORT1(Debug) |
|                | EHCI1   | 2    | Rear IO USB20 Conn          |
|                |         | 3    | Rear IO USB20 Conn          |
|                | UHCI2   | 4    | WLAN                        |
|                |         | 5    | Touch                       |
| RMH2           | UHCI3   | 6    | Disabled on H81             |
|                |         | 7    | Disabled on H81             |
|                | UHCI4   | 8    | Rear IO USB20 Conn          |
|                |         | 9    | Rear IO USB20 Conn(Debug)   |
|                | UHCI5   | 10   | TV                          |
|                |         | 11   | Camera                      |
|                | UHCI6   | 12   | Disabled on H81             |
|                |         | 13   | Disabled on H81             |

| BTO Item                   | BOM Structure  |
|----------------------------|----------------|
| ME components              | CONN@          |
| UMA Only                   | UMA@           |
| DISCRETE Only              | DIS@           |
| EMI Pop components         | EMI@           |
| ESD Pop components         | ESD@           |
| EMI Unpop components       | @EMI@          |
| ESD Unpop components       | @ESD@          |
| HDMI OUT                   | HDMIO@         |
| TV                         | TV@            |
| SSD                        | SSD@           |
| CRT                        | CRT@ (EVTonly) |
| Unpop                      | @              |
| VRAM select                | X76@           |
| EVT for Reserve components | EVT@           |
| PCB                        | PCB@           |
| SKU IO Select              | GPIO68_H@      |
|                            | GPIO68_L@      |
|                            | GPIO69_H@      |
|                            | GPIO69_L@      |
|                            | GPIO70_H@      |
|                            | GPIO70_L@      |
| Touch                      | TOUCH@         |
| Non Charger                | NCHG@          |
| Charger                    | CHG@           |

| SATA Port Table |   |                 |
|-----------------|---|-----------------|
| Port            |   | Device          |
| 6G              | 0 | HDD             |
|                 | 1 | m-SATA          |
| 3G              | 2 | Disabled on H81 |
|                 | 3 | Disabled on H81 |
|                 | 4 | ODD             |
|                 | 5 | NC              |

| PCIe Port Table |                 |
|-----------------|-----------------|
| Port            | Device          |
| 1               | LAN             |
| 2               | Card Reader     |
| 3               | WLAN            |
| 4               | TV              |
| 5               | NC              |
| 6               | NC              |
| 7               | Disabled on H81 |
| 8               | Disabled on H81 |

| BOARD ID Table |              |
|----------------|--------------|
| Board ID       | PCB Revision |
| 0              | 0.1          |
| 3              | 0.2          |
| 4              | 0.3          |
| 5              |              |

| PCH SM Bus Address |            |     |             |
|--------------------|------------|-----|-------------|
| Power              | Device     | HEX | Address     |
| +3VS               | DDR(JDDR2) |     | 1010 000X b |
| +3VS               | DDR(JDDR1) |     | 1010 010X b |

| EC SM Bus1 Address |        |     |            |
|--------------------|--------|-----|------------|
| Power              | Device | HEX | Address    |
|                    | ALC106 | 48H | 0100_100xb |

| Board ID | Rb          | V <sub>min</sub> | V <sub>typ</sub> | V <sub>max</sub> | EC AD3      |
|----------|-------------|------------------|------------------|------------------|-------------|
| 0        | 0           | 0 V              | 0 V              | 0.155 V          | 0x00 - 0x0C |
| 3        | 33K +/- 5%  | 0.634 V          | 0.819 V          | 0.945 V          | 0x31 - 0x49 |
| 4        | 56K +/- 5%  | 0.958 V          | 1.185 V          | 1.359 V          | 0x4A - 0x69 |
| 5        | 100K +/- 5% | 1.372 V          | 1.650 V          | 1.838 V          | 0x6A - 0x8E |
| 6        | 200K +/- 5% | 1.851 V          | 2.200 V          | 2.420 V          | 0x8F - 0xBB |
| 7        | NC          | 2.433 V          | 3.300 V          | 3.300 V          | 0xBC - 0xFF |

| PCH SML1 Bus Address |                                   |     |         |
|----------------------|-----------------------------------|-----|---------|
| Power                | Device                            | HEX | Address |
|                      | VGA Ext. thermal sensor           |     |         |
|                      | VGA Int. thermal sensor (default) |     | 0x9b    |

| SKU ID(Project) Table |                      |                      |                              |                       |                                                                  |
|-----------------------|----------------------|----------------------|------------------------------|-----------------------|------------------------------------------------------------------|
| Project ID2 (GPIO68)  | Project ID1 (GPIO69) | Project ID0 (GPIO70) | SKU                          |                       |                                                                  |
| 0                     | 0                    | 0                    | UMA 4519QH38L04              | TV@ NLDO@             | EMI@ ESD@ GPIO68_L@ GPIO69_L@ GPIO70_L@ PCB@ CHG@                |
| 0                     | 0                    | 1                    | UMA W/HDMI 4519QH38L05       | HDMIO@ NLDO@          | EMI@ ESD@ GPIO68_L@ GPIO69_L@ GPIO70_H@ PCB@ 8111G@ NCHG@ TOUCH@ |
| 0                     | 1                    | 0                    | DIS-MIC1G 4519QH38L06        | DIS@ NLDO@            | EMI@ ESD@ GPIO68_L@ GPIO69_H@ GPIO70_L@ PCB@ 8111G@ TOUCH@       |
| 0                     | 1                    | 1                    | DIS-SAM1G W/HDMI 4519QH38L07 | DIS@ HDMIO@ TV@ NLDO@ | EMI@ ESD@ GPIO68_L@ GPIO69_H@ GPIO70_H@ PCB@ 8111G@ NCHG@        |
| 1                     | 0                    | 0                    | DIS-MIC2G W/HDMI 4519QH38L08 | DIS@ NLDO@            | EMI@ ESD@ GPIO68_H@ GPIO69_L@ GPIO70_L@ PCB@ 8111G@ NCHG@ TOUCH@ |
| 1                     | 0                    | 1                    | DIS-MIC2G W/HDMI 4519QH38L09 | DIS@ HDMIO@ TV@ NLDO@ | EMI@ ESD@ GPIO68_H@ GPIO69_L@ GPIO70_L@ PCB@ 8111G@ NCHG@ TOUCH@ |
| 1                     | 1                    | 0                    |                              |                       |                                                                  |
| 1                     | 1                    | 1                    |                              |                       |                                                                  |
|                       |                      |                      |                              |                       |                                                                  |
|                       |                      |                      |                              |                       |                                                                  |
|                       |                      |                      |                              |                       |                                                                  |
|                       |                      |                      |                              |                       |                                                                  |

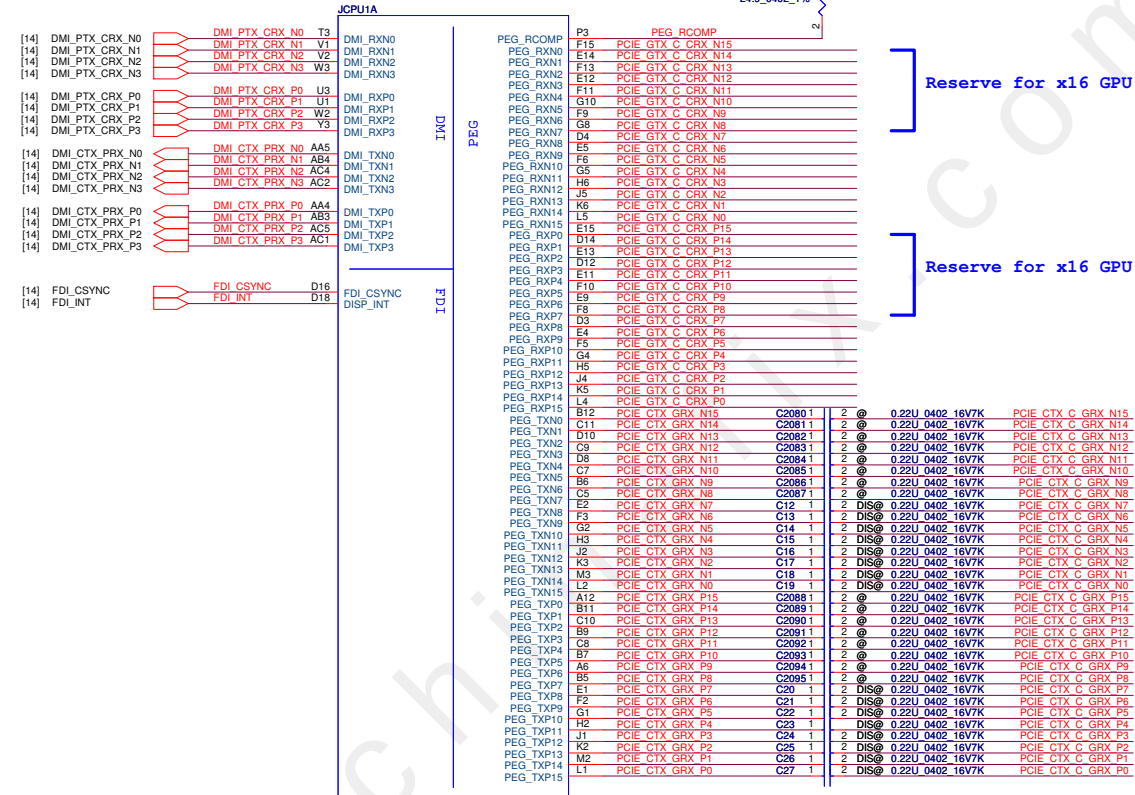
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PCIE GTX\_C CRX\_N0..15] PCIE GTX\_C CRX\_N0..15] [21]  
PCIE GTX\_C CRX\_P0..15] PCIE GTX\_C CRX\_P0..15] [21]  
PCIE CTX\_C GRX\_N0..15] PCIE CTX\_C GRX\_N0..15] [21]  
PCIE CTX\_C GRX\_P0..15] PCIE CTX\_C GRX\_P0..15] [21]

PEG\_RCOMP trace width=12mil  
spacing 15mil length<400mil



Reserve for x16 GPU

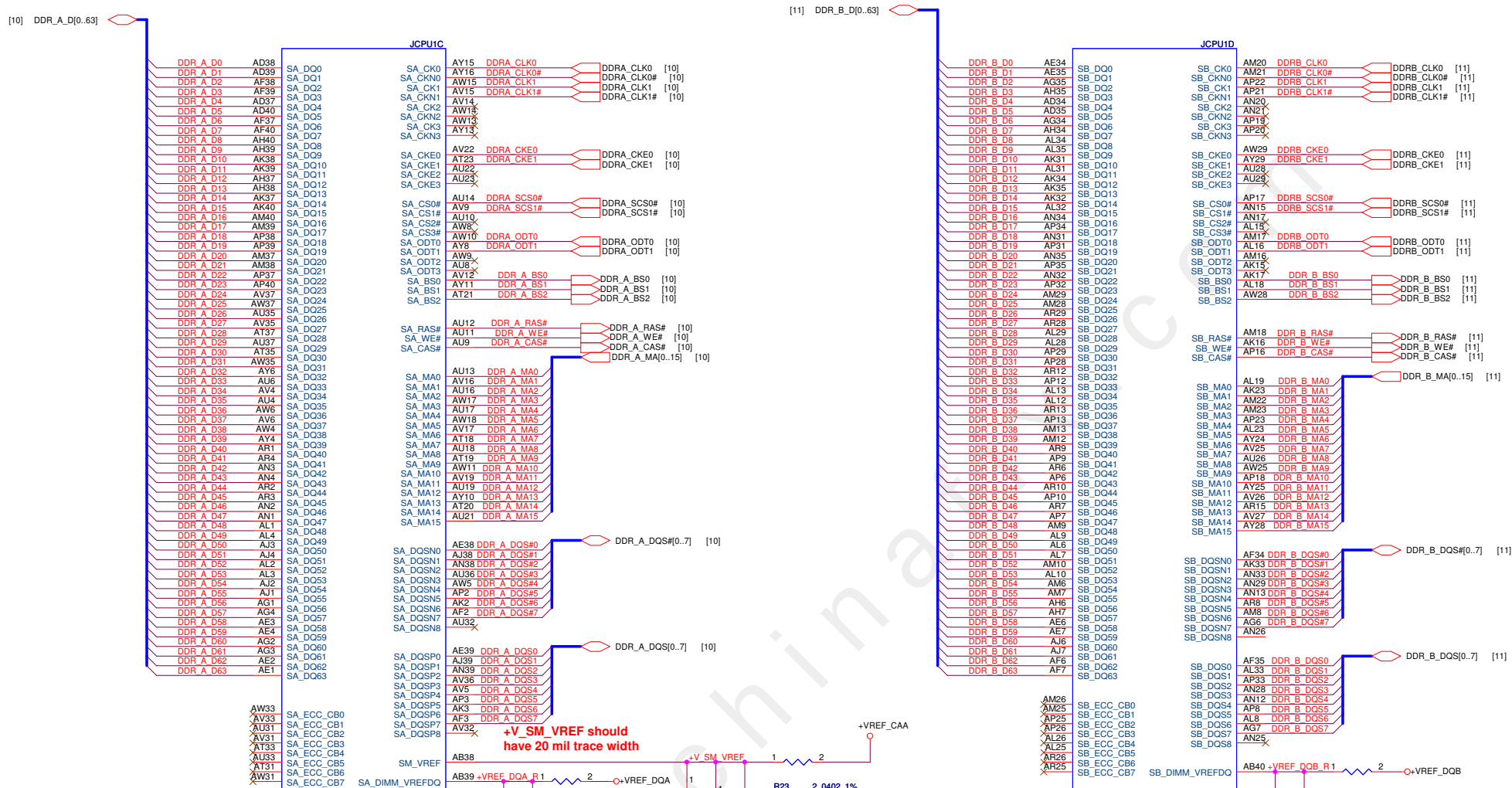
Reserve for x16 GPU

Reserve for x16 GPU

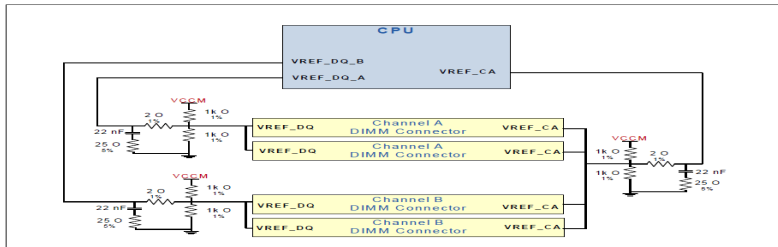
Reserve for x16 GPU

Typ- suggest 220nF. The change in AC capacitor value from 100nF to 220nF is to enable compatibility with future platforms having PCIe Gen3 (8GT/s)

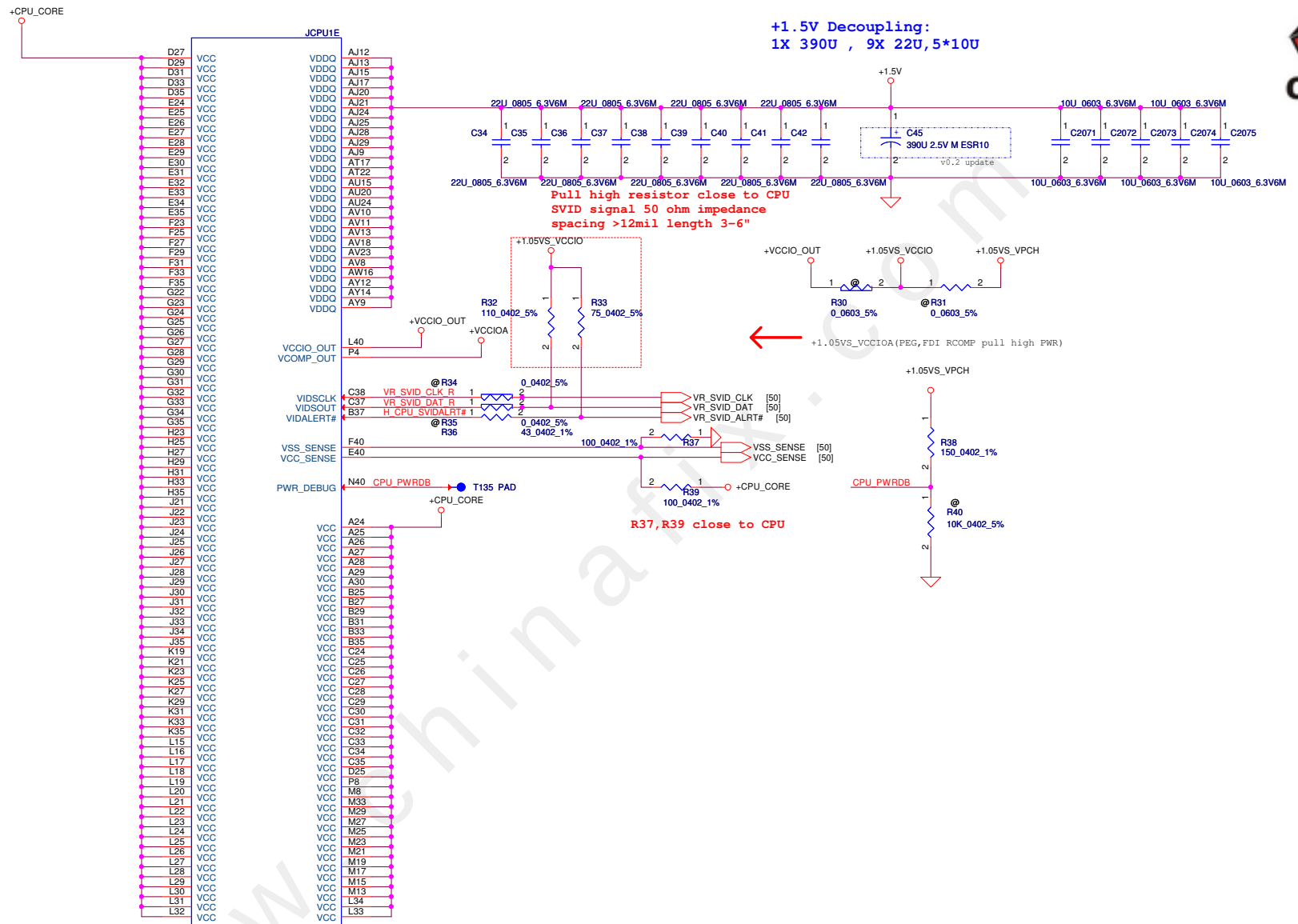
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DDR3 VREF Topologies



|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                             |                 |                          |              |
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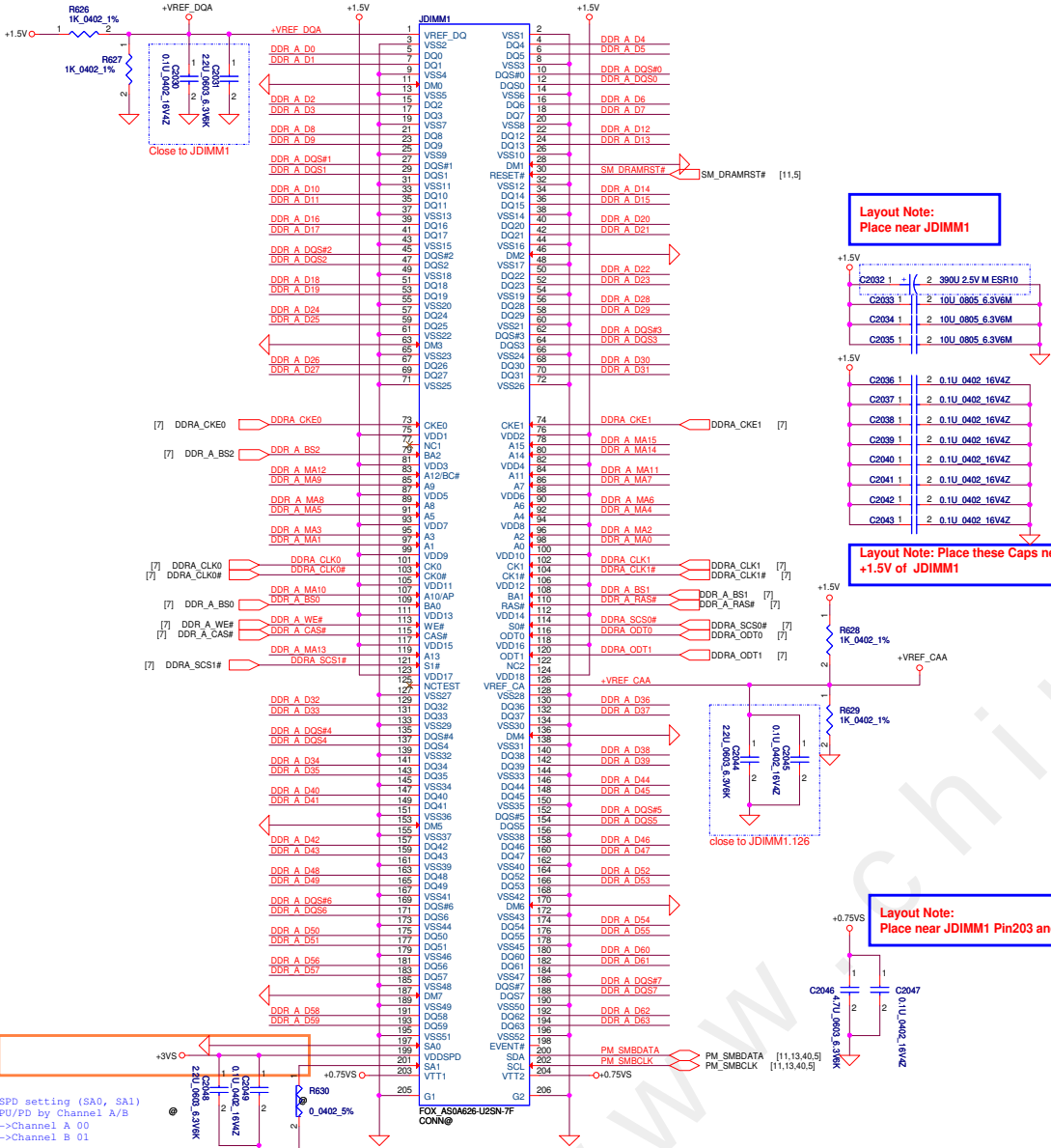


FOX\_3H993827-4M41-01H\_HASWELL

|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            |                          |                             |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Rev                      | 0.3                         |



# CHA SO-DIMM 0(A0)



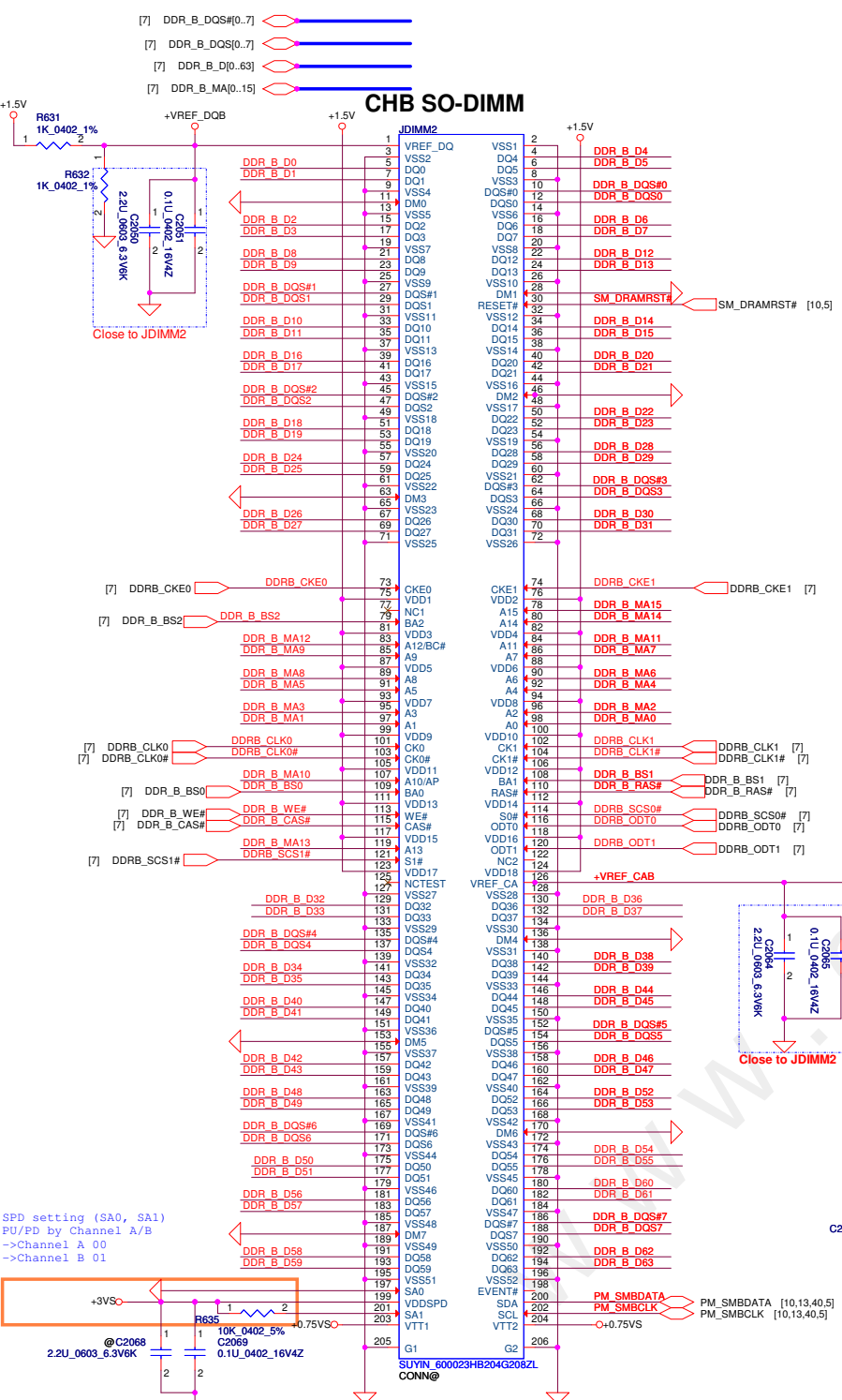
Layout Note:  
Place near JDIMM1

Layout Note: Place these Caps near  
+1.5V of JDIMM1

Layout Note:  
Place near JDIMM1 Pin203 and 204

|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  |                          |  |                             |                |                    |  |         |  |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  | Size                     |  | Document Number             |                | ZEA00 LA-A061P M/B |  | Rev 0.3 |  |
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# CHB SO-DIMM

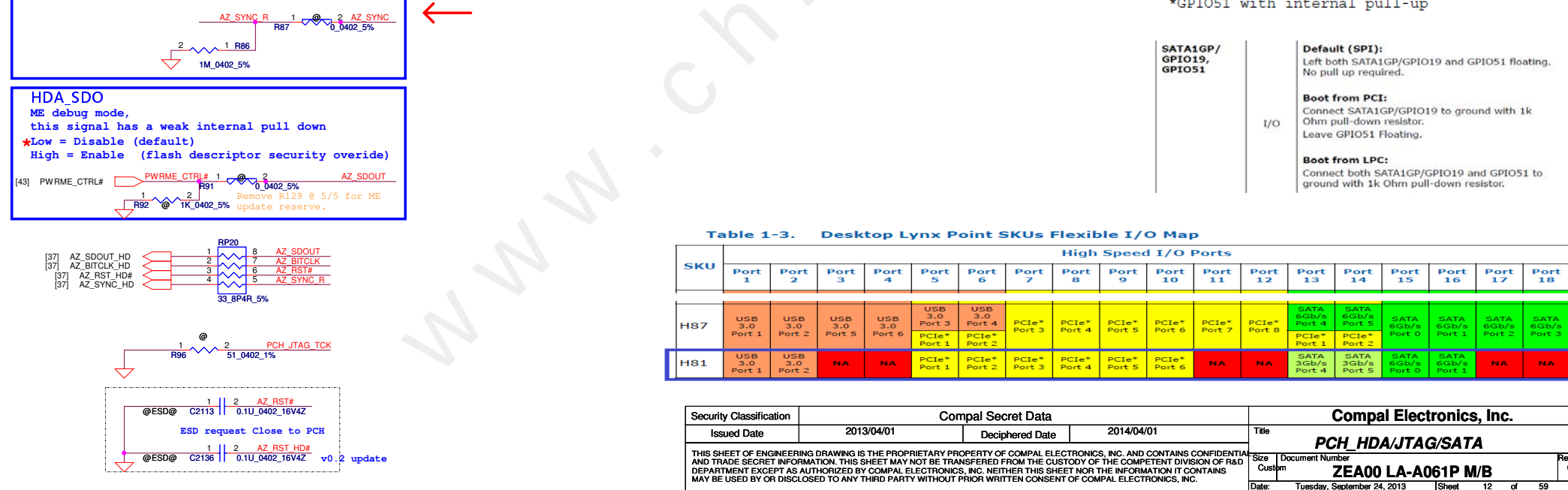
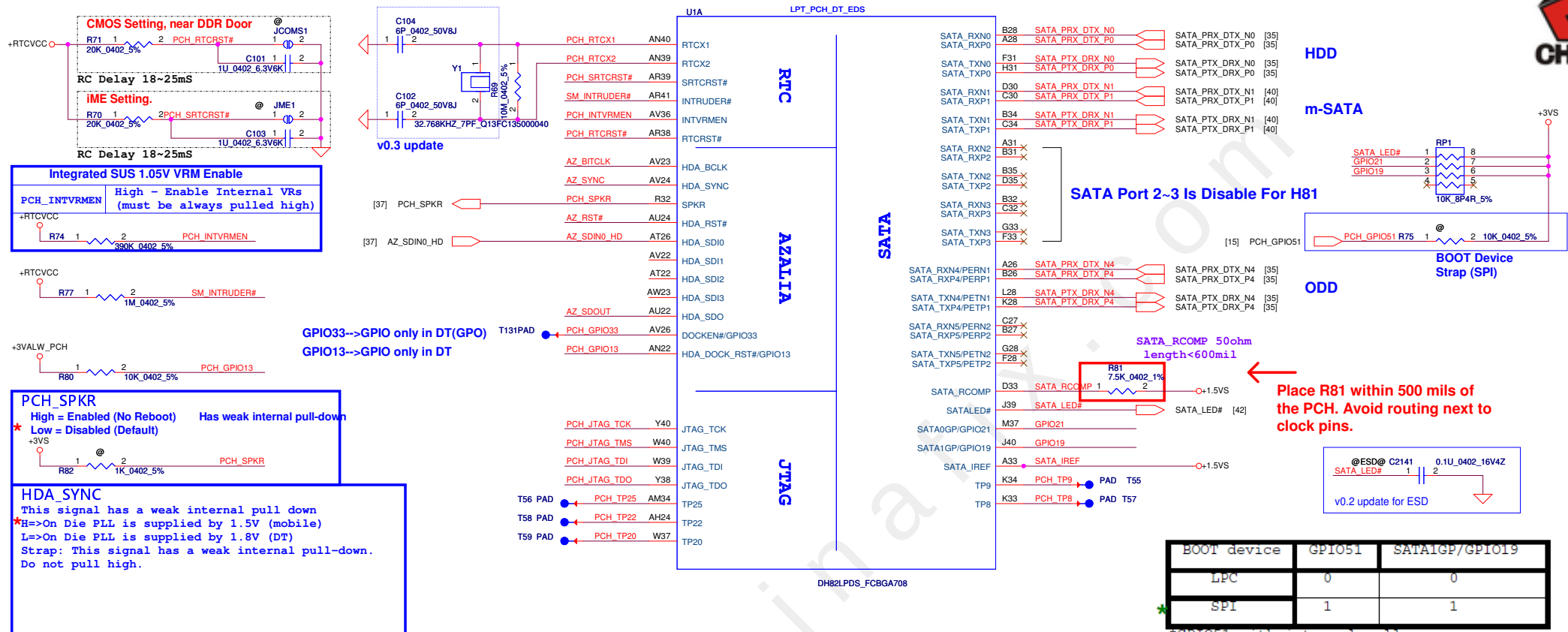


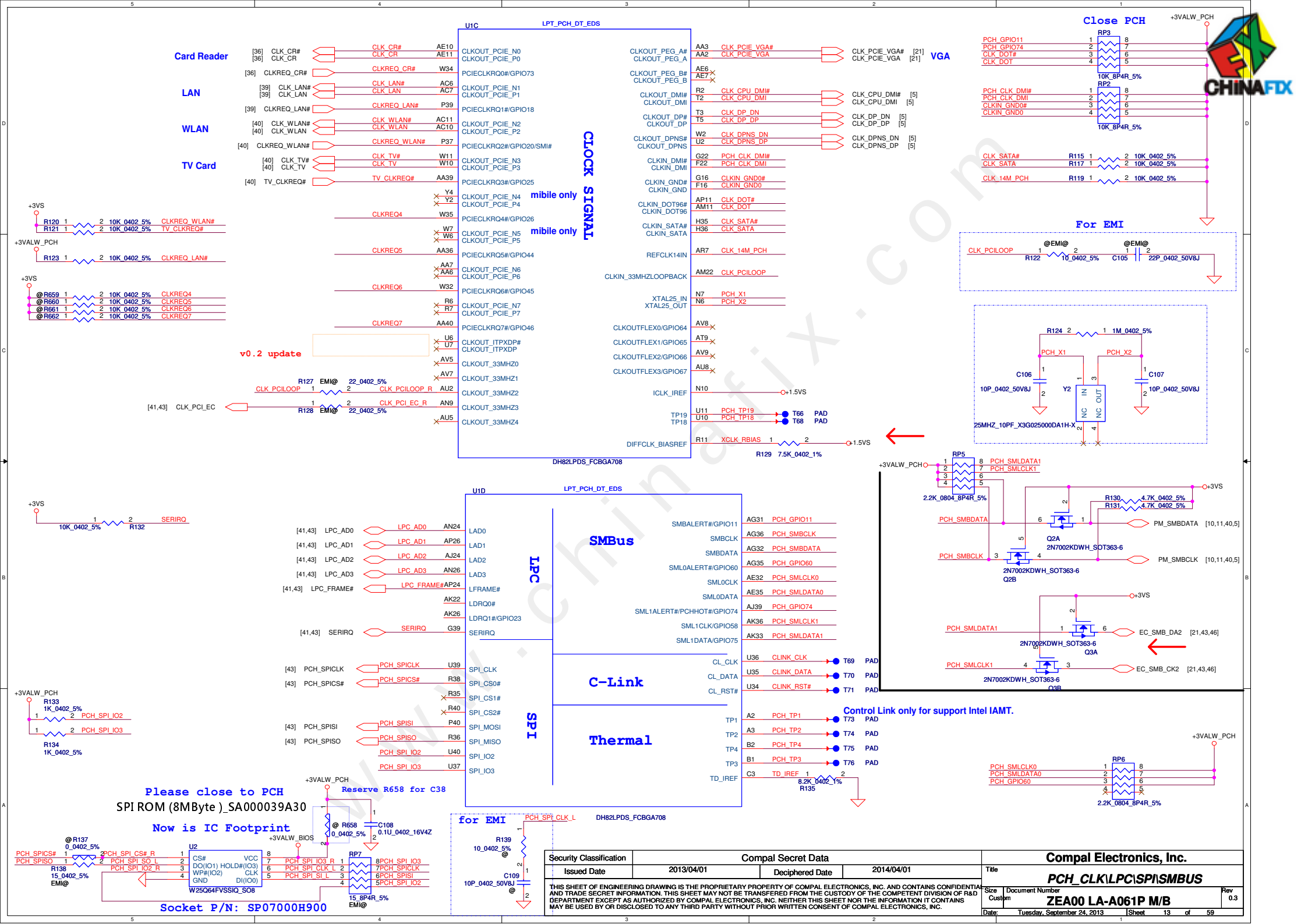
Layout Note:  
Place near JDIMM2

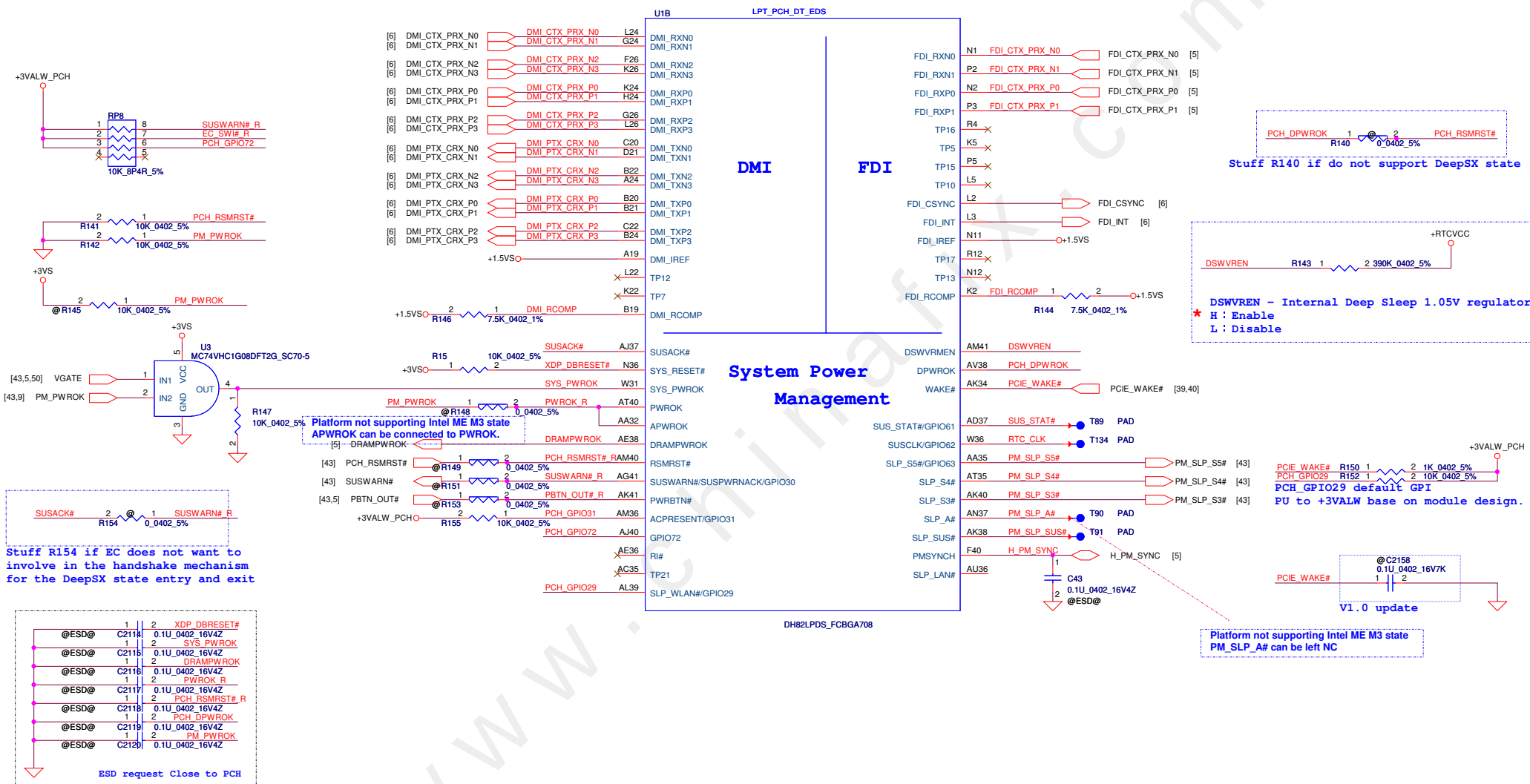
Layout Note: Place these Caps near  
+1.5V of JDIMM2

Layout Note:  
Place near JDIMM2 Pin203 and 204

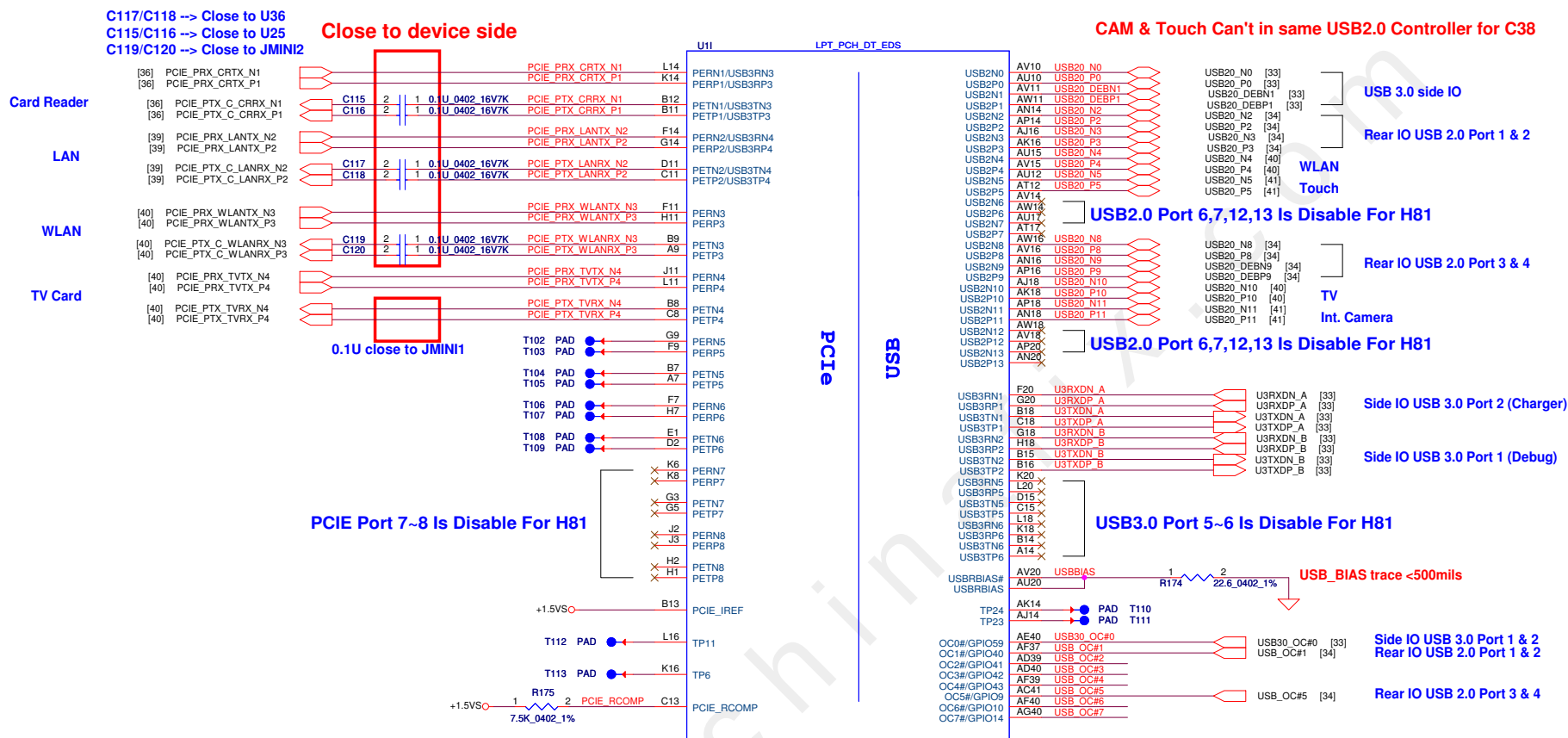
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                    |            | Date: Tuesday, September 24, 2013 |  |
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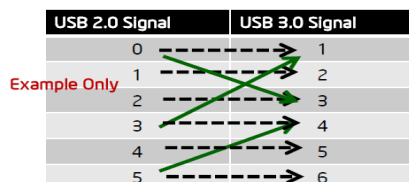








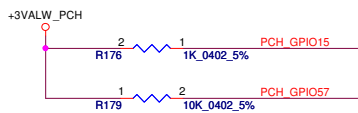
- Port mapping restrictions removed
  - USB 3.0 signals can now be paired with any of USB2.0 signal 0-13
  - Custom mapping through ACPI table/BIOS
  - Default mapping USB 3.0 1-6 to USB 2.0 0-5 ports



**Table 14-4. Overcurrent Pin Default Usage**

| Pin  | Default Port Mapping | Pin  | Default Port Mapping |
|------|----------------------|------|----------------------|
| OC0# | Port 0, Port 1       | OC4# | Port 8, Port 9       |
| OC1# | Port 2, Port 3       | OC5# | Port 10, Port 11     |
| OC2# | Port 4, Port 5       | OC6# | Port 12, Port 13     |
| OC3# | Port 6, Port 7       | OC7# | Not Used             |

- Better USB 2.0 performance than EHCI
- Windows\* 8 is expected to include a native inbox xHCI driver

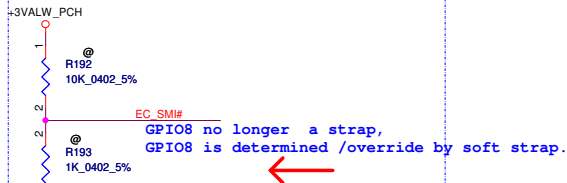


#### GPIO8

Integrated Clock Chip Enable (Removed)

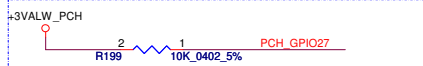
H: Disable

L: Enable



This signal has a weak internal pull-up but requires an external pull down.

The current default is clock enable



In Deep Sleep Power Well. Unmuxed. Defaults to GPI. Not used Weak pull-up 10kΩ to VccDSW3\_3 -->Check list1.5 P402. PD to GND for Huron River!!

#### GPIO28

On-Die PLL Voltage Regulator

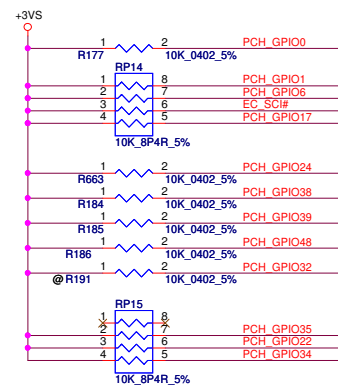
H: Enable

L: Disable



Clock validation strap  
ICG is EN when LOW  
\*GPIO36 with internal pull-down

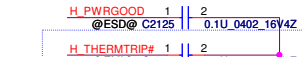
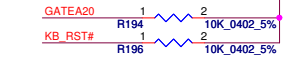
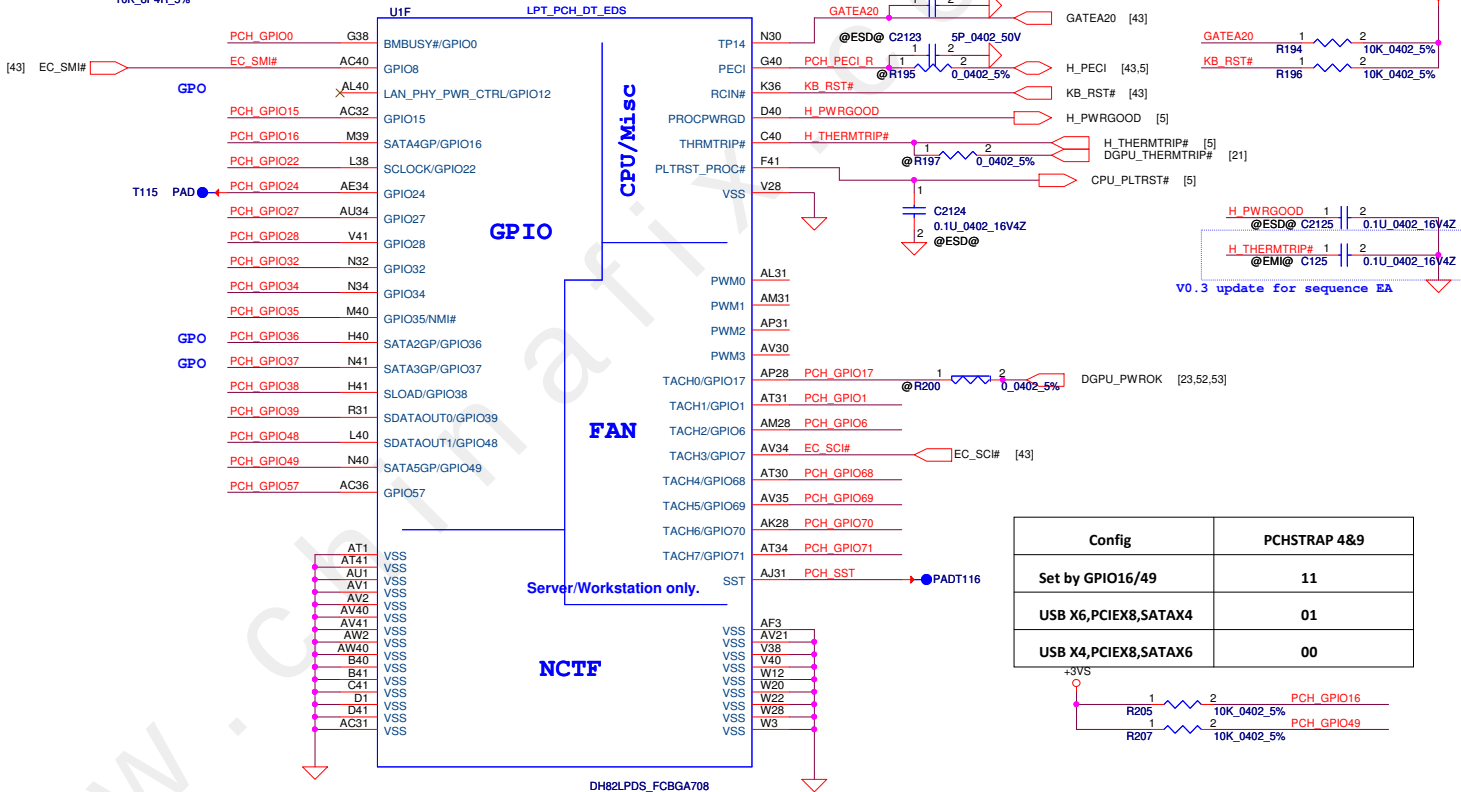
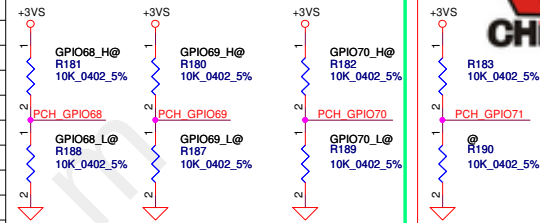
TL5  
Hi:with confidentiality  
Low:with no confidentiality  
\*GPIO37 with internal pull-down



120821

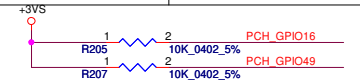
| SKU ID | GPIO68 | GPIO69 | GPIO69 |
|--------|--------|--------|--------|
| SKU1   | 0      | 0      | 0      |
| SKU2   | 0      | 0      | 1      |
| SKU3   | 0      | 1      | 0      |
| SKU4   | 0      | 1      | 1      |
| SKU5   | 1      | 0      | 0      |
| SKU6   | 1      | 0      | 1      |
| SKU7   | 1      | 1      | 0      |
| SKU8   | 1      | 1      | 1      |

SKU ID TABLE



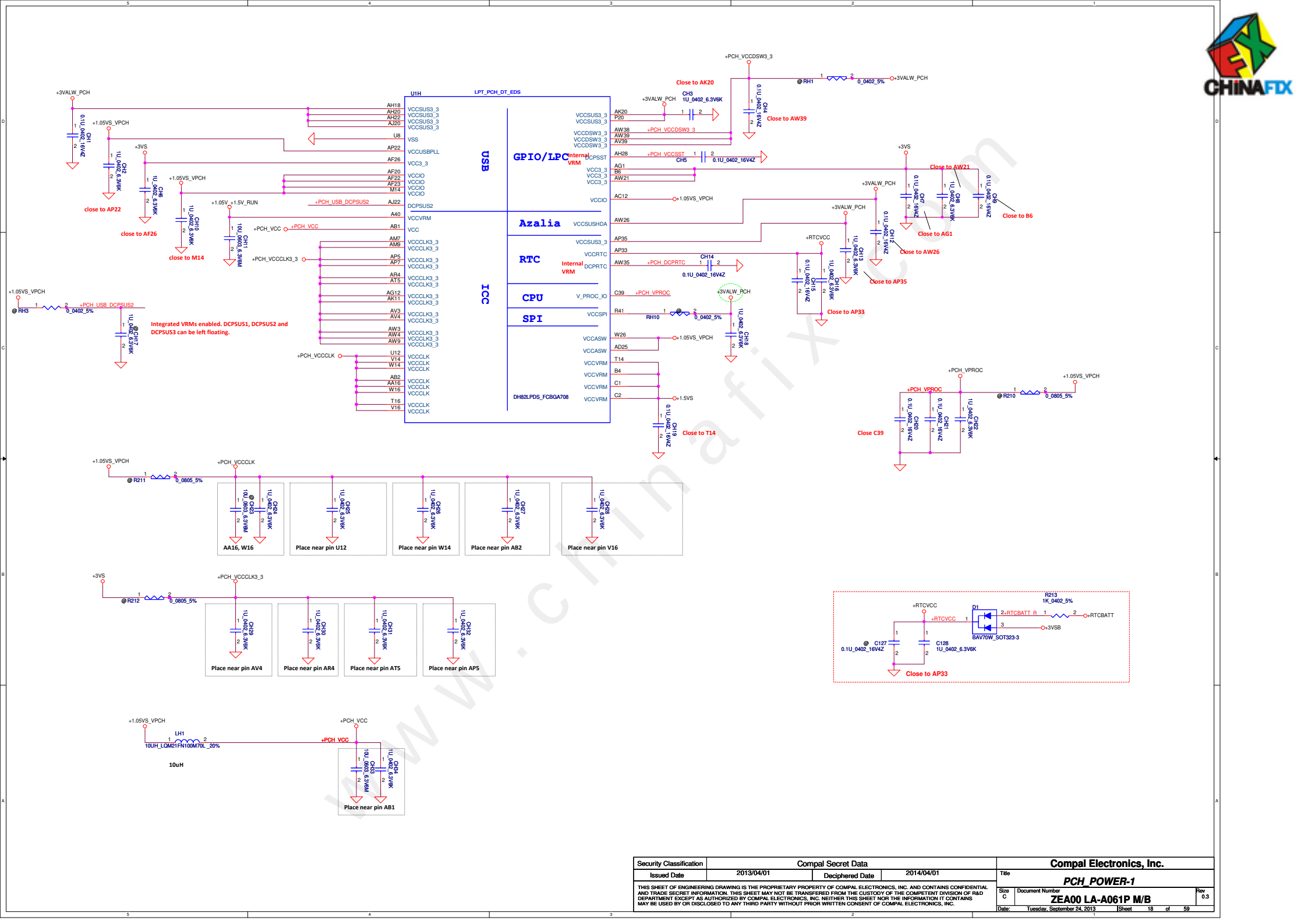
V0.3 update for sequence EA

| Config               | PCHSTRAP 4&9 |
|----------------------|--------------|
| Set by GPIO16/49     | 11           |
| USB X6,PCIEX8,SATAX4 | 01           |
| USB X4,PCIEX8,SATAX6 | 00           |



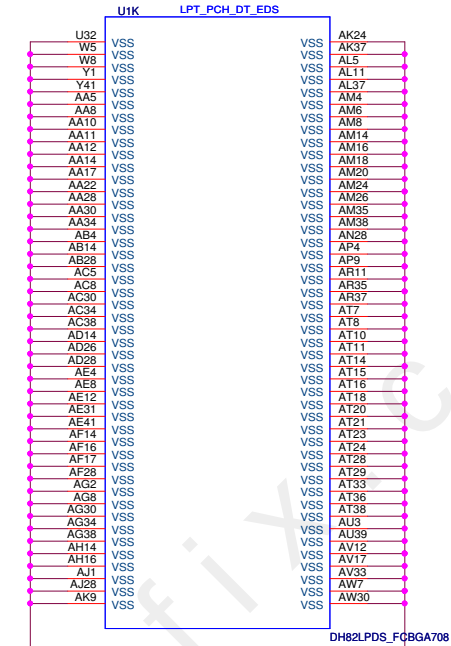
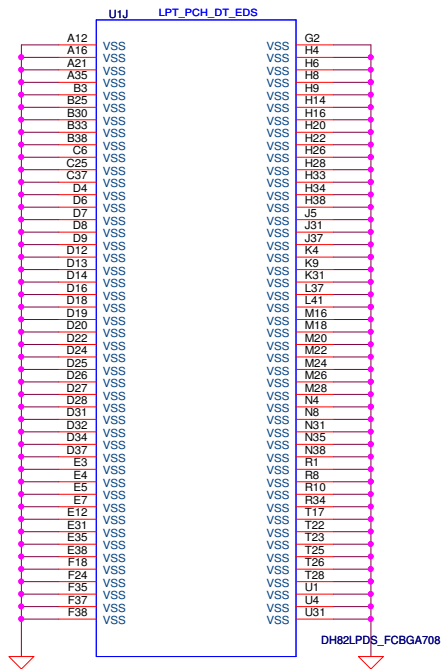
| Fixed Signals |        |        |        | Muxed Signals |        | Fixed Signals |        |        |        |        |        | Muxed Signals |        | Fixed Signals |        |        |        |
|---------------|--------|--------|--------|---------------|--------|---------------|--------|--------|--------|--------|--------|---------------|--------|---------------|--------|--------|--------|
| USB3 1        | USB3 2 | USB3 5 | USB3 6 | PCIE 1        | PCIE 2 | PCIE 3        | PCIE 4 | PCIE 5 | PCIE 6 | PCIE 7 | PCIE 8 | SATA 4        | SATA 5 | SATA 0        | SATA 1 | SATA 2 | SATA 3 |
|               |        |        |        | (00)          | (00)   |               |        |        |        |        |        | (00)          | (00)   |               |        |        |        |
|               |        |        |        | USB3 3        | USB3 4 |               |        |        |        |        |        | PCIE 1        | PCIE 2 |               |        |        |        |
|               |        |        |        | (01)          | (01)   |               |        |        |        |        |        | (01)          | (01)   |               |        |        |        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  |                 |  |                          |  |              |                             |  |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------|--|-----------------|--|--------------------------|--|--------------|-----------------------------|--|-------|
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |  | Compal Secret Data |  |                 |  | Compal Electronics, Inc. |  |              |                             |  |       |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  | 2013/04/01         |  | Deciphered Date |  | 2014/04/01               |  | Title        |                             |  |       |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  |                 |  |                          |  | Size         | Document Number             |  | Rev   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  |                 |  |                          |  | Custom       | ZEA00 LA-A061P M/B          |  | 0.3   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  |                 |  |                          |  | Date:        | Tuesday, September 24, 2013 |  | Sheet |

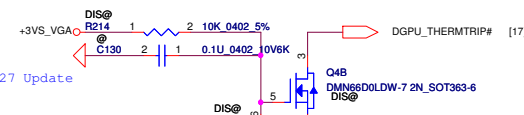
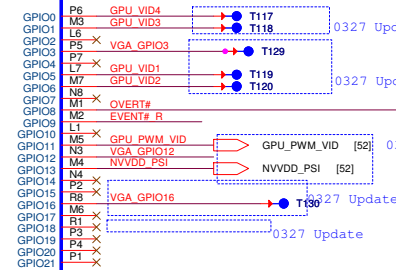
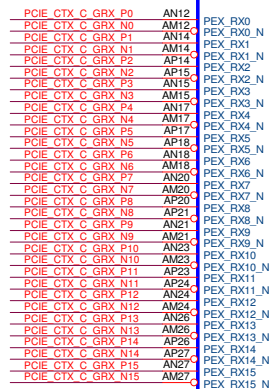
[illegible]



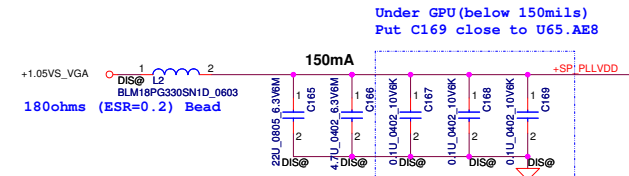
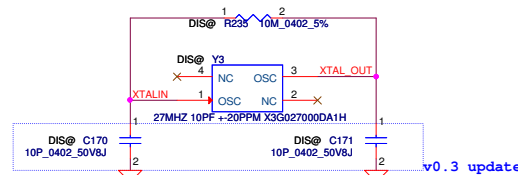
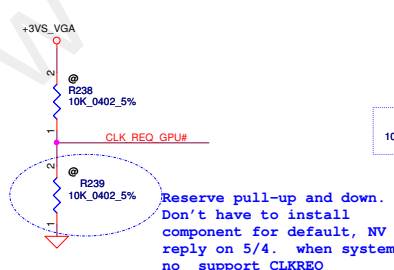
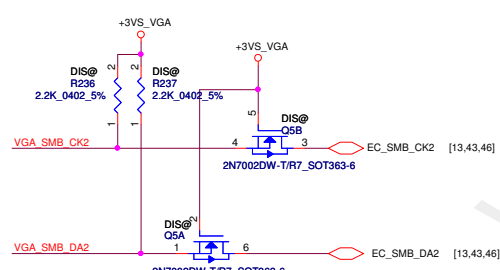
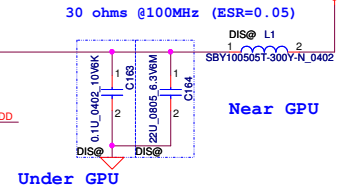
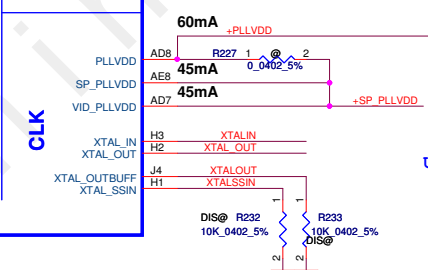
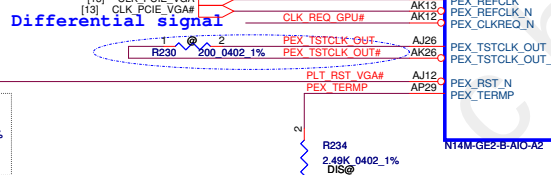
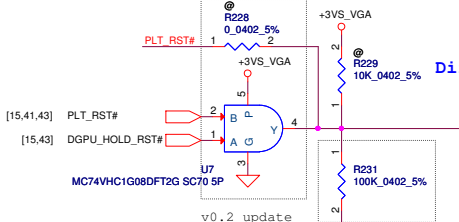
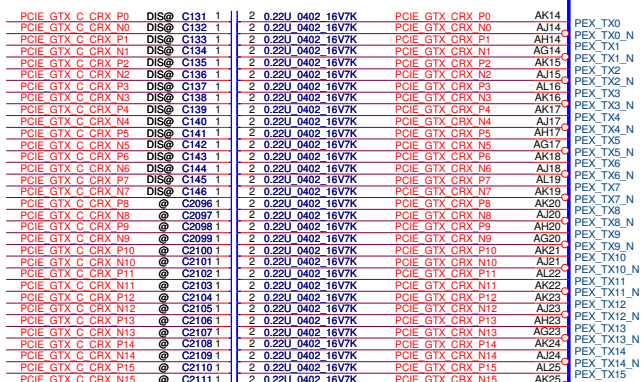
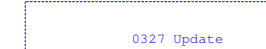
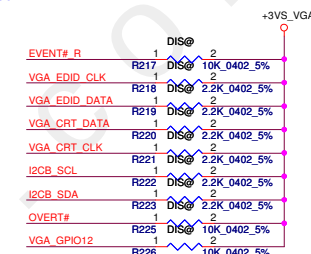
| Voltage Rail | Voltage | S0 Iccmax Current (A) |
|--------------|---------|-----------------------|
| VCC          | 1.05V   | 1.29 A                |
| VCCIO        | 1.05V   | 3.629 A               |
| VCCADAC1_5   | 1.5V    | 0.070 A               |
| VCCADAC3_3   | 3.3V    | 0.0133 A              |
| VCCCLK       | 1.05V   | 0.306 A               |
| VCCCLK3_3    | 3.3V    | 0.055 A               |
| VCCVRM       | 1.5V    | 0.179 A               |
| VCC3_3       | 3.3V    | 0.133 A               |
| VCCASW       | 1.05V   | 0.67 A                |
| VCCSUSHDA    | 3.3V    | 0.01 A                |
| VCCSPI       | 3.3V    | 0.022 A               |
| VCCSUS3_3    | 3.3V    | 0.261 A               |
| VCCDSW3_3    | 3.3V    | 0.015 A               |
| V_PROC_IO    | 1.05V   | 0.004 A               |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                             |            |                          |          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------|------------|--------------------------|----------|
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | Compal Secret Data          |            | Compal Electronics, Inc. |          |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2013/04/01 | Deciphered Date             | 2014/04/01 | Title                    |          |
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| Size                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            | Document Number             |            |                          | Rev      |
| Custom                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            | ZEA00 LA-A061P M/B          |            |                          | 0.3      |
| Date:                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            | Tuesday, September 24, 2013 |            | Sheet                    | 20 of 59 |



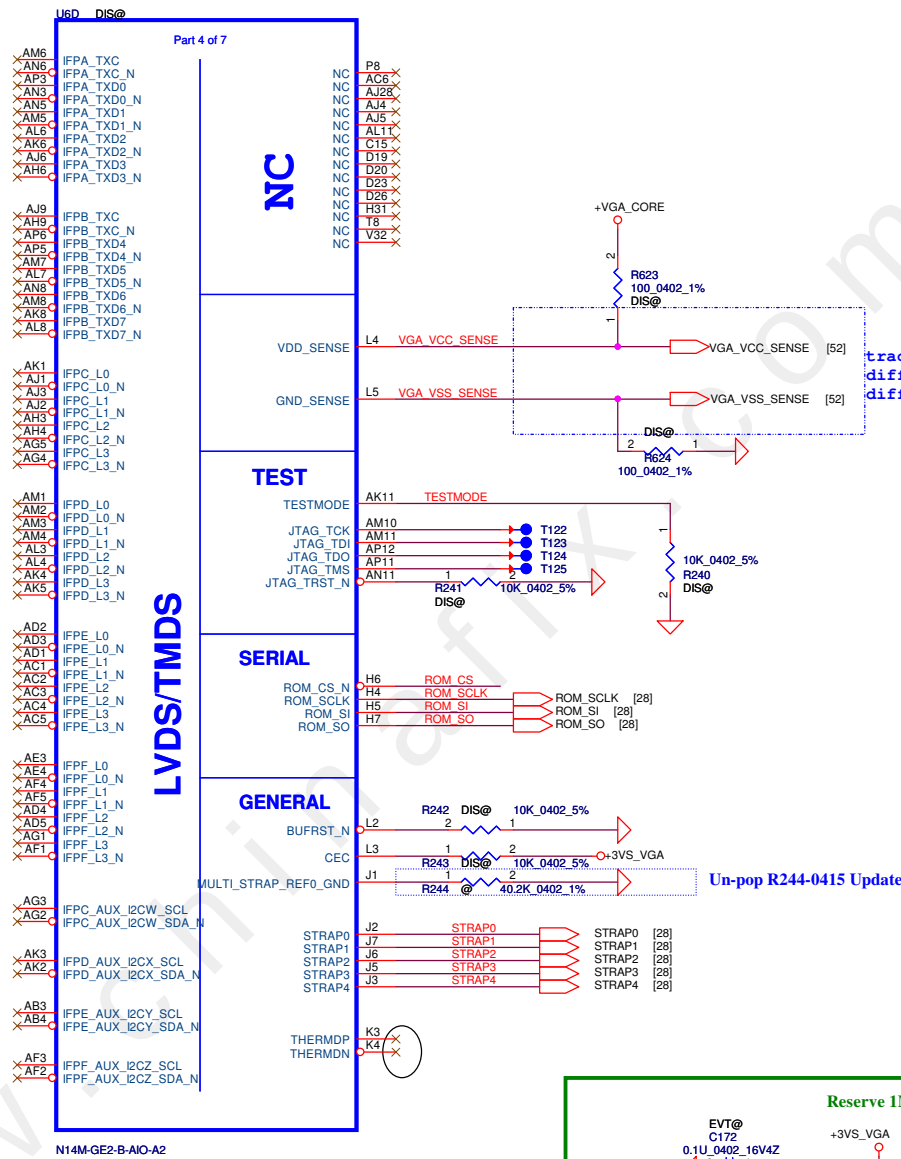
| GPIO pin Name | Normal Function    | I/O | Functional Description                           | Recommended Default Pull-up or Pull-down                                                                                    |
|---------------|--------------------|-----|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| GPIO0         | GPU_VID4           | 0   | GPU Core VDD VID4                                | Strap to boot HVDD                                                                                                          |
| GPIO1         | GPU_VID3           | 0   | GPU Core VDD VID3                                | Strap to boot HVDD                                                                                                          |
| GPIO2         | LCD_BL_PWM         | 0   | Panel Backlight PWM Brightness Control           | 100 k pull-down                                                                                                             |
| GPIO3         | LCD_VCC or PSI     | 0   | Panel Power Enable or Phase Shedding             | LCD_VCC: 100k pull-down<br>PSI: 10k pull-up or pull-down<br>stuff as needed to disable phase shedding by default            |
| GPIO4         | LCD_BLEN           | 0   | Panel Backlight Enable                           | 100 k pull-down                                                                                                             |
| GPIO5         | GPU_VID1           | 0   | GPU Core VDD VID1                                | Strap to boot HVDD                                                                                                          |
| GPIO6         | GPU_VID2           | 0   | GPU Core VDD VID2                                | Strap to boot HVDD                                                                                                          |
| GPIO7         | 3D_Vision          | 0   | 3D Vision Left/Right signal                      | 100 k pull-down                                                                                                             |
| GPIO8         | OV/ERT             | I/O | Active Low Thermal Catastrophic Over Temperature | 100 k pull-up                                                                                                               |
| GPIO9         | ALERT              | I/O | Active Low Thermal Alert                         | 100 k Pull-up                                                                                                               |
| GPIO10        | MEM_VREF_CTL       | 0   | Memory VREF Control                              | 100 k pull-down                                                                                                             |
| GPIO11        | GPU_VID0           | 0   | GPU Core VDD VID0                                | Strap to boot HVDD                                                                                                          |
| GPIO12        | PWR_LEVEL          | 0   | AC power detect or power supply overdraw input   | 100 k pull-up                                                                                                               |
| GPIO13        | GPU_VID5           | 0   | GPU Core VDD VID5                                | Strap to boot HVDD                                                                                                          |
| GPIO14        | HPD_AB             | I   | Hot Plug Detect for IFPB                         | See Figure 76                                                                                                               |
| GPIO15        | HPD_C              | I   | Hot Plug Detect for IFPC                         | See Figure 76                                                                                                               |
| GPIO16        | PSI or MEM_VDD_CTL | 0   | Phase Shedding or Memory VDD VID                 | PSI: 10k pull-up or pull-down<br>stuff as needed to disable phase shedding by default<br>MEM_VDD_CTL: Strap to boot FBVDD/Q |
| GPIO17        | HPD_D              | I   | Hot Plug Detect for IFPD                         | See Figure 76                                                                                                               |
| GPIO18        | HPD_E              | I   | Hot Plug Detect for IFPE                         | See Figure 76                                                                                                               |
| GPIO19        | HPD_F              | I   | Hot Plug Detect for IFPF                         | See Figure 76                                                                                                               |
| GPIO20        | Reserved           |     |                                                  |                                                                                                                             |
| GPIO21        | Reserved           |     |                                                  |                                                                                                                             |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |                    |            |                                                                  |                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------|------------|------------------------------------------------------------------|-----------------------------|
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                               |            | Compal Secret Data |            | <b>Compal Electronics, Inc.</b><br><b>N14M-GE2-PCIE/DAC/GPIO</b> |                             |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2013/04/01 | Deciphered Date    | 2014/04/01 | Title                                                            |                             |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |                    |            |                                                                  | <b>2EA00 LA-A061P M/B</b>   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |                    |            | Date:                                                            | Tuesday, September 24, 2013 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |                    |            | Sheet                                                            | 21 of 59                    |

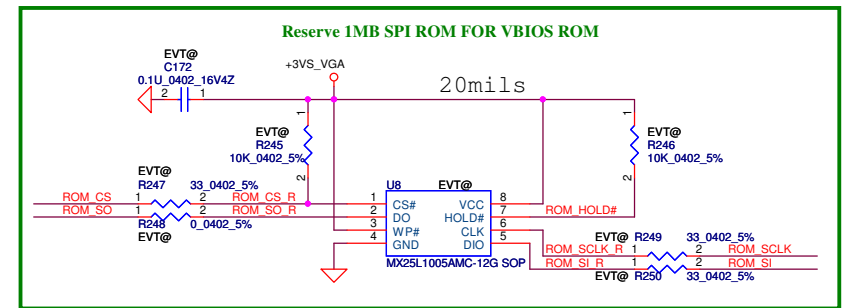
Table 66. N13x Family Display Link Summary

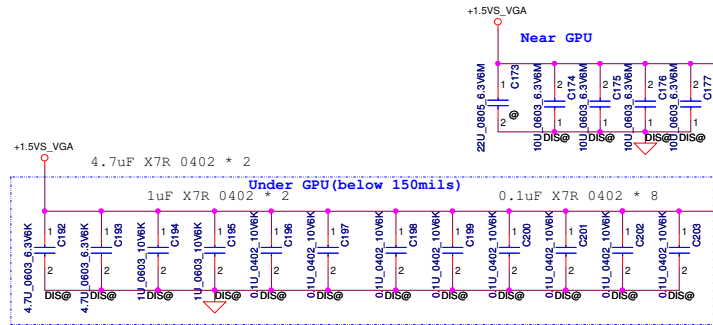
| Link   | Description                                                       |
|--------|-------------------------------------------------------------------|
| Link A | LVDS (Single Link or Dual Link with IFPB)                         |
| Link B | LVDS (Dual Link with IFPA)                                        |
| Link C | DisplayPort,<br>HDMI                                              |
| Link D | DisplayPort,<br>eDP                                               |
| Link E | DisplayPort,<br>DVI (Single Link or Dual Link with IFPF),<br>HDMI |
| Link F | DisplayPort,<br>DVI (Dual Link with IFPE),<br>HDMI                |



trace width: 16mils  
differential voltage sensing.  
differential signal routing.

Un-pop R244-0415 Update





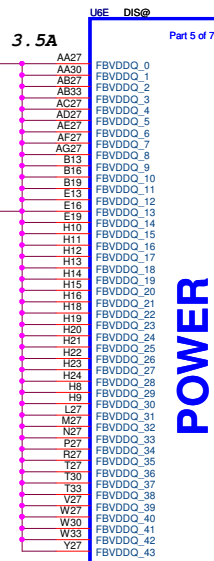
rise 1.5v system source voltage to 1.55-1.57V

Follow PUN-05893

|                  |         |
|------------------|---------|
| CALIBRATION PIN  | DDR3    |
| FB_CAL_x_PD_VDDQ | 40.2ohm |
| FB_CAL_x_PU_GND  | 42.2ohm |
| FB_CAL_xTERM_GND | 51.1ohm |

+1.5V to +1.5VS\_VGA

Supply from Power-0411 update.



## POWER

F1 FB\_VDDQ\_SENSE

F2 FB\_GND\_SENSE

J27 FB\_CAL\_PD\_VDDQ

H27 FB\_CAL\_PU\_GND

H25 FB\_CAL\_TERM\_GND

N14M-GE2-B-AIO-A2

4/3 Update:  
Un-stuff R253, R256, R259 and R265.

+5VALW

DIS@ R268 100K\_0402\_5%

DGPU\_PWROK#

R272 0.0402\_5%

Q8 2N7002\_SOT23

R273 100K\_0402\_5%

[17.52.53] DGPU\_PWROK

DIS@ Q13 2N7002KW\_SOT323-3

R277 22\_0603\_5%

DIS@ C247 10U\_0603\_10V6K

DIS@ C248 10U\_0603\_10V6K

DIS@ C251 0.1U\_0603\_25V7K

DIS@ R282 2K\_0402\_1%

DIS@ R283 2K\_0402\_1%

DIS@ Q14 2N7002KW\_SOT323-3

DIS@ Q15 2N7002KW\_SOT323-3

DIS@ Q16 2N7002KW\_SOT323-3

DIS@ Q17 2N7002KW\_SOT323-3

DIS@ Q18 2N7002KW\_SOT323-3

DIS@ Q19 2N7002KW\_SOT323-3

DIS@ Q20 2N7002KW\_SOT323-3

DIS@ Q21 2N7002KW\_SOT323-3

DIS@ Q22 2N7002KW\_SOT323-3

DIS@ Q23 2N7002KW\_SOT323-3

DIS@ Q24 2N7002KW\_SOT323-3

DIS@ Q25 2N7002KW\_SOT323-3

DIS@ Q26 2N7002KW\_SOT323-3

DIS@ Q27 2N7002KW\_SOT323-3

DIS@ Q28 2N7002KW\_SOT323-3

DIS@ Q29 2N7002KW\_SOT323-3

DIS@ Q30 2N7002KW\_SOT323-3

DIS@ Q31 2N7002KW\_SOT323-3

DIS@ Q32 2N7002KW\_SOT323-3

DIS@ Q33 2N7002KW\_SOT323-3

DIS@ Q34 2N7002KW\_SOT323-3

DIS@ Q35 2N7002KW\_SOT323-3

DIS@ Q36 2N7002KW\_SOT323-3

DIS@ Q37 2N7002KW\_SOT323-3

DIS@ Q38 2N7002KW\_SOT323-3

DIS@ Q39 2N7002KW\_SOT323-3

DIS@ Q40 2N7002KW\_SOT323-3

DIS@ Q41 2N7002KW\_SOT323-3

DIS@ Q42 2N7002KW\_SOT323-3

DIS@ Q43 2N7002KW\_SOT323-3

DIS@ Q44 2N7002KW\_SOT323-3

DIS@ Q45 2N7002KW\_SOT323-3

DIS@ Q46 2N7002KW\_SOT323-3

DIS@ Q47 2N7002KW\_SOT323-3

DIS@ Q48 2N7002KW\_SOT323-3

DIS@ Q49 2N7002KW\_SOT323-3

DIS@ Q50 2N7002KW\_SOT323-3

DIS@ Q51 2N7002KW\_SOT323-3

DIS@ Q52 2N7002KW\_SOT323-3

DIS@ Q53 2N7002KW\_SOT323-3

DIS@ Q54 2N7002KW\_SOT323-3

DIS@ Q55 2N7002KW\_SOT323-3

DIS@ Q56 2N7002KW\_SOT323-3

DIS@ Q57 2N7002KW\_SOT323-3

DIS@ Q58 2N7002KW\_SOT323-3

DIS@ Q59 2N7002KW\_SOT323-3

DIS@ Q60 2N7002KW\_SOT323-3

DIS@ Q61 2N7002KW\_SOT323-3

DIS@ Q62 2N7002KW\_SOT323-3

DIS@ Q63 2N7002KW\_SOT323-3

DIS@ Q64 2N7002KW\_SOT323-3

DIS@ Q65 2N7002KW\_SOT323-3

DIS@ Q66 2N7002KW\_SOT323-3

DIS@ Q67 2N7002KW\_SOT323-3

DIS@ Q68 2N7002KW\_SOT323-3

DIS@ Q69 2N7002KW\_SOT323-3

DIS@ Q70 2N7002KW\_SOT323-3

DIS@ Q71 2N7002KW\_SOT323-3

DIS@ Q72 2N7002KW\_SOT323-3

DIS@ Q73 2N7002KW\_SOT323-3

DIS@ Q74 2N7002KW\_SOT323-3

DIS@ Q75 2N7002KW\_SOT323-3

DIS@ Q76 2N7002KW\_SOT323-3

DIS@ Q77 2N7002KW\_SOT323-3

DIS@ Q78 2N7002KW\_SOT323-3

DIS@ Q79 2N7002KW\_SOT323-3

DIS@ Q80 2N7002KW\_SOT323-3

DIS@ Q81 2N7002KW\_SOT323-3

DIS@ Q82 2N7002KW\_SOT323-3

DIS@ Q83 2N7002KW\_SOT323-3

DIS@ Q84 2N7002KW\_SOT323-3

DIS@ Q85 2N7002KW\_SOT323-3

DIS@ Q86 2N7002KW\_SOT323-3

DIS@ Q87 2N7002KW\_SOT323-3

DIS@ Q88 2N7002KW\_SOT323-3

DIS@ Q89 2N7002KW\_SOT323-3

DIS@ Q90 2N7002KW\_SOT323-3

DIS@ Q91 2N7002KW\_SOT323-3

DIS@ Q92 2N7002KW\_SOT323-3

DIS@ Q93 2N7002KW\_SOT323-3

DIS@ Q94 2N7002KW\_SOT323-3

DIS@ Q95 2N7002KW\_SOT323-3

DIS@ Q96 2N7002KW\_SOT323-3

DIS@ Q97 2N7002KW\_SOT323-3

DIS@ Q98 2N7002KW\_SOT323-3

DIS@ Q99 2N7002KW\_SOT323-3

DIS@ Q100 2N7002KW\_SOT323-3

DIS@ Q101 2N7002KW\_SOT323-3

DIS@ Q102 2N7002KW\_SOT323-3

DIS@ Q103 2N7002KW\_SOT323-3

DIS@ Q104 2N7002KW\_SOT323-3

DIS@ Q105 2N7002KW\_SOT323-3

DIS@ Q106 2N7002KW\_SOT323-3

DIS@ Q107 2N7002KW\_SOT323-3

DIS@ Q108 2N7002KW\_SOT323-3

DIS@ Q109 2N7002KW\_SOT323-3

DIS@ Q110 2N7002KW\_SOT323-3

DIS@ Q111 2N7002KW\_SOT323-3

DIS@ Q112 2N7002KW\_SOT323-3

DIS@ Q113 2N7002KW\_SOT323-3

DIS@ Q114 2N7002KW\_SOT323-3

DIS@ Q115 2N7002KW\_SOT323-3

DIS@ Q116 2N7002KW\_SOT323-3

DIS@ Q117 2N7002KW\_SOT323-3

DIS@ Q118 2N7002KW\_SOT323-3

DIS@ Q119 2N7002KW\_SOT323-3

DIS@ Q120 2N7002KW\_SOT323-3

DIS@ Q121 2N7002KW\_SOT323-3

DIS@ Q122 2N7002KW\_SOT323-3

DIS@ Q123 2N7002KW\_SOT323-3

DIS@ Q124 2N7002KW\_SOT323-3

DIS@ Q125 2N7002KW\_SOT323-3

DIS@ Q126 2N7002KW\_SOT323-3

DIS@ Q127 2N7002KW\_SOT323-3

DIS@ Q128 2N7002KW\_SOT323-3

DIS@ Q129 2N7002KW\_SOT323-3

DIS@ Q130 2N7002KW\_SOT323-3

DIS@ Q131 2N7002KW\_SOT323-3

DIS@ Q132 2N7002KW\_SOT323-3

DIS@ Q133 2N7002KW\_SOT323-3

DIS@ Q134 2N7002KW\_SOT323-3

DIS@ Q135 2N7002KW\_SOT323-3

DIS@ Q136 2N7002KW\_SOT323-3

DIS@ Q137 2N7002KW\_SOT323-3

DIS@ Q138 2N7002KW\_SOT323-3

DIS@ Q139 2N7002KW\_SOT323-3

DIS@ Q140 2N7002KW\_SOT323-3

DIS@ Q141 2N7002KW\_SOT323-3

DIS@ Q142 2N7002KW\_SOT323-3

DIS@ Q143 2N7002KW\_SOT323-3

DIS@ Q144 2N7002KW\_SOT323-3

DIS@ Q145 2N7002KW\_SOT323-3

DIS@ Q146 2N7002KW\_SOT323-3

DIS@ Q147 2N7002KW\_SOT323-3

DIS@ Q148 2N7002KW\_SOT323-3

DIS@ Q149 2N7002KW\_SOT323-3

DIS@ Q150 2N7002KW\_SOT323-3

DIS@ Q151 2N7002KW\_SOT323-3

DIS@ Q152 2N7002KW\_SOT323-3

DIS@ Q153 2N7002KW\_SOT323-3

DIS@ Q154 2N7002KW\_SOT323-3

DIS@ Q155 2N7002KW\_SOT323-3

DIS@ Q156 2N7002KW\_SOT323-3

DIS@ Q157 2N7002KW\_SOT323-3

DIS@ Q158 2N7002KW\_SOT323-3

DIS@ Q159 2N7002KW\_SOT323-3

DIS@ Q160 2N7002KW\_SOT323-3

DIS@ Q161 2N7002KW\_SOT323-3

DIS@ Q162 2N7002KW\_SOT323-3

DIS@ Q163 2N7002KW\_SOT323-3

DIS@ Q164 2N7002KW\_SOT323-3

DIS@ Q165 2N7002KW\_SOT323-3

DIS@ Q166 2N7002KW\_SOT323-3

DIS@ Q167 2N7002KW\_SOT323-3

DIS@ Q168 2N7002KW\_SOT323-3

DIS@ Q169 2N7002KW\_SOT323-3

DIS@ Q170 2N7002KW\_SOT323-3

DIS@ Q171 2N7002KW\_SOT323-3

DIS@ Q172 2N7002KW\_SOT323-3

DIS@ Q173 2N7002KW\_SOT323-3

DIS@ Q174 2N7002KW\_SOT323-3

DIS@ Q175 2N7002KW\_SOT323-3

DIS@ Q176 2N7002KW\_SOT323-3

DIS@ Q177 2N7002KW\_SOT323-3

DIS@ Q178 2N7002KW\_SOT323-3

DIS@ Q179 2N7002KW\_SOT323-3

DIS@ Q180 2N7002KW\_SOT323-3

DIS@ Q181 2N7002KW\_SOT323-3

DIS@ Q182 2N7002KW\_SOT323-3

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DIS@ Q184 2N7002KW\_SOT323-3

DIS@ Q185 2N7002KW\_SOT323-3

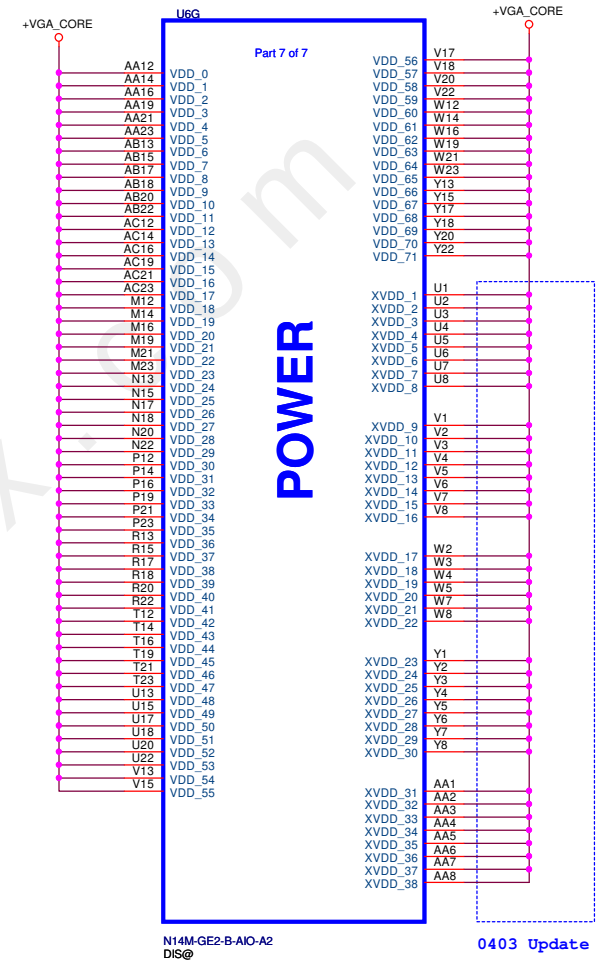
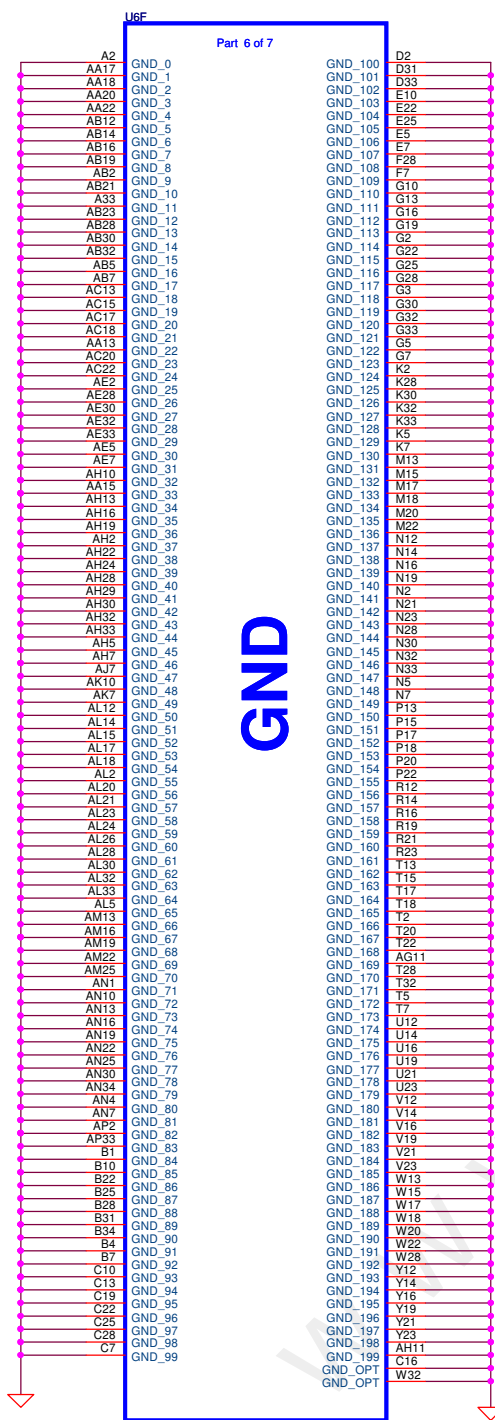
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DIS@ Q189 2N7002KW\_SOT323-3

DIS@ Q190 2N7002KW\_SOT323-3

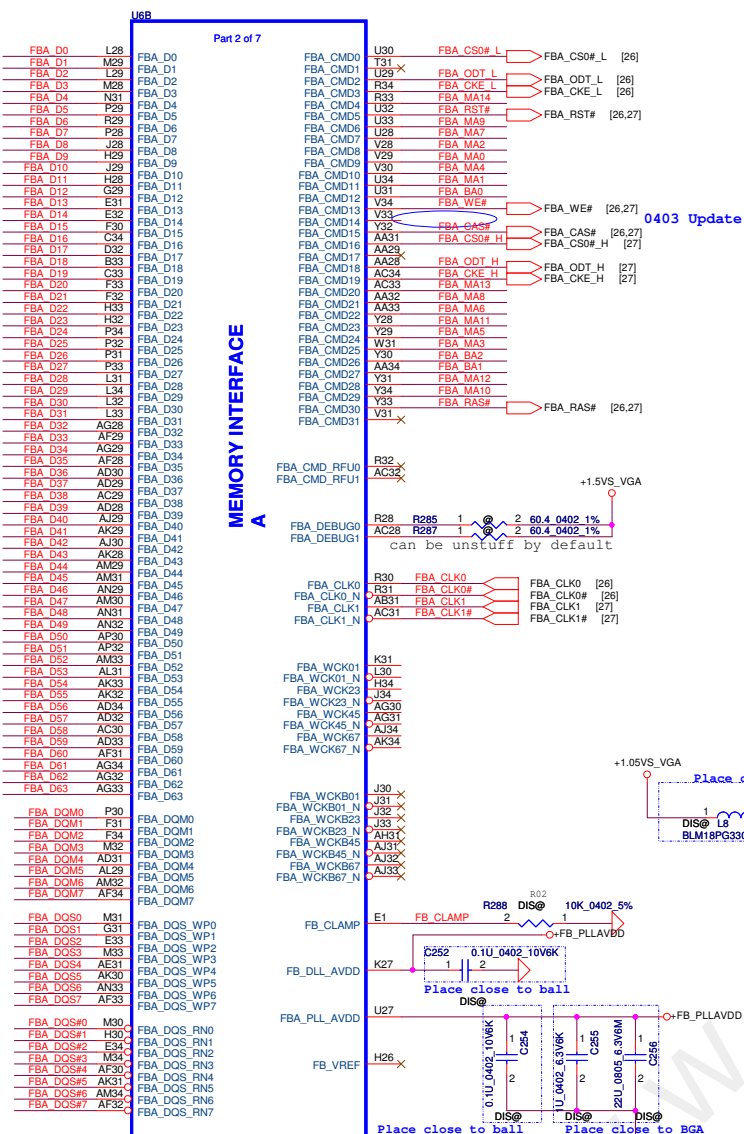


0403 Update

|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  |                                 |  |                                   |  |                               |                           |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------|--|---------------------------------|--|-----------------------------------|--|-------------------------------|---------------------------|-----|
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |  | Compal Secret Data |  | <b>Compal Electronics, Inc.</b> |  |                                   |  |                               |                           |     |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  | 2013/04/01         |  | Deciphered Date                 |  | 2014/04/01                        |  | Title                         |                           |     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  |                                 |  |                                   |  | <b>N14M-GE2-VGA CORE, GND</b> |                           |     |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  |                                 |  |                                   |  |                               | <b>ZEA00 LA-A061P M/B</b> | 0.3 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  |                                 |  |                                   |  |                               |                           |     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  | 3                  |  | 2                               |  | Date: Tuesday, September 24, 2013 |  | Sheet 24 of 59                |                           |     |

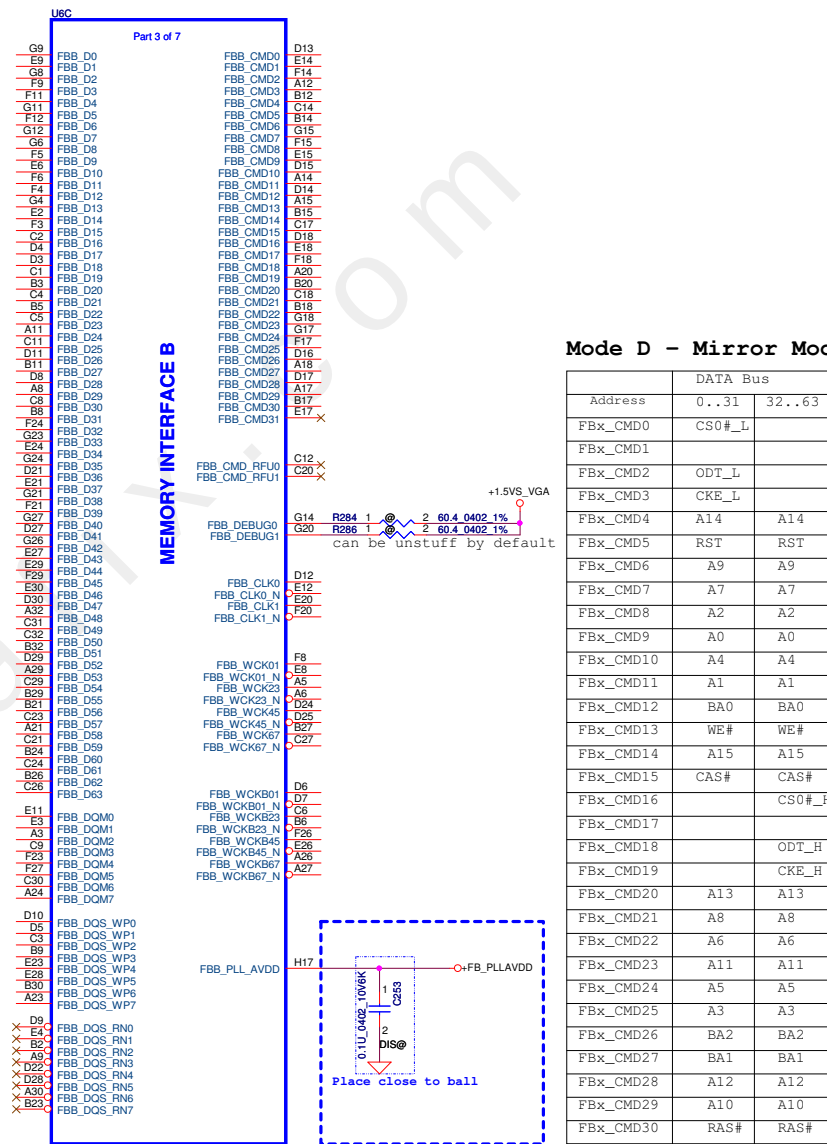


[26,27] FBA\_D[0..63] ← FBA\_D[0..63] [26,27]  
FBA\_MA[14..0] [26,27]  
FBA\_BA[2..0] [26,27]



[26,27] FBA\_DOM[7..0]  
[26,27] FBA\_DQS[7..0]  
[26,27] FBA\_DQS# [7..0]

30ohms (ESR=0.01) Bead  
P/N: SM010007W00



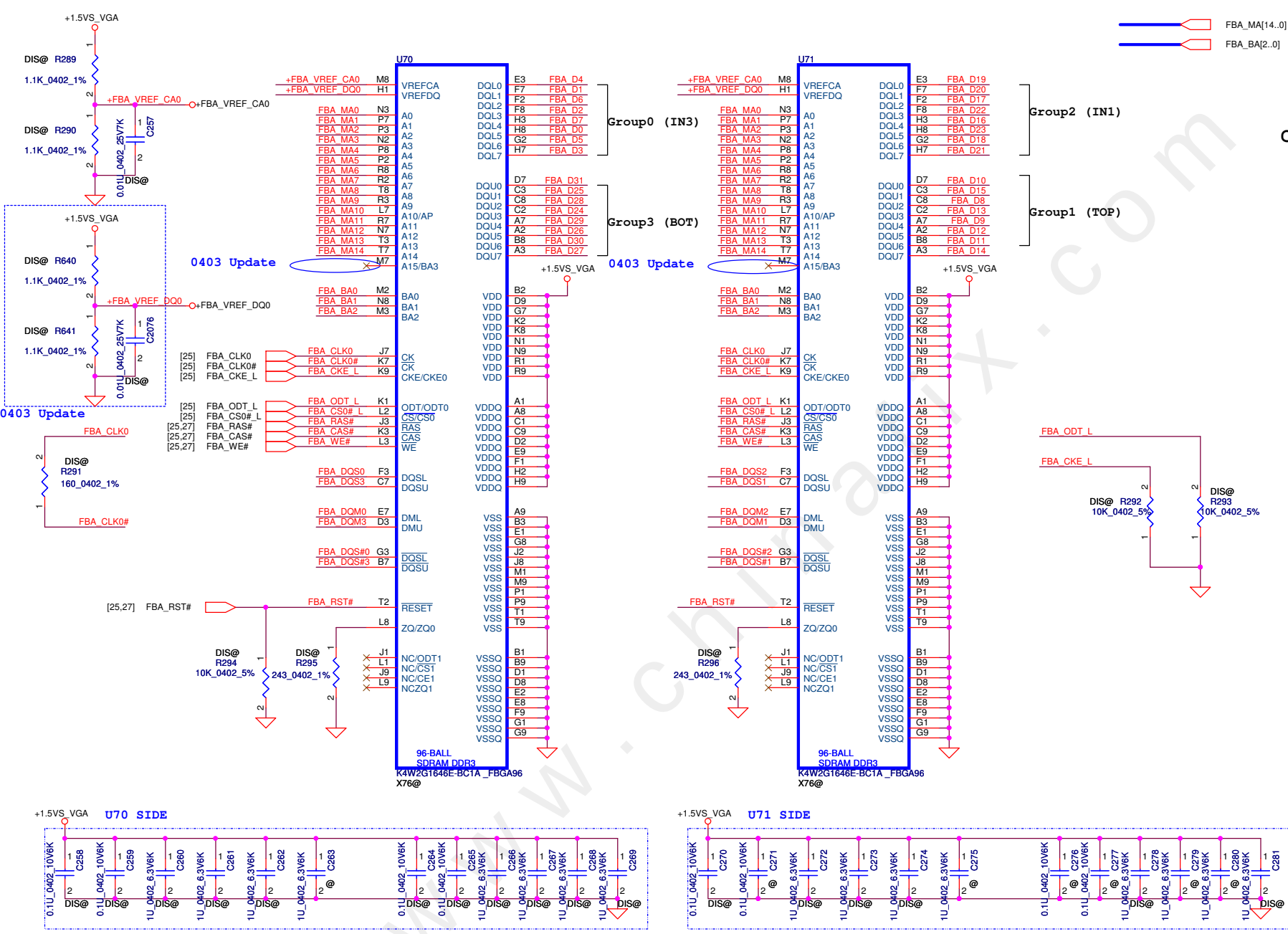
N14M-GE2-B-AIO-A2  
DIS@  
Must connect to power when partition B unused!

## Mode D - Mirror Mode Mapping

| Address   | DATA Bus |
|-----------|----------|
| FBx_CMD0  | CS0#_L   |
| FBx_CMD1  | ODT_L    |
| FBx_CMD2  | CKE_L    |
| FBx_CMD3  | A14      |
| FBx_CMD4  | A9       |
| FBx_CMD5  | A7       |
| FBx_CMD6  | A2       |
| FBx_CMD7  | A0       |
| FBx_CMD8  | A4       |
| FBx_CMD9  | A1       |
| FBx_CMD10 | BA0      |
| FBx_CMD11 | WE#      |
| FBx_CMD12 | CAS#     |
| FBx_CMD13 | CS0#_H   |
| FBx_CMD14 | ODT_H    |
| FBx_CMD15 | CKE_H    |
| FBx_CMD16 | A13      |
| FBx_CMD17 | A8       |
| FBx_CMD18 | A6       |
| FBx_CMD19 | A11      |
| FBx_CMD20 | A5       |
| FBx_CMD21 | A3       |
| FBx_CMD22 | BA2      |
| FBx_CMD23 | BA1      |
| FBx_CMD24 | A12      |
| FBx_CMD25 | A10      |
| FBx_CMD26 | RAS#     |
| FBx_CMD27 |          |
| FBx_CMD28 |          |
| FBx_CMD29 |          |
| FBx_CMD30 |          |
| FBx_CMD31 |          |

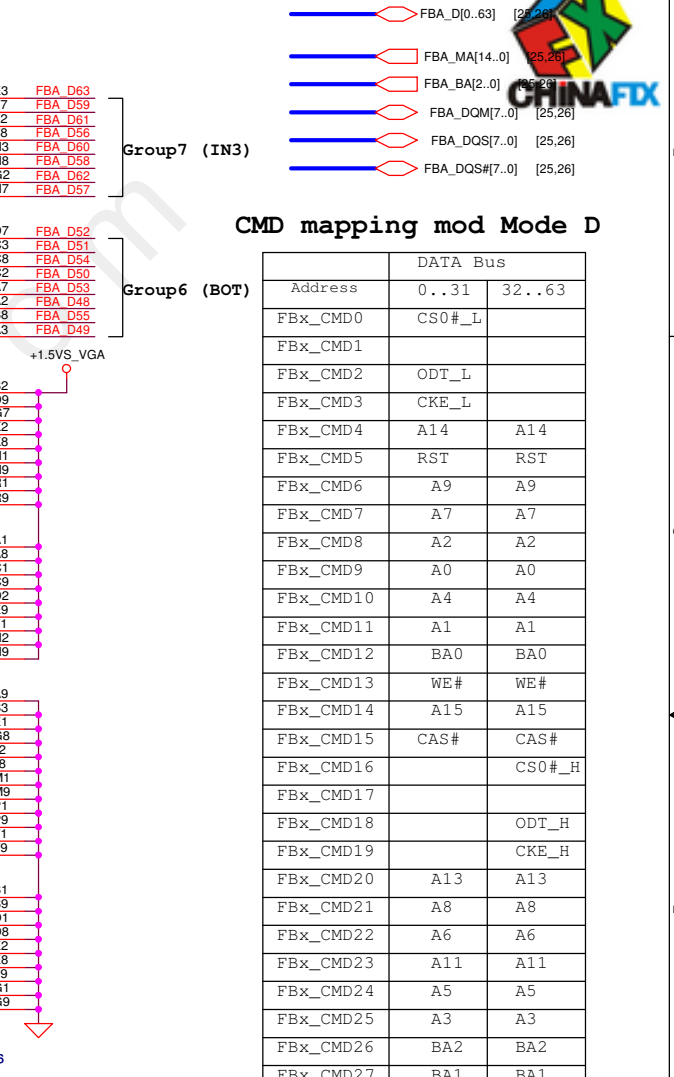
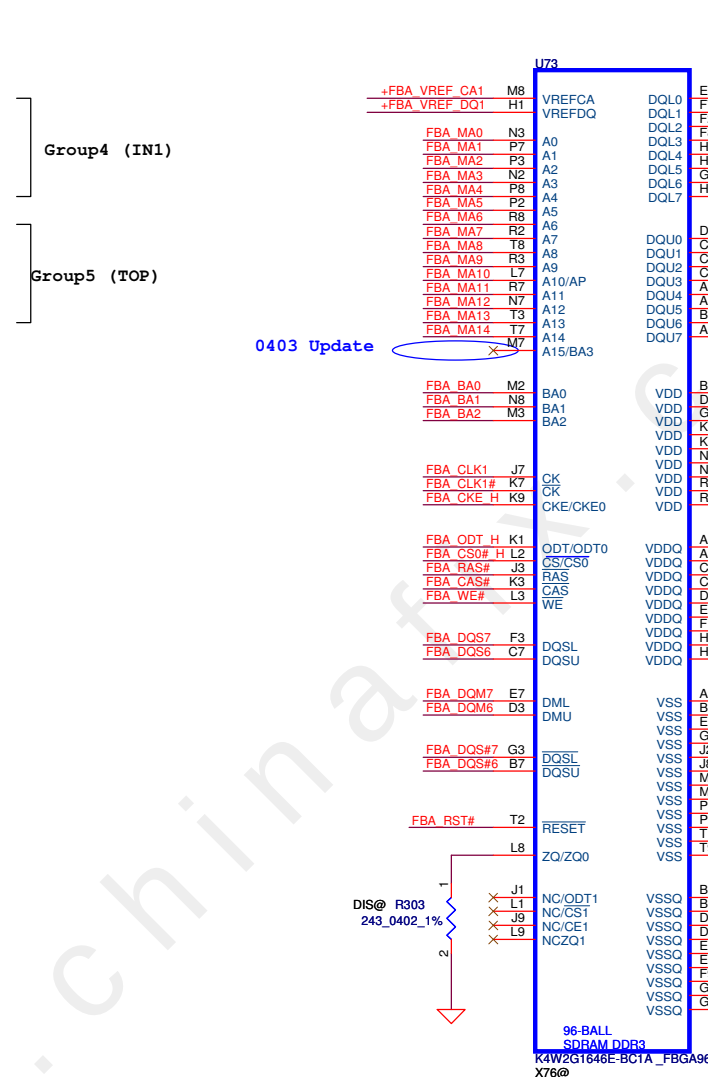
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       | Compal Secret Data          | Title           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------|
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2013/04/01                  | Deciphered Date |
| 2014/04/01                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                             |                 |
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| Compal Electronics, Inc.                                                                                                                                                                                                                                                                                                                                                                                                                                      |                             |                 |
| N14M-GE2-MEM Interface                                                                                                                                                                                                                                                                                                                                                                                                                                        |                             |                 |
| Size                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Document Number             | Rev             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ZEA00 LA-A061P M/B          | 0.3             |
| Date:                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Tuesday, September 24, 2013 | Sheet 25 of 59  |

Memory Partition A - Lower 32 bits

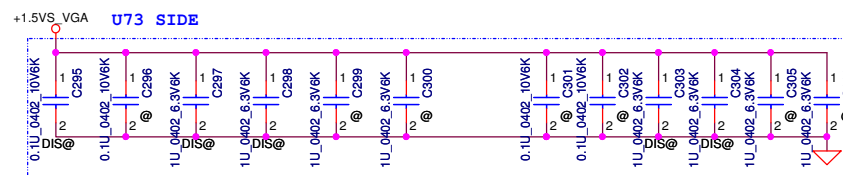


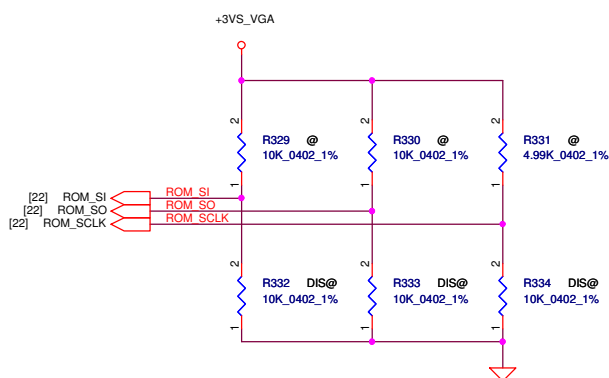
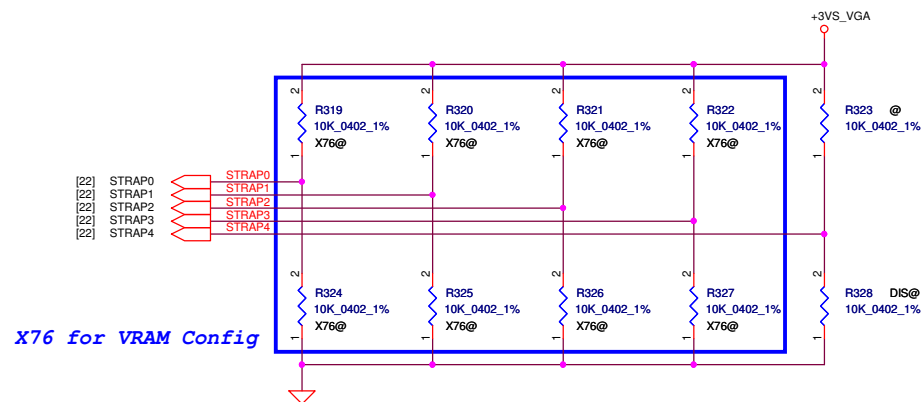
CMD mapping mod Mode D

| DATA Bus  |              |
|-----------|--------------|
| Address   | 0..31 32..63 |
| FBx_CMD0  | CS0#_L       |
| FBx_CMD1  |              |
| FBx_CMD2  | ODT_L        |
| FBx_CMD3  | CKE_L        |
| FBx_CMD4  | A14 A14      |
| FBx_CMD5  | RST RST      |
| FBx_CMD6  | A9 A9        |
| FBx_CMD7  | A7 A7        |
| FBx_CMD8  | A2 A2        |
| FBx_CMD9  | A0 A0        |
| FBx_CMD10 | A4 A4        |
| FBx_CMD11 | A1 A1        |
| FBx_CMD12 | BA0 BA0      |
| FBx_CMD13 | WE# WE#      |
| FBx_CMD14 | A15 A15      |
| FBx_CMD15 | CAS# CAS#    |
| FBx_CMD16 | CS0#_H       |
| FBx_CMD17 |              |
| FBx_CMD18 | ODT_H        |
| FBx_CMD19 | CKE_H        |
| FBx_CMD20 | A13 A13      |
| FBx_CMD21 | A8 A8        |
| FBx_CMD22 | A6 A6        |
| FBx_CMD23 | A11 A11      |
| FBx_CMD24 | A5 A5        |
| FBx_CMD25 | A3 A3        |
| FBx_CMD26 | BA2 BA2      |
| FBx_CMD27 | BA1 BA1      |
| FBx_CMD28 | A12 A12      |
| FBx_CMD29 | A10 A10      |
| FBx_CMD30 | RAS# RAS#    |



|  |           | DATA Bus |        |
|--|-----------|----------|--------|
|  | Address   | 0..31    | 32..63 |
|  | FBx_CMD0  | CS0#_L   |        |
|  | FBx_CMD1  |          |        |
|  | FBx_CMD2  | ODT_L    |        |
|  | FBx_CMD3  | CKE_L    |        |
|  | FBx_CMD4  | A14      | A14    |
|  | FBx_CMD5  | RST      | RST    |
|  | FBx_CMD6  | A9       | A9     |
|  | FBx_CMD7  | A7       | A7     |
|  | FBx_CMD8  | A2       | A2     |
|  | FBx_CMD9  | A0       | A0     |
|  | FBx_CMD10 | A4       | A4     |
|  | FBx_CMD11 | A1       | A1     |
|  | FBx_CMD12 | BA0      | BA0    |
|  | FBx_CMD13 | WE#      | WE#    |
|  | FBx_CMD14 | A15      | A15    |
|  | FBx_CMD15 | CAS#     | CAS#   |
|  | FBx_CMD16 |          | CS0#_H |
|  | FBx_CMD17 |          |        |
|  | FBx_CMD18 |          | ODT_H  |
|  | FBx_CMD19 |          | CKE_H  |
|  | FBx_CMD20 | A13      | A13    |
|  | FBx_CMD21 | A8       | A8     |
|  | FBx_CMD22 | A6       | A6     |
|  | FBx_CMD23 | A11      | A11    |
|  | FBx_CMD24 | A5       | A5     |
|  | FBx_CMD25 | A3       | A3     |
|  | FBx_CMD26 | BA2      | BA2    |
|  | FBx_CMD27 | BA1      | BA1    |
|  | FBx_CMD28 | A12      | A12    |
|  | FBx_CMD29 | A10      | A10    |
|  | FBx_CMD30 | RAS#     | RAS#   |





[PUN-06026-001]

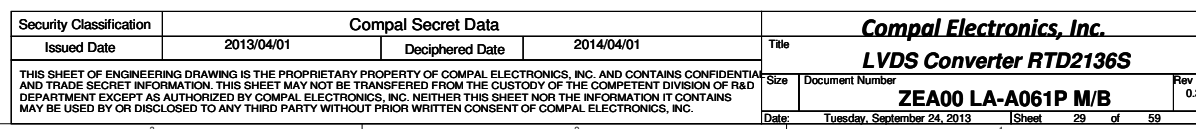
Table 4. Binary Strap Mode Mapping

| Strap Pin Name | Strap Mapping  | Resistance   | Polarity                                                               |
|----------------|----------------|--------------|------------------------------------------------------------------------|
| ROM_SCLK       | SMB_ALT_ADDR   | 10k $\Omega$ | Pull-down to GND                                                       |
| ROM_SI         | SUB_VENDOR     | 10k $\Omega$ | Pull-up to 3V3 if VBIOS ROM exists<br>Pull-down to GND if no VBIOS ROM |
| ROM_SO         | VGA_DEVICE     | 10k $\Omega$ | Pull-down to GND (no display)                                          |
| STRAP0         | RAM_CFG[0]     | 10k $\Omega$ | See Note                                                               |
| STRAP1         | RAM_CFG[1]     | 10k $\Omega$ | See Note                                                               |
| STRAP2         | RAM_CFG[2]     | 10k $\Omega$ | See Note                                                               |
| STRAP3         | RAM_CFG[3]     | 10k $\Omega$ | See Note                                                               |
| STRAP4         | PCIE_MAX_SPEED | 10k $\Omega$ | Pull-down to GND                                                       |

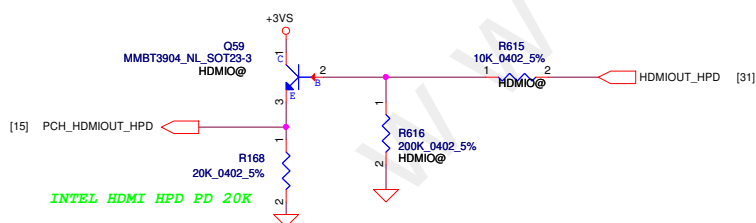
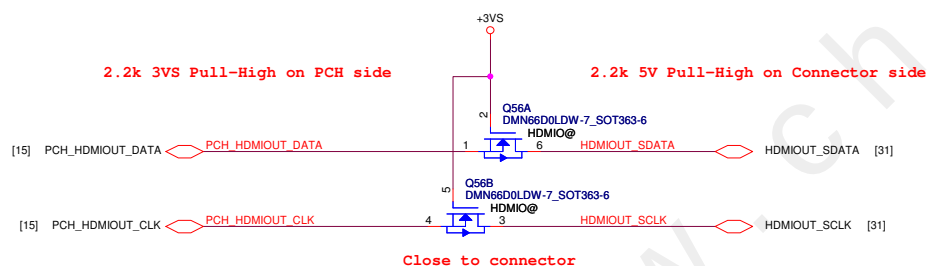
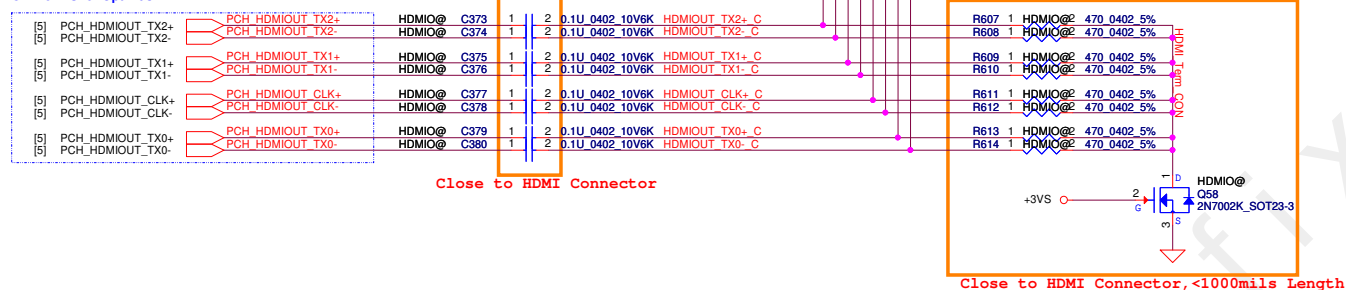
[VRAM Config-RVL-06366-001]

| GPU      | Frenq.  | Memory Size        | Memory Config                                       | strap3              | strap2              | strap1              | strap0              |
|----------|---------|--------------------|-----------------------------------------------------|---------------------|---------------------|---------------------|---------------------|
| N14M-GE2 | 900 MHz | 128M* 16* 4<br>1GB | Hynix (0x6)<br>H5TQ2G63BFR-11C<br>SA00003YO10       | 0<br>R327<br>PD 10K | 1<br>R321<br>PU 10K | 1<br>R320<br>PU 10K | 0<br>R324<br>PD 10K |
|          |         |                    | Samsung (0x5)<br>K4W2G1646E-BC11<br>SA00005SH00     | 0<br>R327<br>PD 10K | 1<br>R321<br>PU 10K | 0<br>R325<br>PD 10K | 1<br>R319<br>PU 10K |
|          |         |                    | Micron (0x1)<br>MT41J128M16JT-107G:K<br>SA00005SM30 | 0<br>R327<br>PD 10K | 0<br>R326<br>PD 10K | 0<br>R325<br>PD 10K | 1<br>R319<br>PU 10K |
| N14M-GE2 | 900 MHz | 256M* 16* 4<br>2GB | Micron (0xD)<br>MT41K256M16HA-107G:E<br>SA000065D20 | 1<br>R322<br>PU 10K | 1<br>R321<br>PU 10K | 0<br>R325<br>PD 10K | 1<br>R319<br>PU 10K |
|          |         |                    | Samsung (0xB)<br>K4W4G1646B-HC11<br>SA000068R10     | 1<br>R322<br>PU 10K | 0<br>R326<br>PD 10K | 1<br>R320<br>PU 10K | 1<br>R319<br>PU 10K |
|          |         |                    | Hynix (0x4)<br>H5TC4G63AFR-11C<br>SA00006E800       | 0<br>R327<br>PD 10K | 1<br>R321<br>PU 10K | 0<br>R325<br>PD 10K | 0<br>R324<br>PD 10K |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  |                 |  |                          |  |               |  |                             |  |                |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------|--|-----------------|--|--------------------------|--|---------------|--|-----------------------------|--|----------------|--|
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |  | Compal Secret Data |  |                 |  | Compal Electronics, Inc. |  |               |  |                             |  |                |  |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  | 2013/04/01         |  | Deciphered Date |  | 2014/04/01               |  | Title         |  |                             |  |                |  |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  |                 |  |                          |  | Size          |  | Document Number             |  | Rev            |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  |                 |  |                          |  |               |  | ZEA00 LA-A061P M/B          |  | 0.3            |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |                    |  |                 |  |                          |  | Date:         |  | Tuesday, September 24, 2013 |  | Sheet 28 of 59 |  |

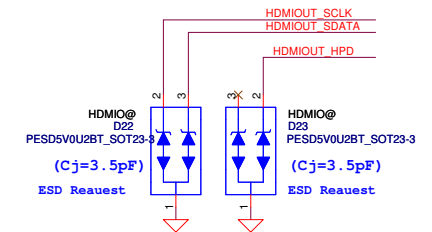
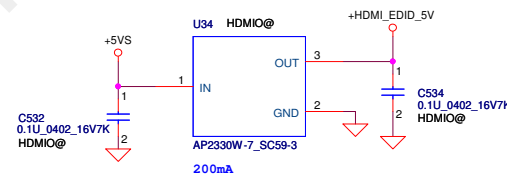
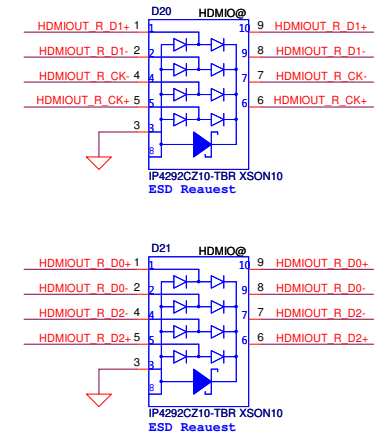
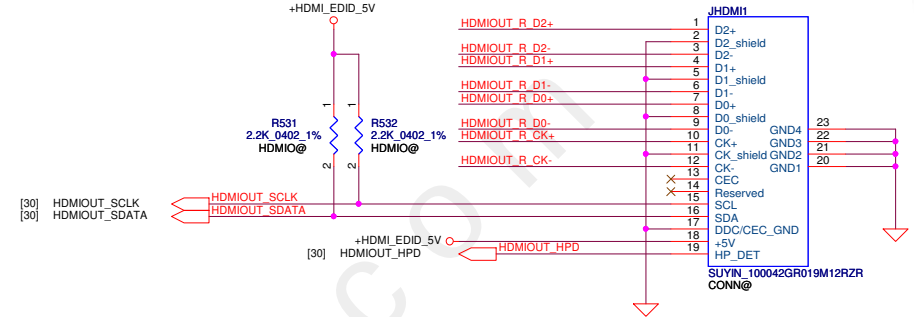
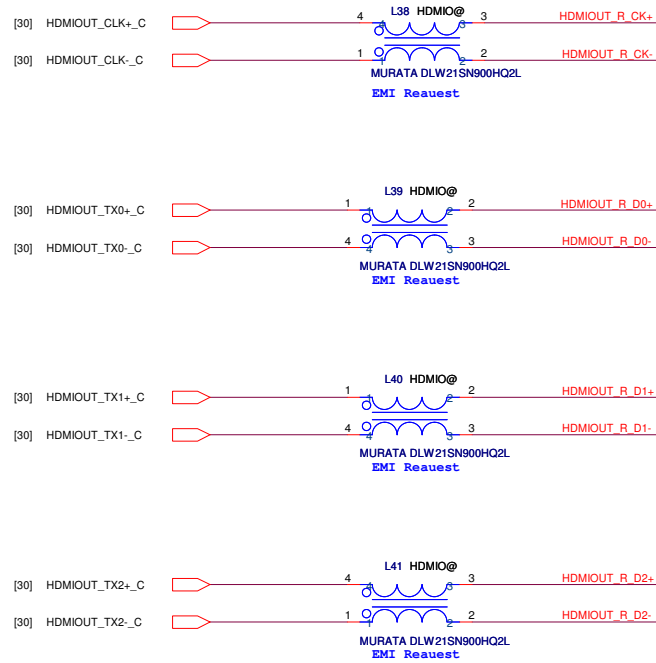


# UMA & DIS for Optimus

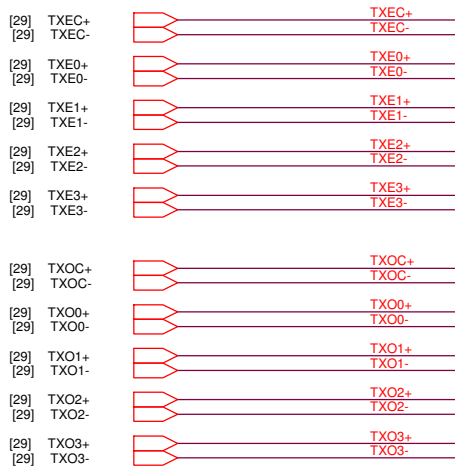


|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                             |                    |                    |                          |     |
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| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |                             | Compal Secret Data |                    | Compal Electronics, Inc. |     |
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| Size                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Custom                      | Document Number    | ZEA00 LA-A061P M/B |                          | Rev |
| Date:                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Tuesday, September 24, 2013 | Sheet              | 30                 | of                       | 59  |

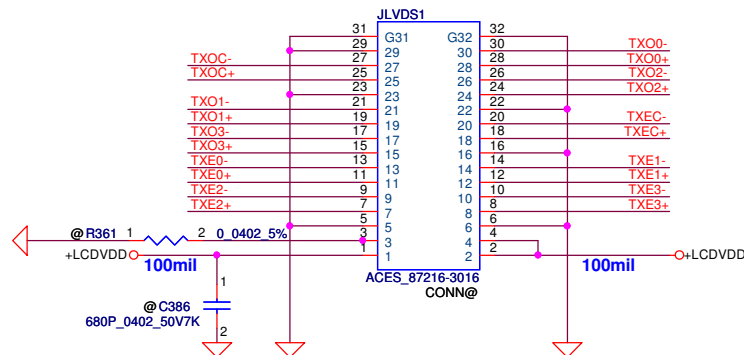
# HDMI-OUT Connector



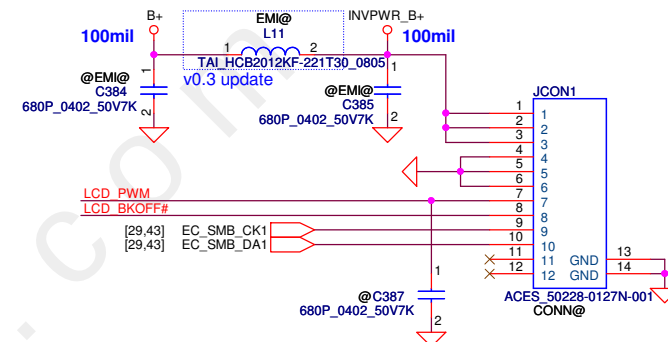
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            |                             |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------|------------|-----------------------------|-----|
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       | Compal Secret Data |                 |            | Compal Electronics, Inc.    |     |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2013/04/01         | Deciphered Date | 2014/04/01 | Title                       |     |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | Custom                      | 0.3 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | Document Number             |     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | ZEA00 LA-A061P M/B          |     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | Date                        |     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | Tuesday, September 24, 2013 |     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | Sheet                       |     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | 31                          |     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | of                          |     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | 59                          |     |



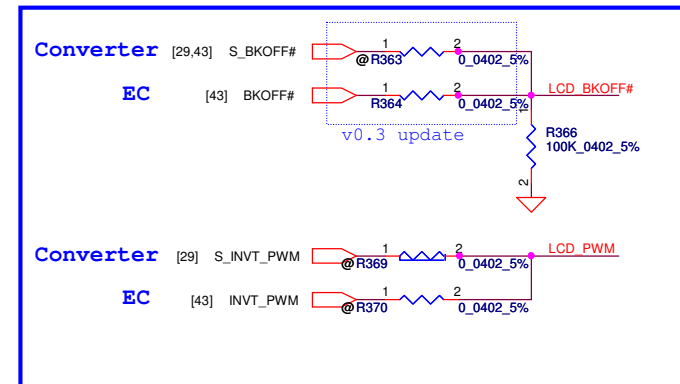
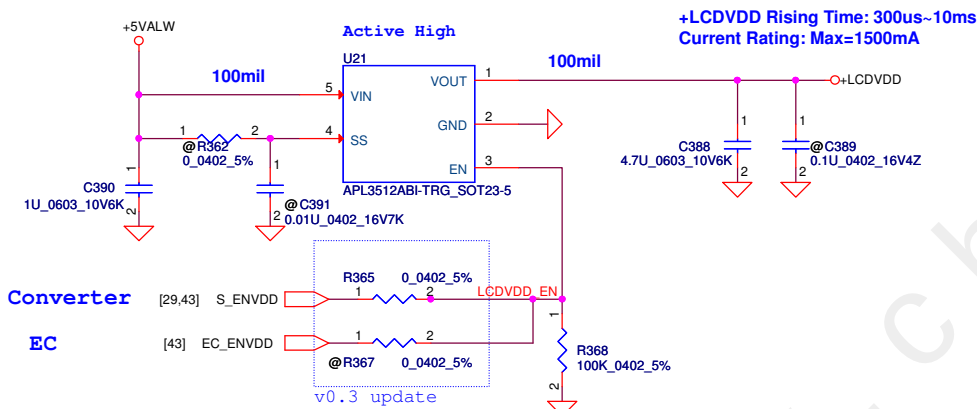
## LVDS Conn.



## Converter

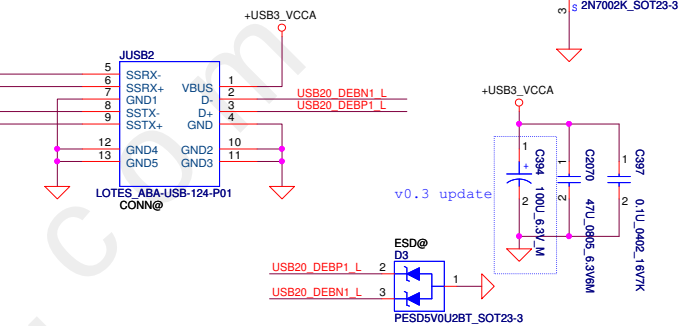
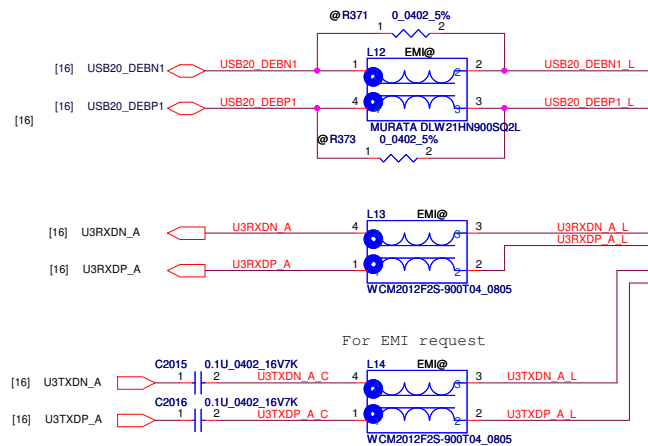
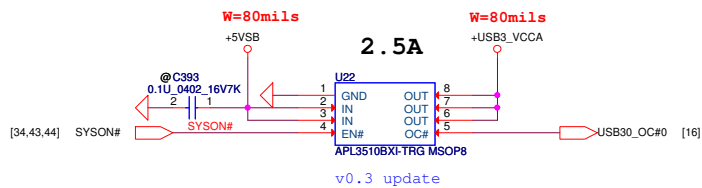


INVPWR\_B+:  
Current Rating: Max=375mA

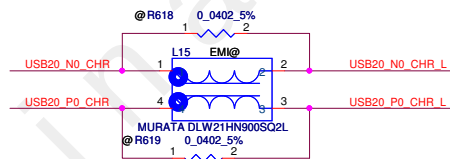
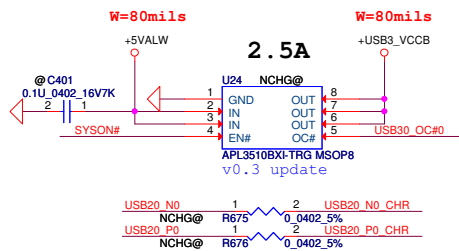


| Security Classification |  |  |  | Compal Secret Data |  |                 |  | Compal Electronics, Inc. |  |                                   |  |
|-------------------------|--|--|--|--------------------|--|-----------------|--|--------------------------|--|-----------------------------------|--|
| Issued Date             |  |  |  | 2013/04/01         |  | Deciphered Date |  | 2014/04/01               |  | Title                             |  |
|                         |  |  |  |                    |  |                 |  |                          |  | LVDS                              |  |
|                         |  |  |  |                    |  |                 |  |                          |  | ZEA00 LA-A061P M/B                |  |
|                         |  |  |  |                    |  |                 |  |                          |  | Rev 0.3                           |  |
|                         |  |  |  |                    |  |                 |  |                          |  | Date: Tuesday, September 24, 2013 |  |
|                         |  |  |  |                    |  |                 |  |                          |  | Sheet 32 of 59                    |  |

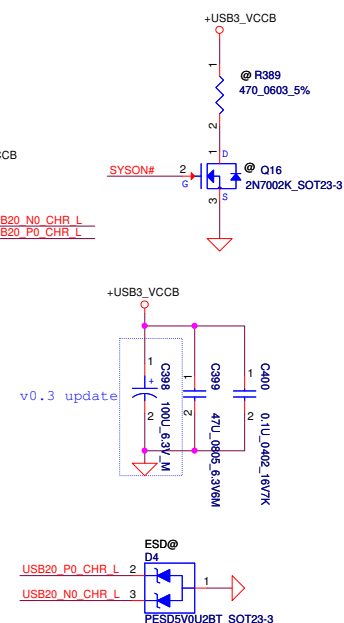
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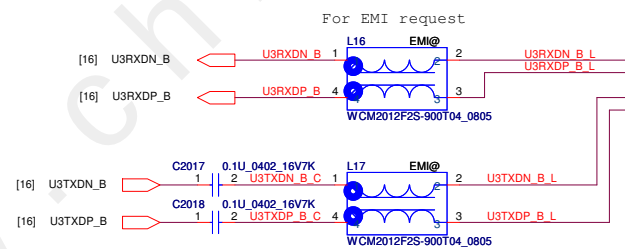
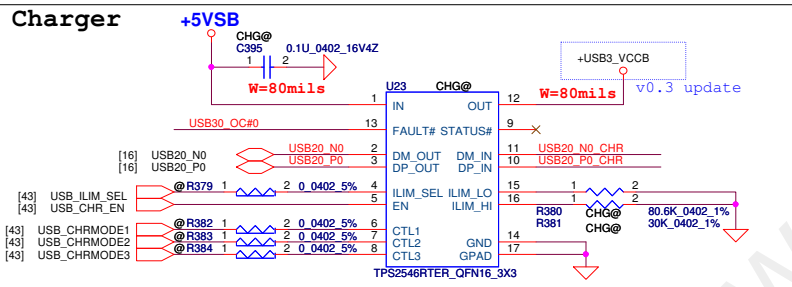
## Non Changer



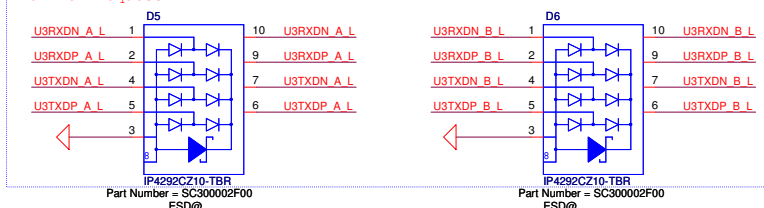
## Charge USB Port



## Charger



## For ESD request



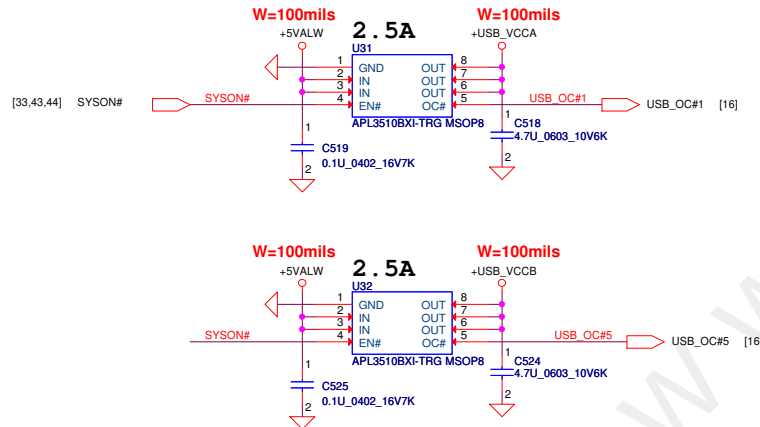
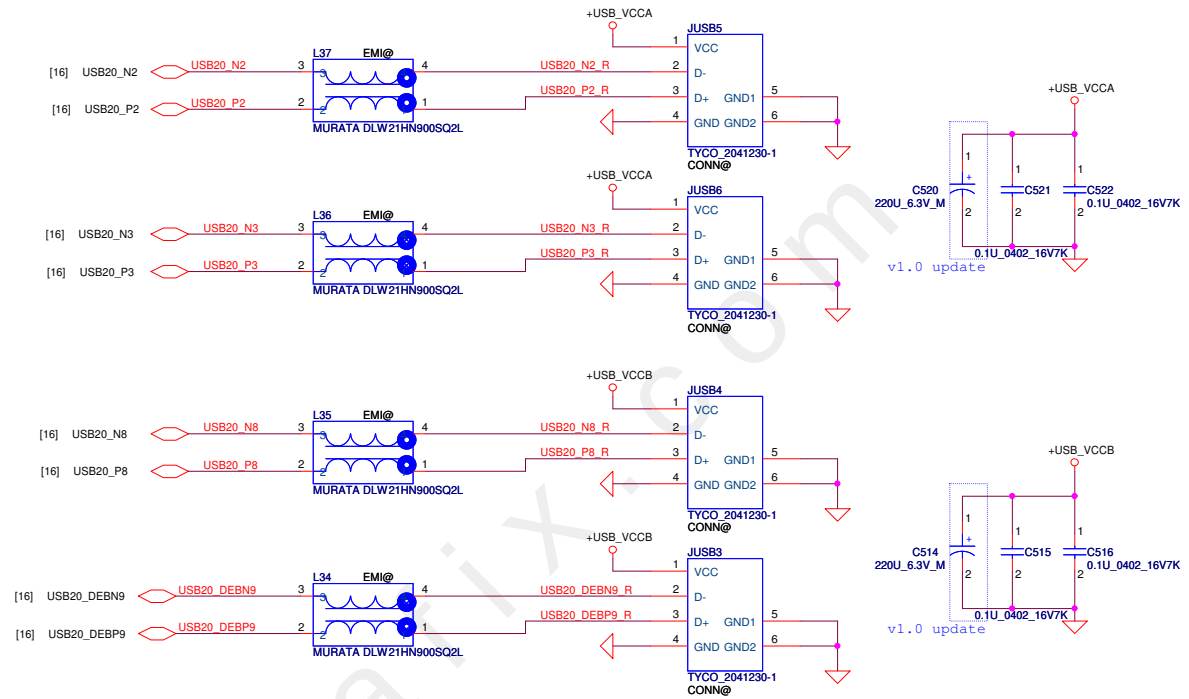
| Charger CT  | CTL1              | CTL2           | CTL3              | ILIM_SEL       |
|-------------|-------------------|----------------|-------------------|----------------|
| EC GPIO     | GPXIOA07 (pin104) | GPIO22 (pin41) | GPXIOA11 (pin108) | GPIO21 (pin40) |
| S0 (CDP)    | 1                 | 1              | 1                 | 1              |
| S3 (SDP)    | 1                 | 1              | 1                 | 0              |
| S4/S5 (DCP) | 0                 | 0              | 1                 | 1              |

| System Global Power State | TPS2546/TPS2544 Mode                                                                      | Charging | CTL1 | CTL2 | CTL3 | ILIM_SEL | Current Limit Setting |
|---------------------------|-------------------------------------------------------------------------------------------|----------|------|------|------|----------|-----------------------|
| S3                        | SDP, no discharge to / from CDP                                                           |          | 1    | 1    | 1    | 0        | ILIM_LO               |
| S0                        | CDP, load detection with ILIM_LO + 60mA thresholds or if a BC1.2 primary detection occurs |          | 1    | 1    | 1    | 1        | ILIM_HI               |
| S4/S5                     | Auto mode, load detection with power wake thresholds, no mouse wake                       |          | 0    | 0    | 1    | 1        | ILIM_HI               |

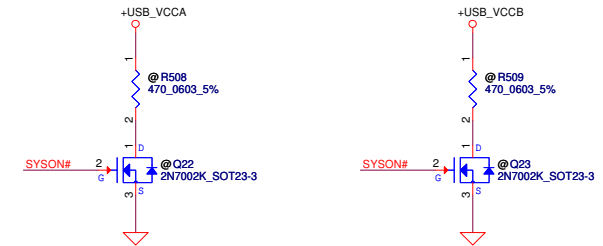
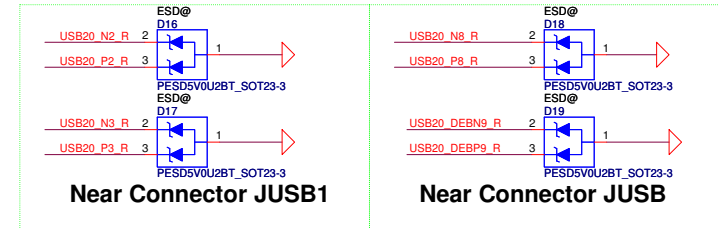
| Security Classification |  |  | Compal Secret Data |  |  | Compal Electronics, Inc. |  |  |
|-------------------------|--|--|--------------------|--|--|--------------------------|--|--|
| Issued Date             |  |  | Deciphered Date    |  |  | Title                    |  |  |
|                         |  |  |                    |  |  | USB 3.0 CONN             |  |  |
|                         |  |  |                    |  |  | Size Custom              |  |  |
|                         |  |  |                    |  |  | Document Number          |  |  |
|                         |  |  |                    |  |  | ZEA00 LA-A061P M/B       |  |  |
|                         |  |  |                    |  |  | Rev                      |  |  |
|                         |  |  |                    |  |  | 0.3                      |  |  |
|                         |  |  |                    |  |  | Date:                    |  |  |
|                         |  |  |                    |  |  | Sheet 33 of 59           |  |  |

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# USB20

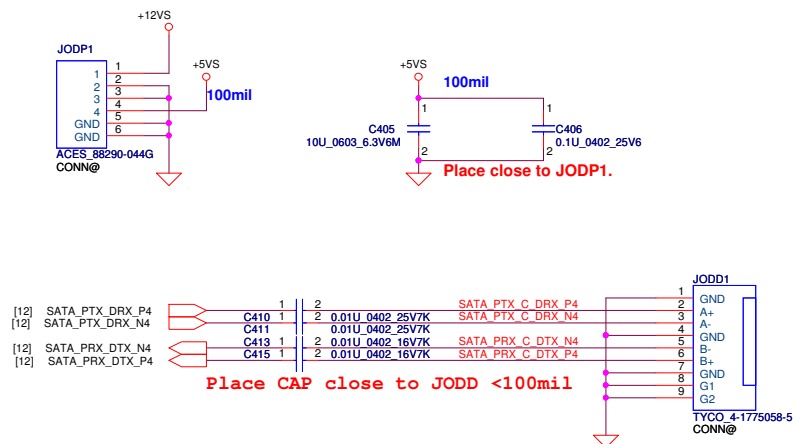


## For USB2.0 ESD diode

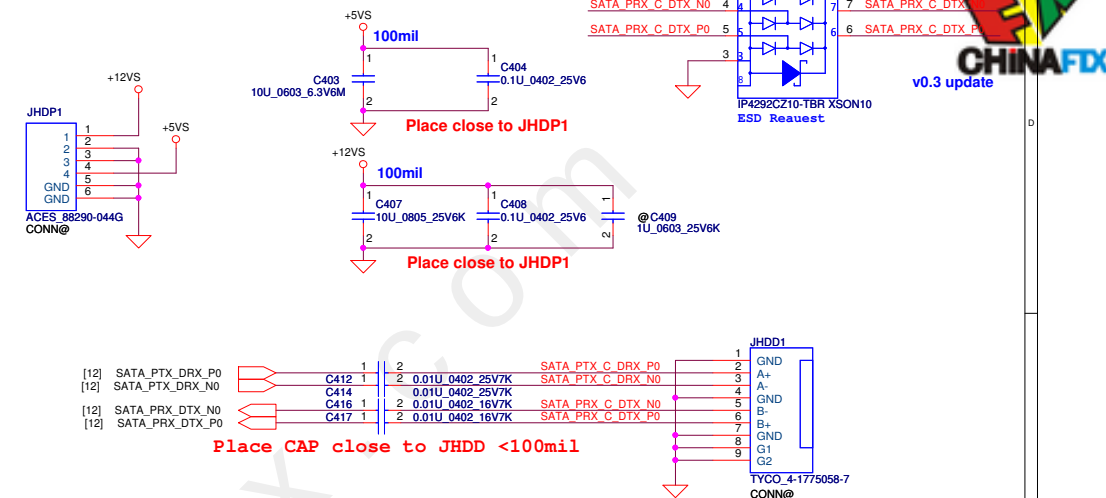


|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            |                          |                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------|------------|--------------------------|-----------------------------|
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | Compal Secret Data |            | Compal Electronics, Inc. |                             |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Date                     | Tuesday, September 24, 2013 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Sheet                    | 34 of 59                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Rev                      | 0.3                         |

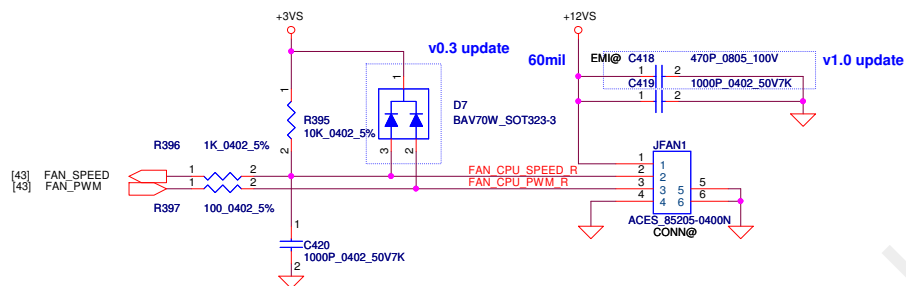
## SATA ODD Conn



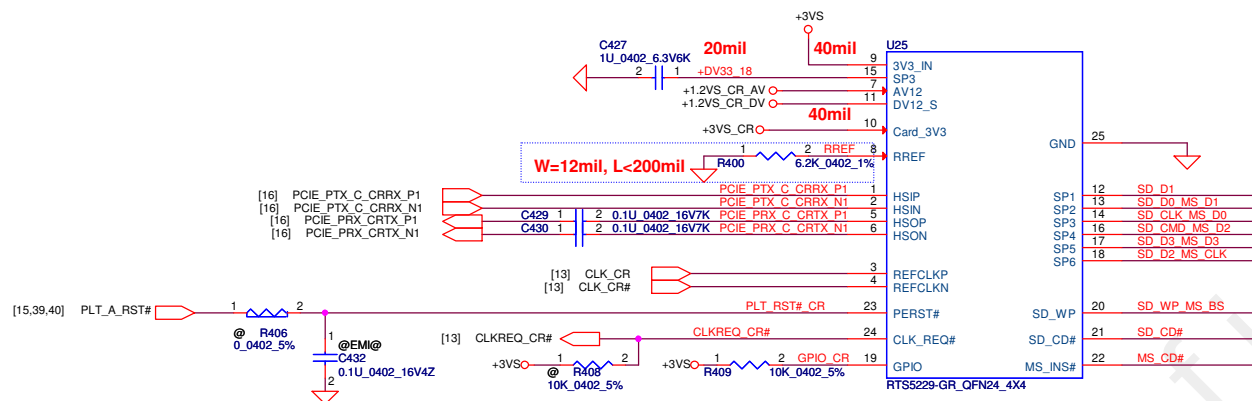
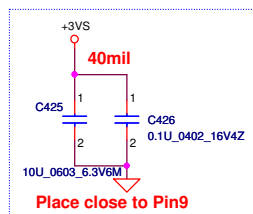
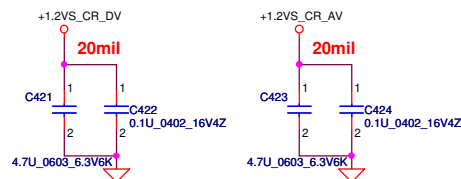
## SATA HDD Conn.



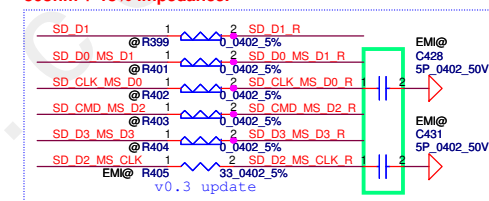
## FAN Control Circuit



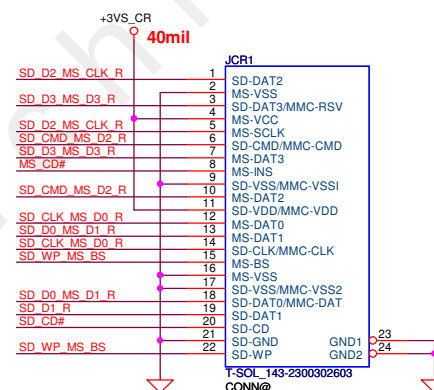
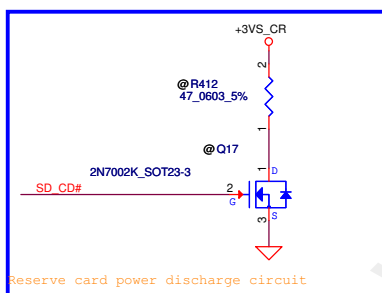
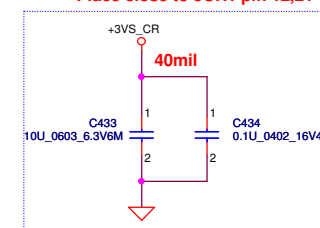
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            |                          |                             |                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------|------------|--------------------------|-----------------------------|----------------|
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       | Compal Secret Data |                 |            | Compal Electronics, Inc. |                             |                |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2013/04/01         | Deciphered Date | 2014/04/01 | Title                    | SATA-HDD/ODD/USB            |                |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            |                          | ZEA00 LA-A061P M/B          | 0.3            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | Date:                    | Tuesday, September 24, 2013 | Sheet 35 of 59 |



Length of per trace 2inch no more 2 via  
mismatch trace length <100mil  
50ohm +-15% impedance.

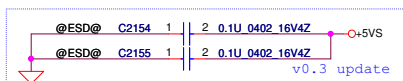


Place close to JCR1 pin 12,21

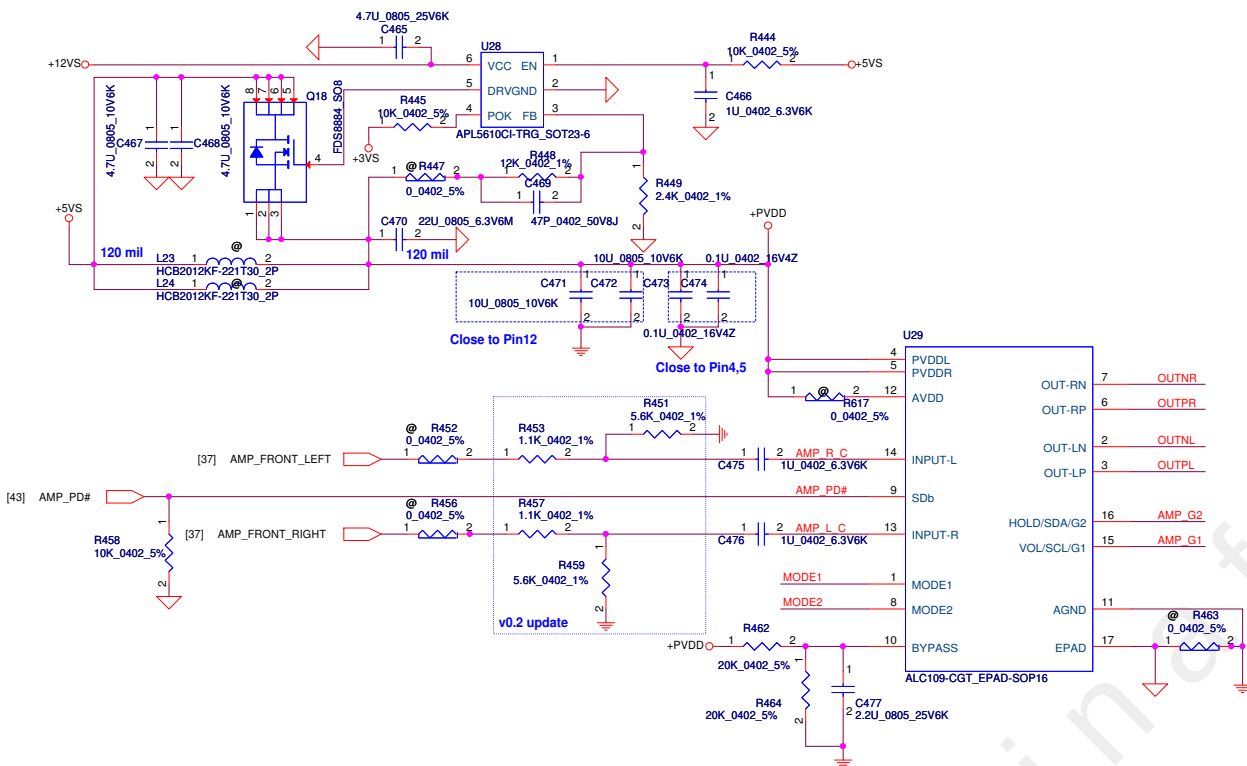


|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            |                               |                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------|------------|-------------------------------|-----------------------------|
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | Compal Secret Data |            | Compal Electronics, Inc.      |                             |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2013/04/01 | Deciphered Date    | 2014/04/01 | RTS5229 Media Card Controller |                             |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Custom                        | 0.3                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Document Number               | ZEA00 LA-A061P M/B          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Date:                         | Tuesday, September 24, 2013 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Sheet                         | 36 of 59                    |

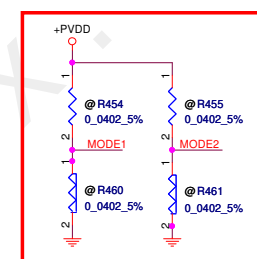




$V_o = 0.8 (1 + R_{606} / R_{607})$   
Output: 4.8V  
Max I: 7.5A

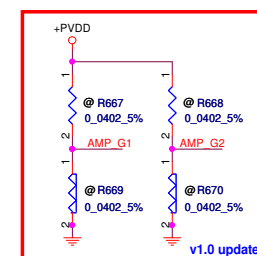


Mode selet: Fix Gain

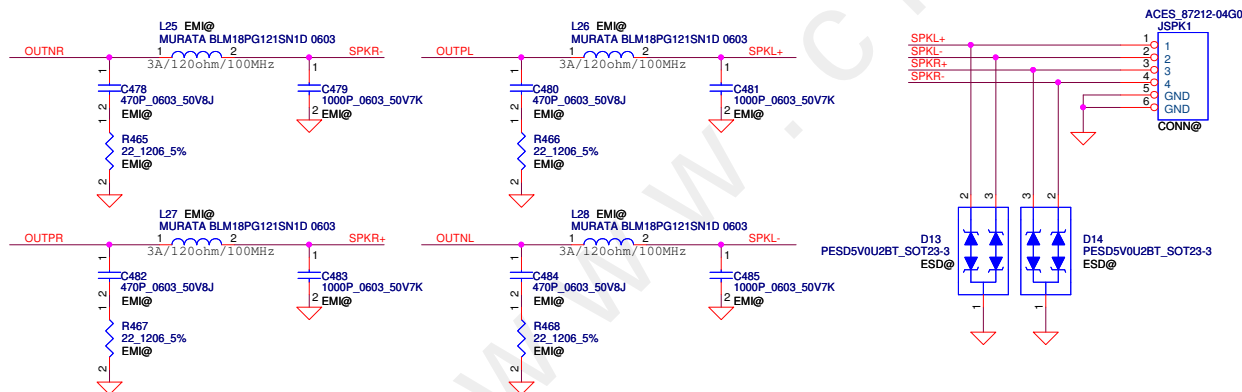


| Model1 | Model2 | Option     | Pin15 | Pin16 |
|--------|--------|------------|-------|-------|
| 0      | 0      | Fixed Gain | G1    | G2    |
| 0      | 1      | I2C        | SCL   | SDA   |
| 1      | 0      | PWM        | PWM   | Hold  |
| 1      | 1      | DC         | DC    | Hold  |

Gain Select



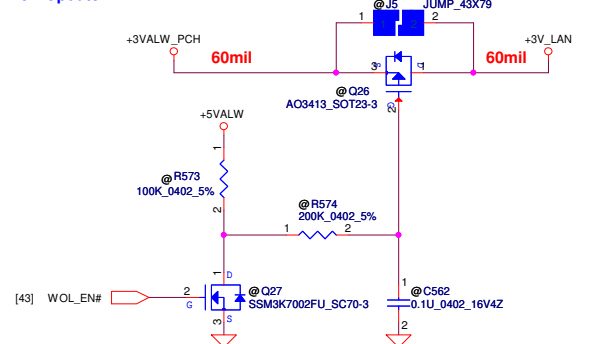
| AMP_G1 | AMP_G2 | Gain           |
|--------|--------|----------------|
| 0      | 0      | 11dB (Default) |
| 0      | 1      | 14dB           |
| 1      | 0      | 19dB           |
| 1      | 1      | 25dB           |



## WOL circuit

**Short J5 for WOL support (Enable/Disable by BIOS setup)**

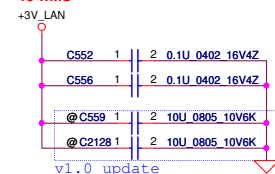
v0.2 update



**+3V LAN rising time (10%~90%) need > 0.5ms and <100ms.**

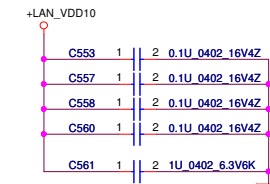
## Power ( Decoupling Cap. )

40 mils



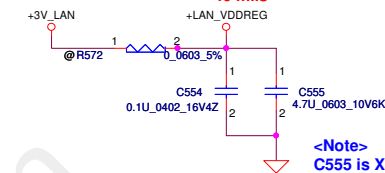
C552, C556, C559, C2128 close to Pin 11,32

**60 mils**

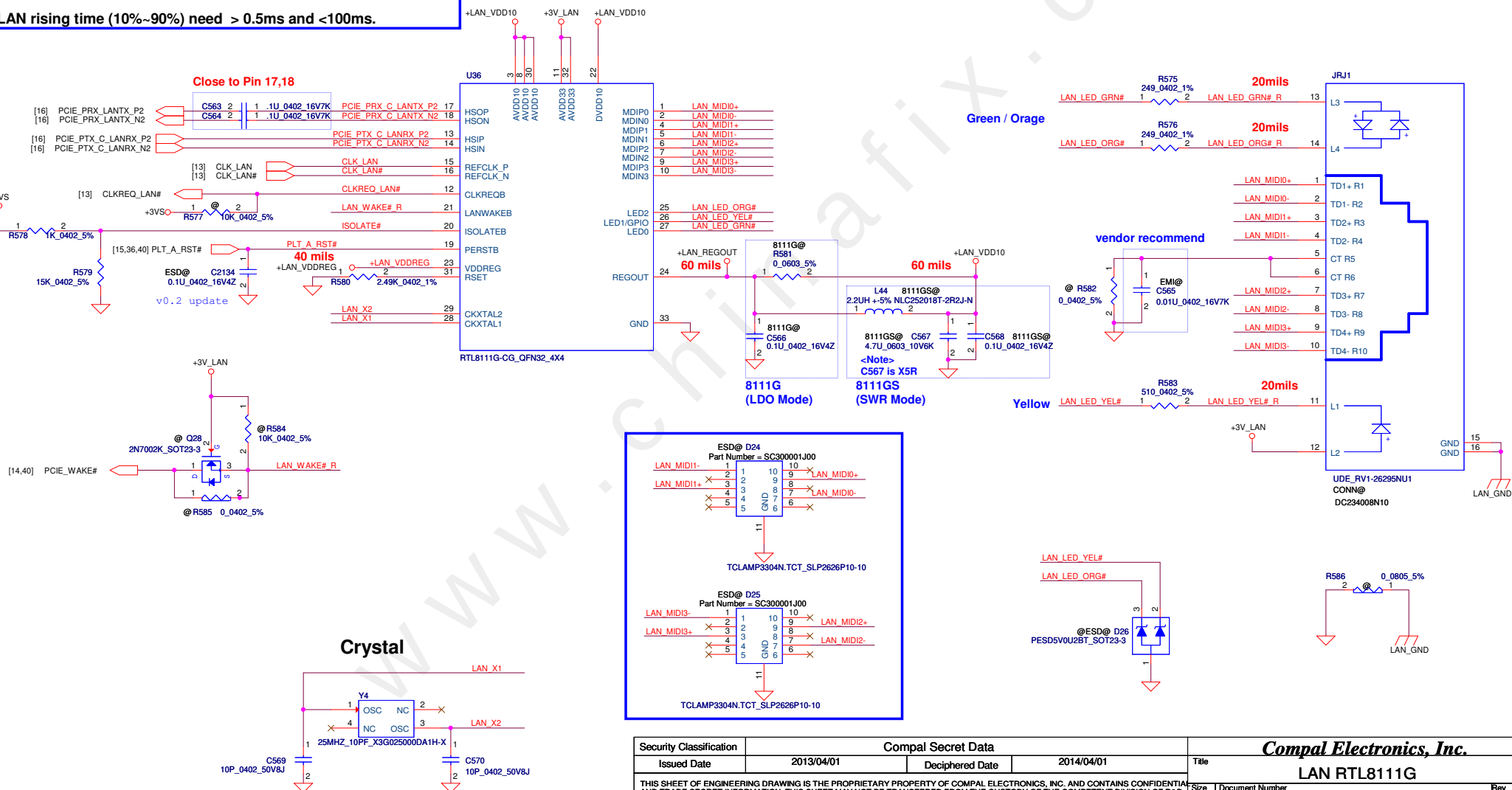


C553, C557, C558, C560, close to Pin 3,8,22,30  
C561(1uF) close to Pin22

40 mils

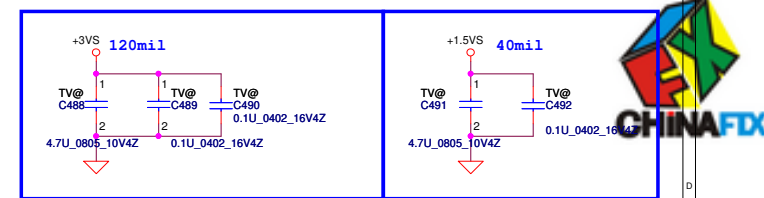


**<Note>**  
C554, C555 close to U26  
Pin22



|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            |                                                        |                             |                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------|------------|--------------------------------------------------------|-----------------------------|----------------|
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | Compal Secret Data |            | <b>Compal Electronics, Inc.</b><br><b>LAN RTL8111G</b> |                             |                |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2013/04/01 | Deciphered Date    | 2014/04/01 | Title                                                  |                             |                |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Custom                                                 | <b>ZEAO00 LA-A061P M/B</b>  | 0.3            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Date:                                                  | Tuesday, September 24, 2013 | Sheet 39 of 59 |

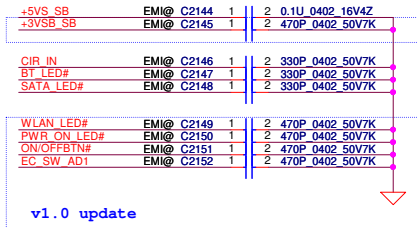
Mini Card Slot 1---TV tuner Current: +3VS : 2750mA, 1.5V: 500mA



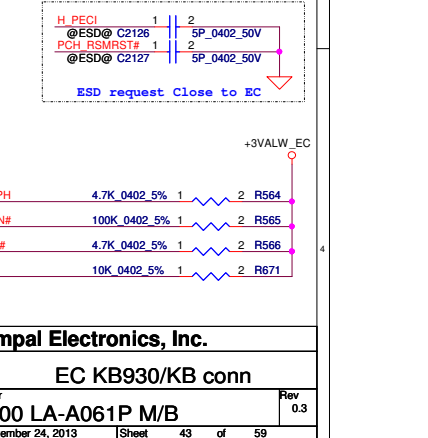
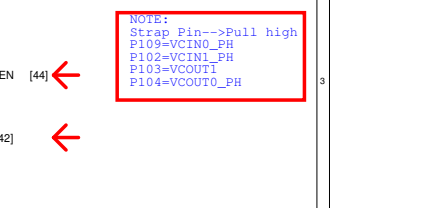
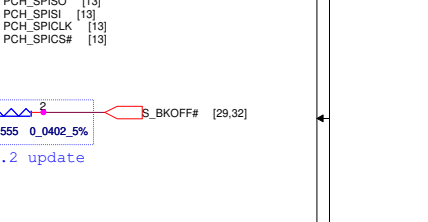
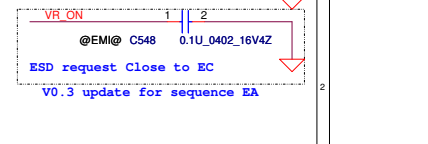
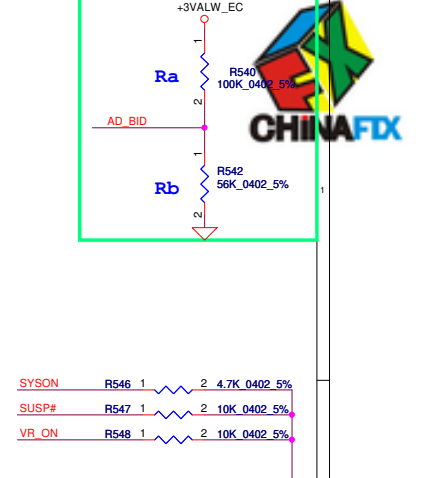
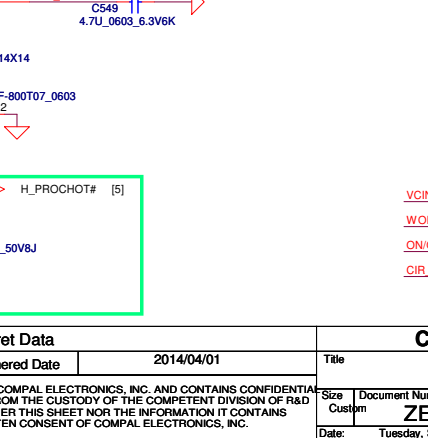
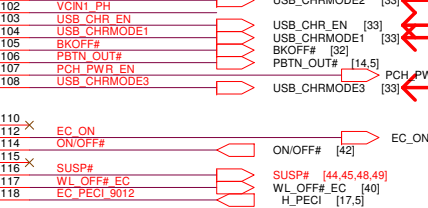
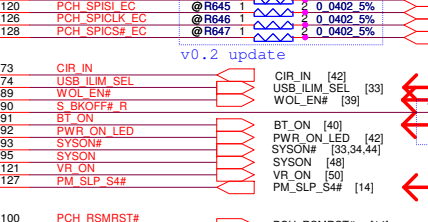
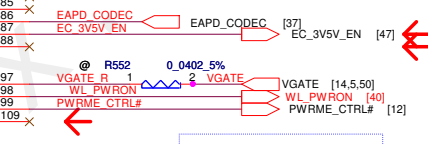
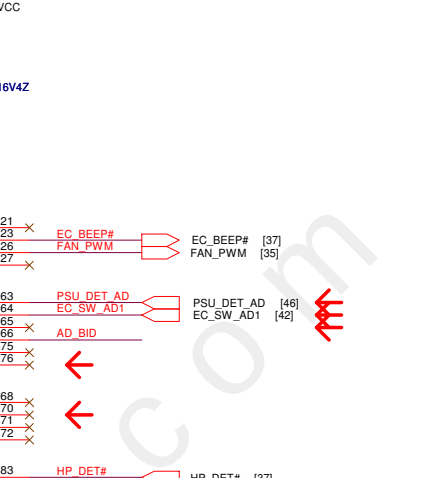
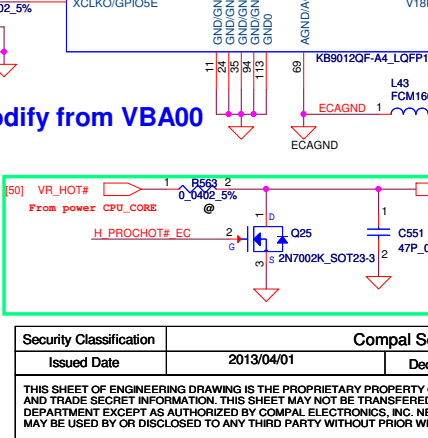
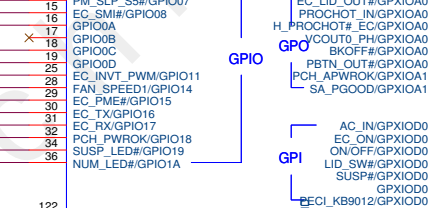
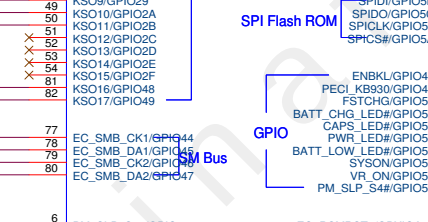
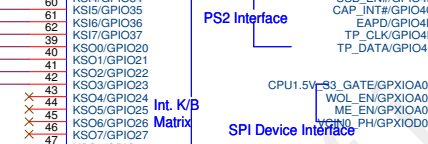
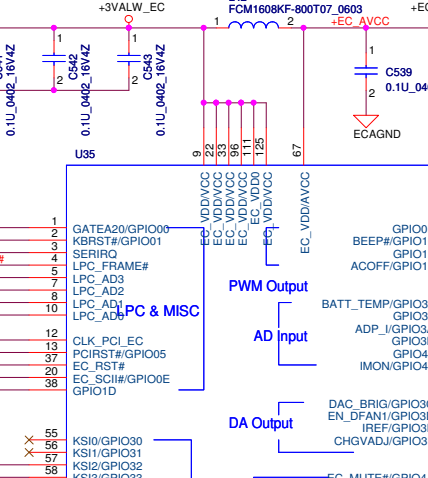
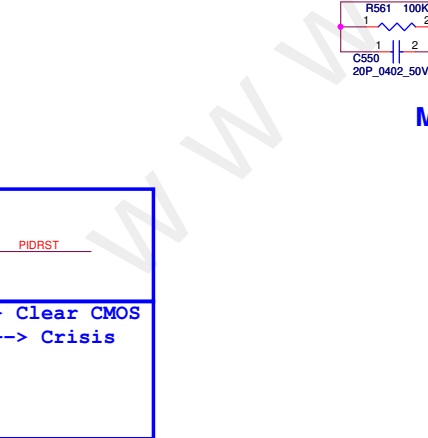
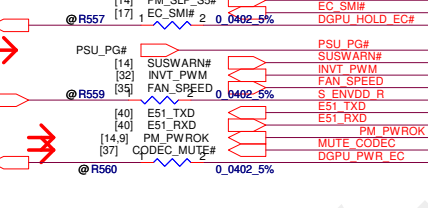
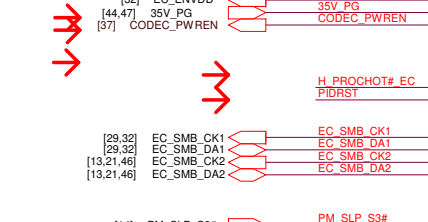
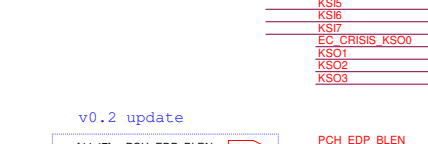
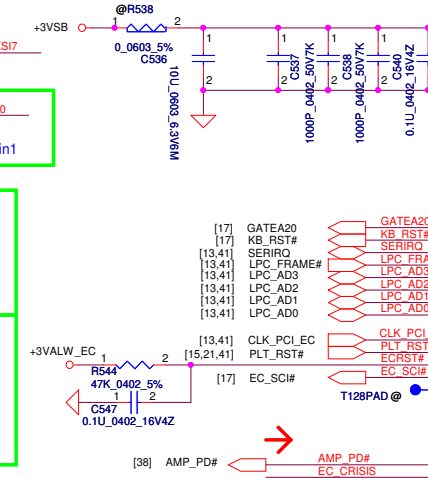
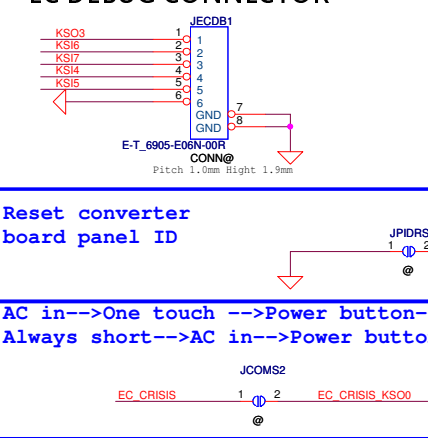
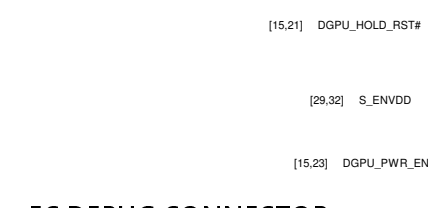
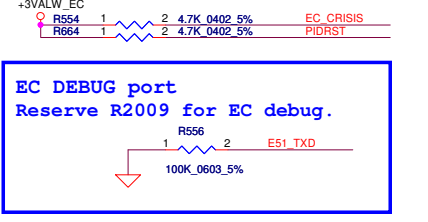
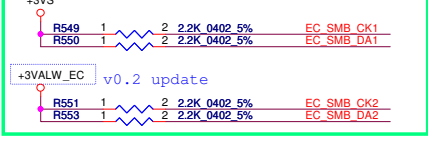
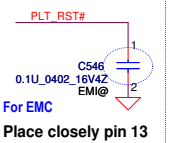
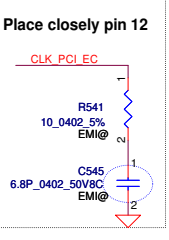
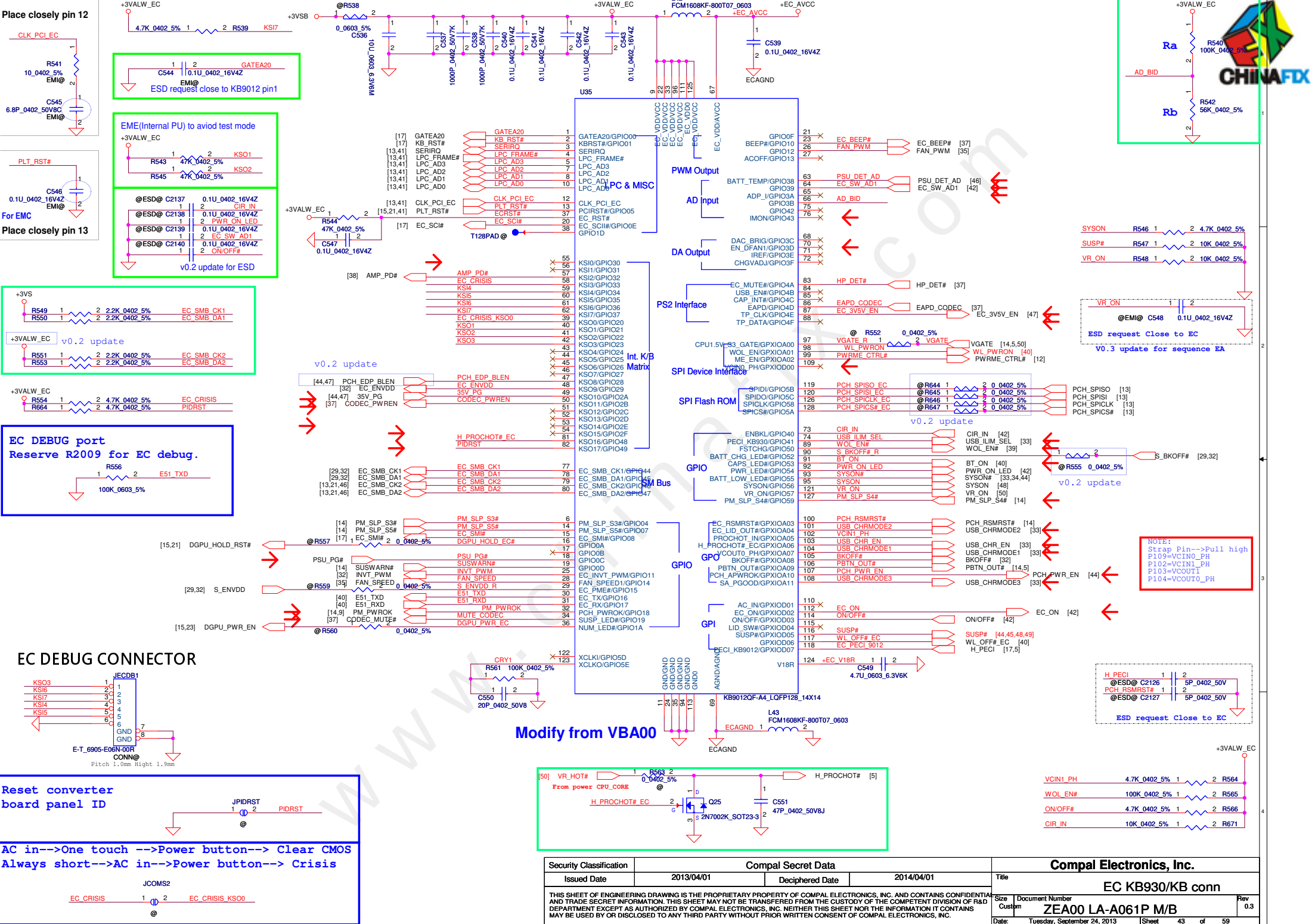
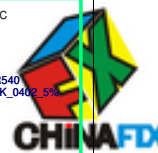
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| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | Compal Secret Data |            |                                    |                             |       |          |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2013/04/01 | Deciphered Date    | 2014/04/01 | Title<br><b>PCle-WLAN/TV_B-CAS</b> |                             |       |          |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | <b>ZEA00 LA-A061P M/B</b>          |                             |       | 0.3      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Date:                              | Tuesday, September 24, 2013 | Sheet | 40 of 59 |



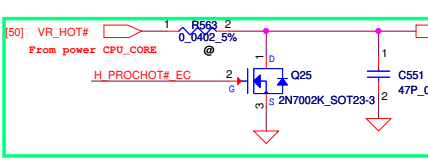
### 8Pin sub-board Connecetor



|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            |                                                               |                 |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------|------------|---------------------------------------------------------------|-----------------|-----|
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | Compal Secret Data |            | <b>Compal Electronics, Inc.</b><br><b>Sub-board Connector</b> |                 |     |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2013/04/01 | Deciphered Date    | 2014/04/01 | Title                                                         |                 |     |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | <b>ZEA00 LA-A061P M/B</b>                                     |                 |     |
| Date: Tuesday, September 24, 2013                                                                                                                                                                                                                                                                                                                                                                                                                             |            | Sheet 42 of 59     |            |                                                               |                 |     |



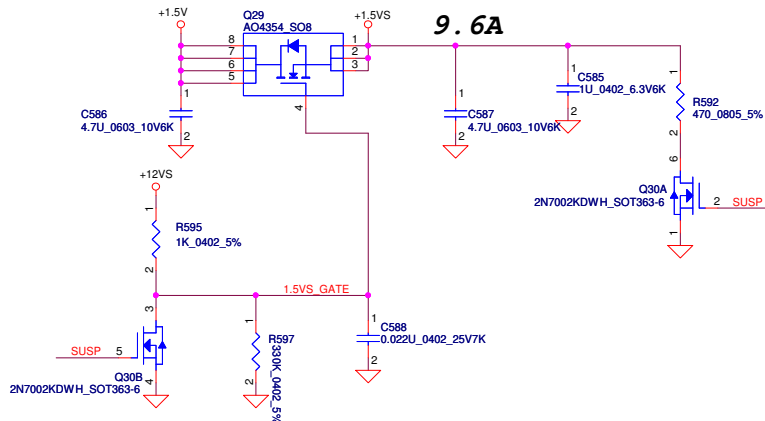
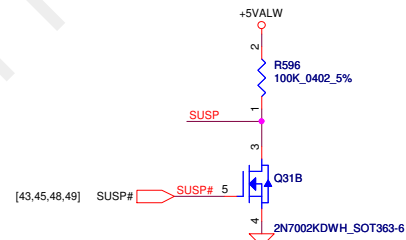
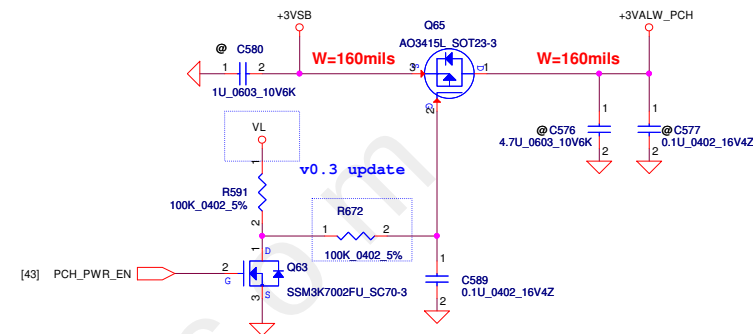
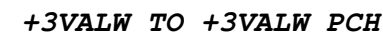
Modify from VBA00



| Security Classification | Compal Secret Data |
|-------------------------|--------------------|
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| Deciphered Date         | 2014/04/01         |

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| Compal Electronics, Inc. |                             |
| EC KB930/KB conn         |                             |
| Size                     | Document Number             |
| Custom                   | ZEA00 LA-A061P M/B          |
| Date                     | Tuesday, September 24, 2013 |
| Sheet                    | 43 of 59                    |



Four circuit diagrams showing the connection of SC59-3 MOSFETs to various voltage sources. Each diagram includes a MOSFET (Q32, Q33, Q34, Q35), a resistor (R599, R600, R601, R602), and a gate signal (SUSP, SYSON#). A fifth diagram shows a more complex connection involving a 12V source, a resistor (R603), and multiple MOSFETs (Q36, Q37).

Diagram 1: MOSFET Q32 (SSM3K7002BF 1N SC59-3) with gate resistor R599 (470\_0805\_5%) connected to .05VS\_VPCH. Gate signal is SUSP.

Diagram 2: MOSFET Q33 (SSM3K7002BF 1N SC59-3) with gate resistor R600 (22\_0805\_5%) connected to +0.75VS. Gate signal is SUSP.

Diagram 3: MOSFET Q34 (SSM3K7002BF 1N SC59-3) with gate resistor R601 (470\_0805\_5%) connected to +1.05VS\_VCCIO. Gate signal is SUSP.

Diagram 4: MOSFET Q35 (SSM3K7002BF 1N SC59-3) with gate resistor R602 (470\_0805\_5%) connected to +1.5V. Gate signal is SYSON#.

Diagram 5: MOSFET Q36 (SSM3K7002BF 1N SC59-3) with gate resistor R603 (100\_0805\_5%) connected to +12VS. Gate signal is SUSP. MOSFET Q37 (SSM3K7002BF 1N SC59-3) is also shown with gate resistor R604 (330K\_0402\_5%) connected to B+. Gate signal is 35V\_PG.

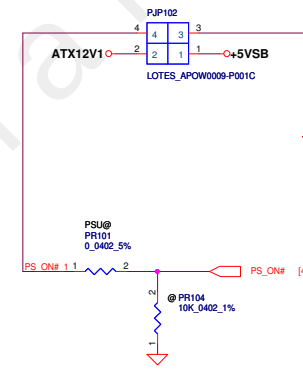
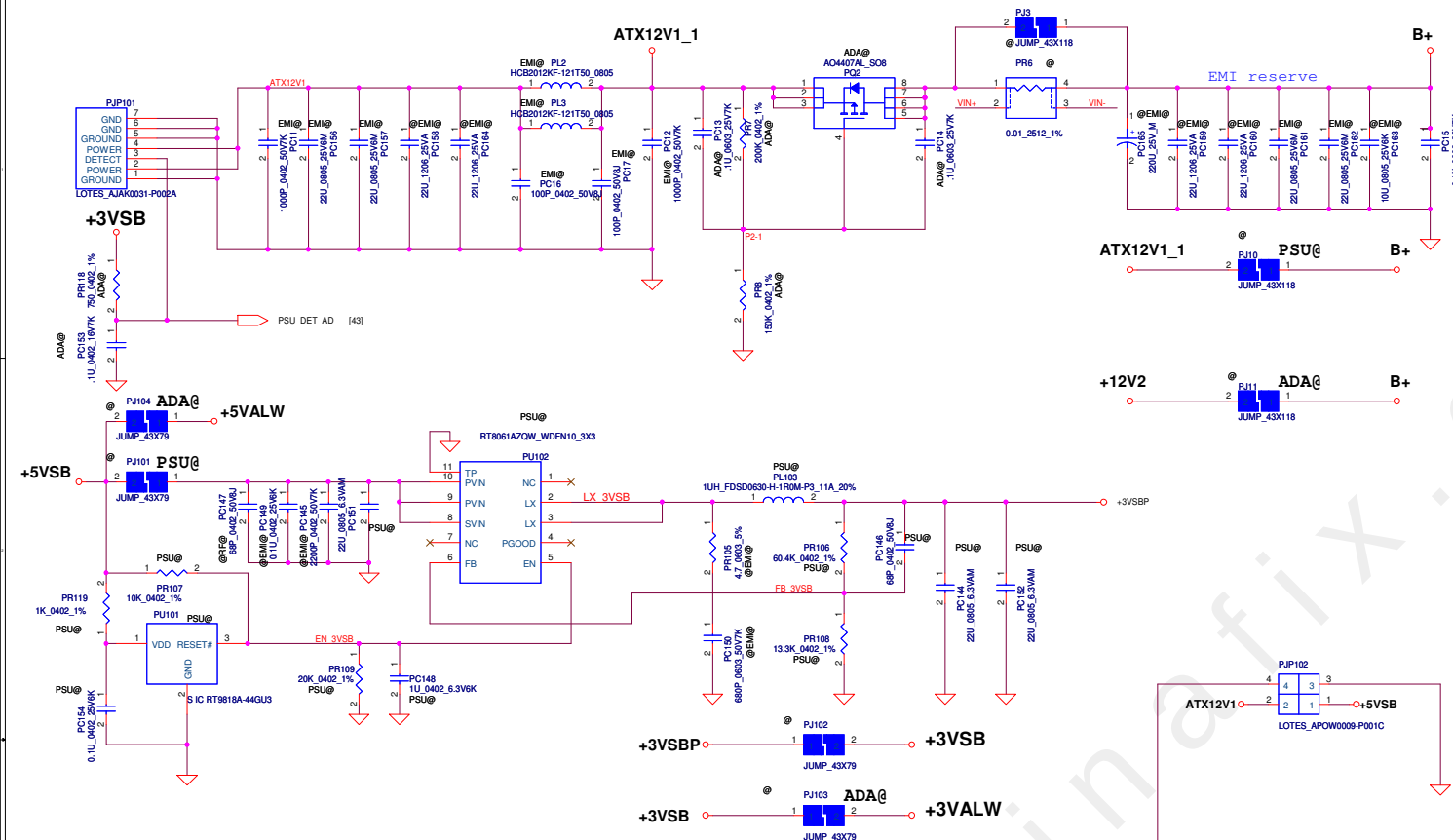
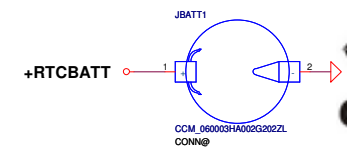
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| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            | Compal Secret Data |            | <i>Compal Electronics, Inc.</i> |                           |            |       |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |                    |            | Size                            | Document Number           | Rev        |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |                    |            |                                 | <b>ZEA00 LA-A061P M/B</b> | <b>0.3</b> |       |
| Date:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                    |            | Tuesday, September 24, 2013     | Sheet                     | 44         | of 59 |



Power dissipation: 0.06W

**VFB=0.8V**

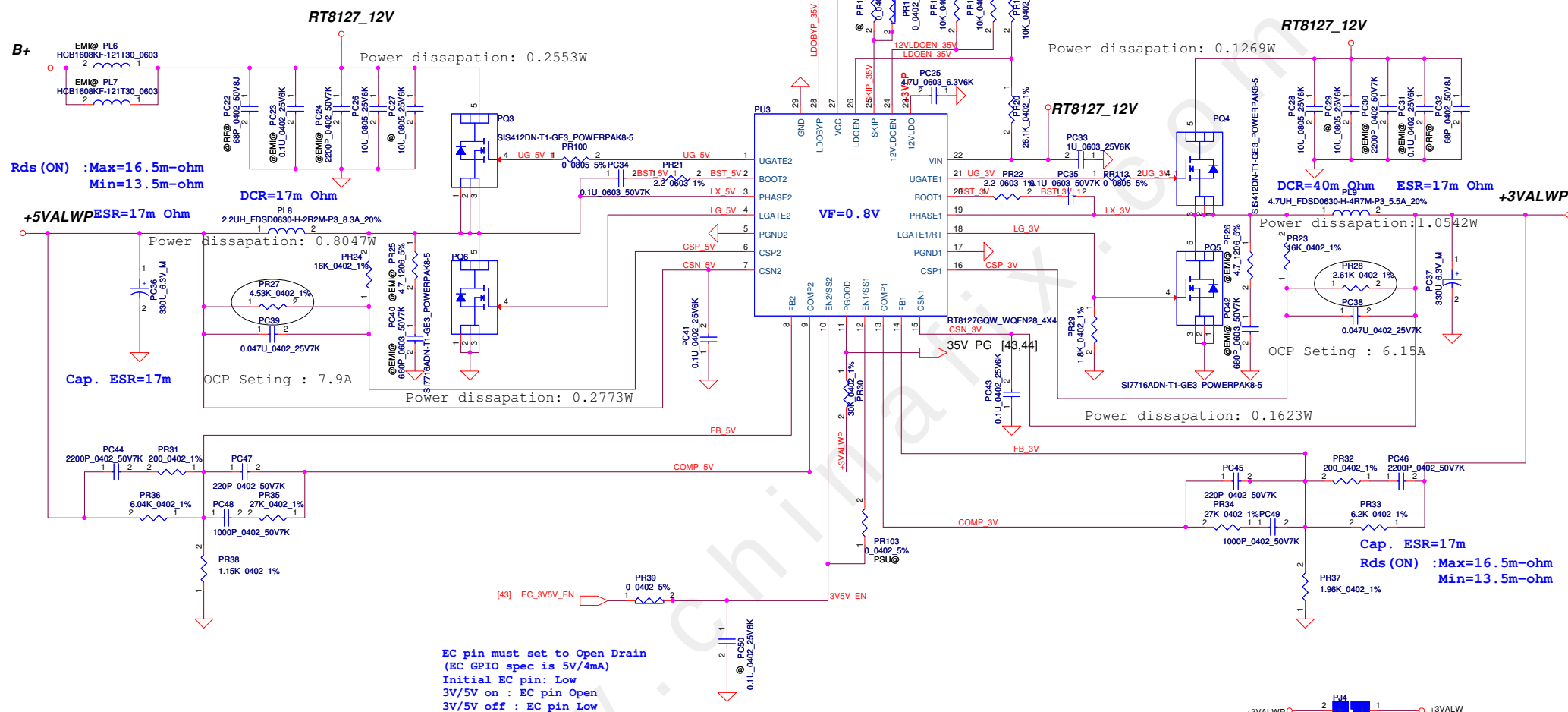
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| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Compal Secret Data |                 |            | Compal Electronics, Inc. |                             |       |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2012/04/27         | Deciphered Date | 2013/04/27 | Title                    | PWR- 12VSP                  |       |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | Custom                   | VBA11 LA-A111 M/B           | 0     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | Date                     | Tuesday, September 24, 2013 | Sheet |



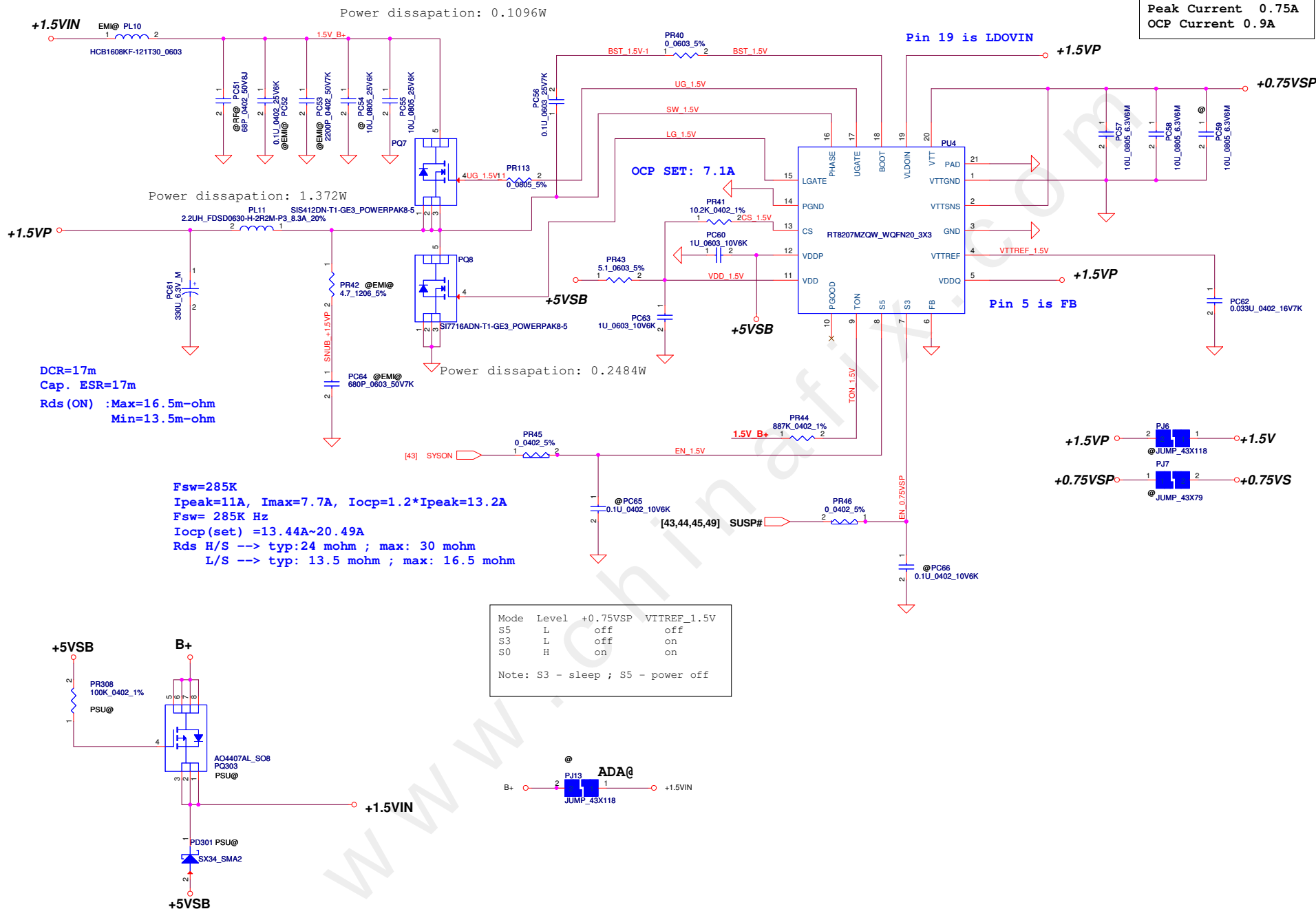
## Current sense solution 2

Ventura for CPU side  
slave address : 1000001  
please placemnet near R-sense

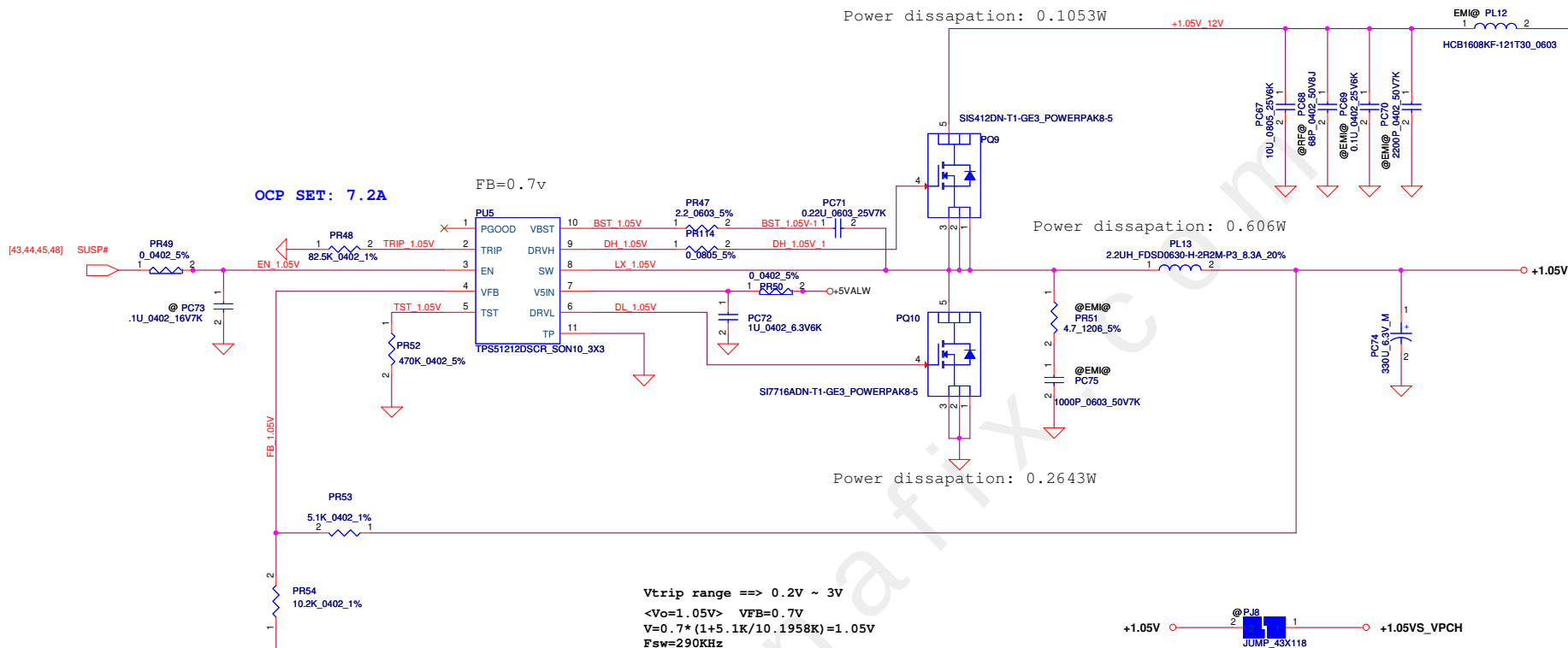
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | Compal Secret Data |            | Compal Electronics, Inc. |                             |
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| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2012/09/01 | Deciphered Date    | 2012/11/12 | Title                    |                             |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Sheet                    | 46 of 56                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Rev                      | 0.1                         |



0.75V<sub>olt</sub> +/- 5%  
TDC 0.525A  
Peak Current 0.75A  
OCP Current 0.9A



| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       | Compal Secret Data |                 |            | Compal Electronics, Inc.          |                |
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| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2012/09/01         | Deciphered Date | 2013/12/31 | Title                             |                |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | 1.5VP                             | 0.1            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | Date: Tuesday, September 24, 2013 | Sheet 48 of 56 |



OCP SET: 7.2A

FB=0.7v

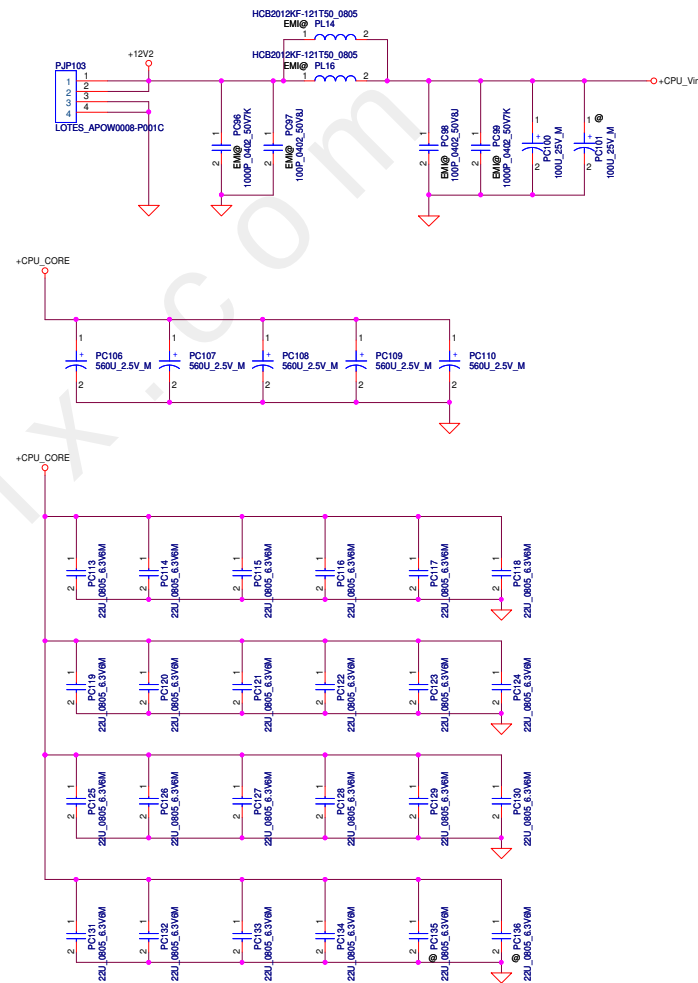
Vtrip range ==> 0.2V ~ 3V  
<Vo=1.05V> VFB=0.7V  
V=0.7\*(1+5.1K/10.1958K)=1.05V  
Fsw=290KHz

Rds (ON) :Max=16.5m-ohm  
Typ=13.5m-ohm

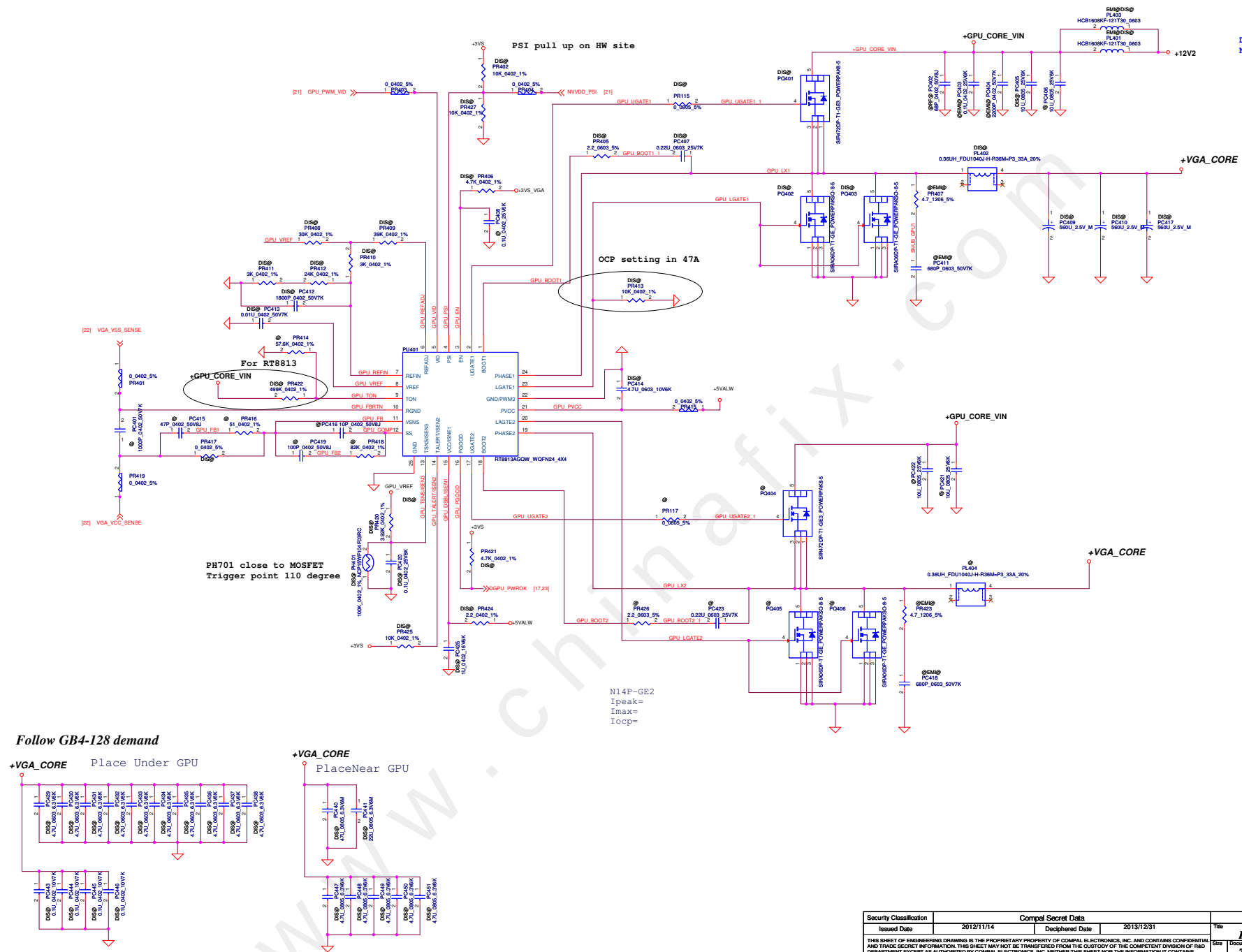
Ipeak=6.526A, Imax=4.568A, 1.2\*Ipeak=7.831A  
Iocp(set)=7.851A~11.555A

|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            |                          |                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------|------------|--------------------------|-----------------------------|
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | Compal Secret Data |            | Compal Electronics, Inc. |                             |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2012/09/01 | Deciphered Date    | 2013/12/31 | Title                    | +1.05VSP                    |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Date                     | Tuesday, September 24, 2013 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Sheet                    | 49 of 56                    |

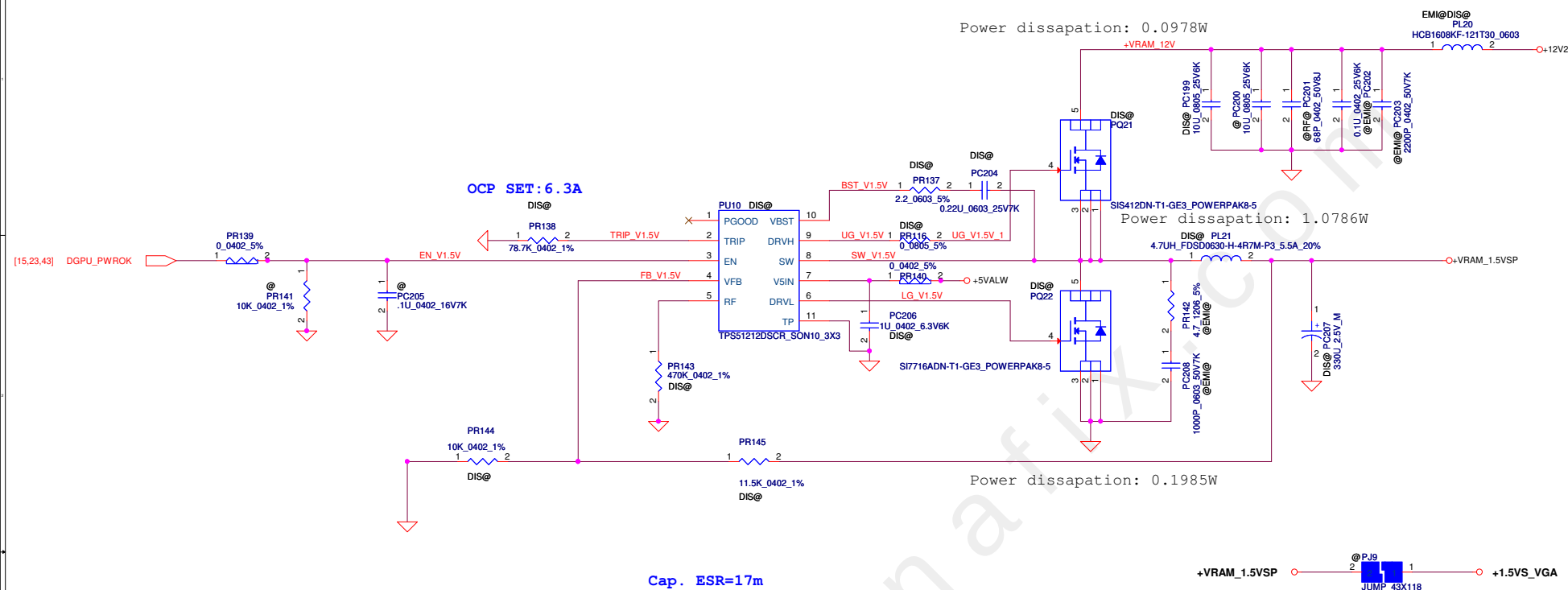




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|-------------|---------------------------------------|-------|----------|
|             |                                       |       |          |
| Title       |                                       |       |          |
| Power Stage |                                       |       |          |
| Size        | Document Number<br>VCA00 LA-9792P M/B |       |          |
| Date:       | Tuesday, September 24, 2013           | Sheet | 51 of 56 |



| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | Compal Secret Data |            | Compal Electronics, Inc. |                       |
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| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2012/11/14 | Deciphered Date    | 2013/12/31 | Title                    | PWR VGA CORE/VGA PCIE |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Version                  | September 24, 2013    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Sheet                    | 52 of 56              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Rev                      | 0.1                   |

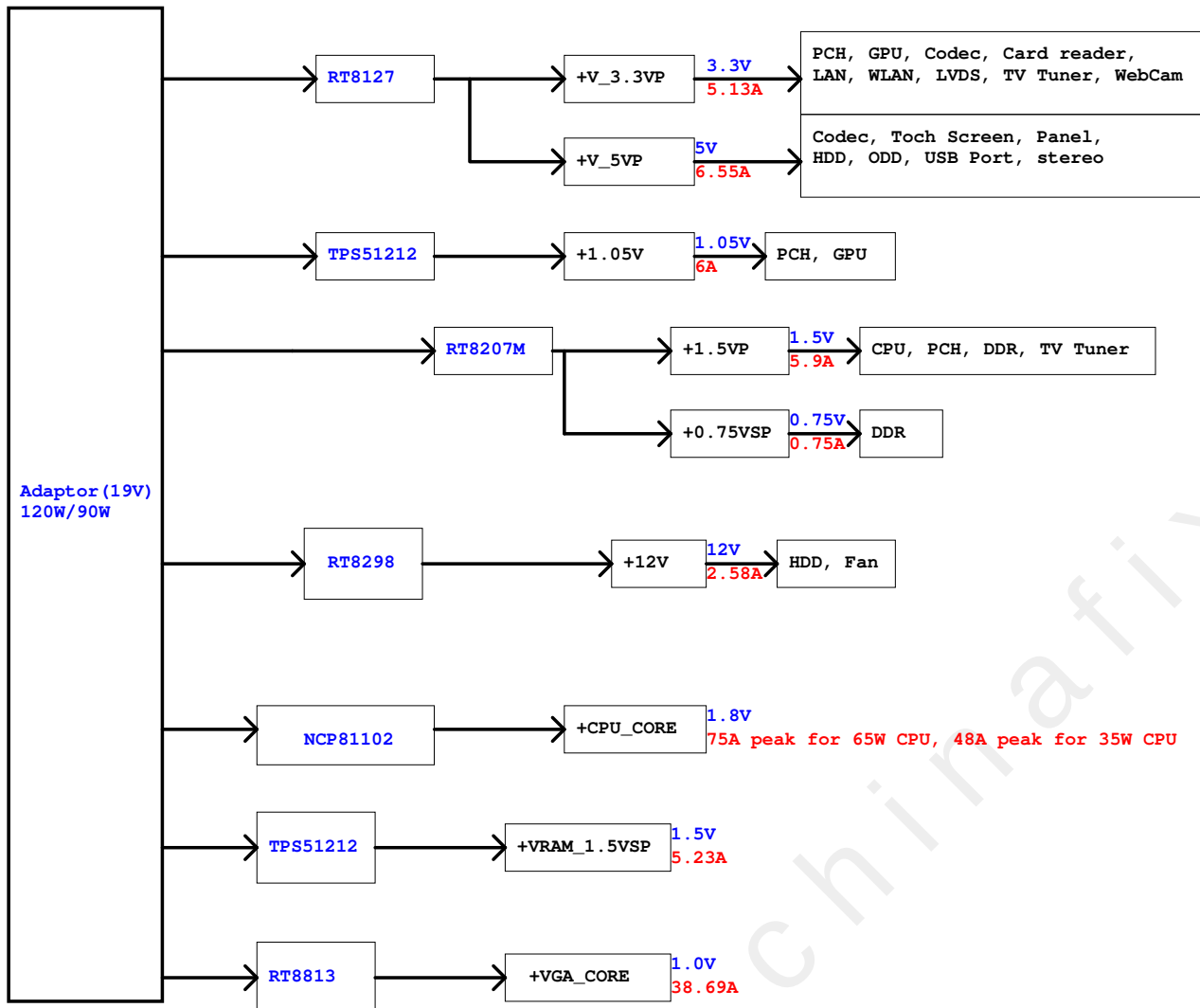


Cap. ESR=17m  
 Rds (ON) :Max=15m-ohm  
 Typ=12m-ohm  
 Vtrip range ==> 0.2V ~ 3V  
 <Vo=1.5V> VFB=0.7V  
 V=0.7\*(1+11.5K/10K)=1.505V  
 Fsw=290KHz

Ipeak=4.7A, Imax=3.29A, Iocp=1.2\*Ipeak=5.64A

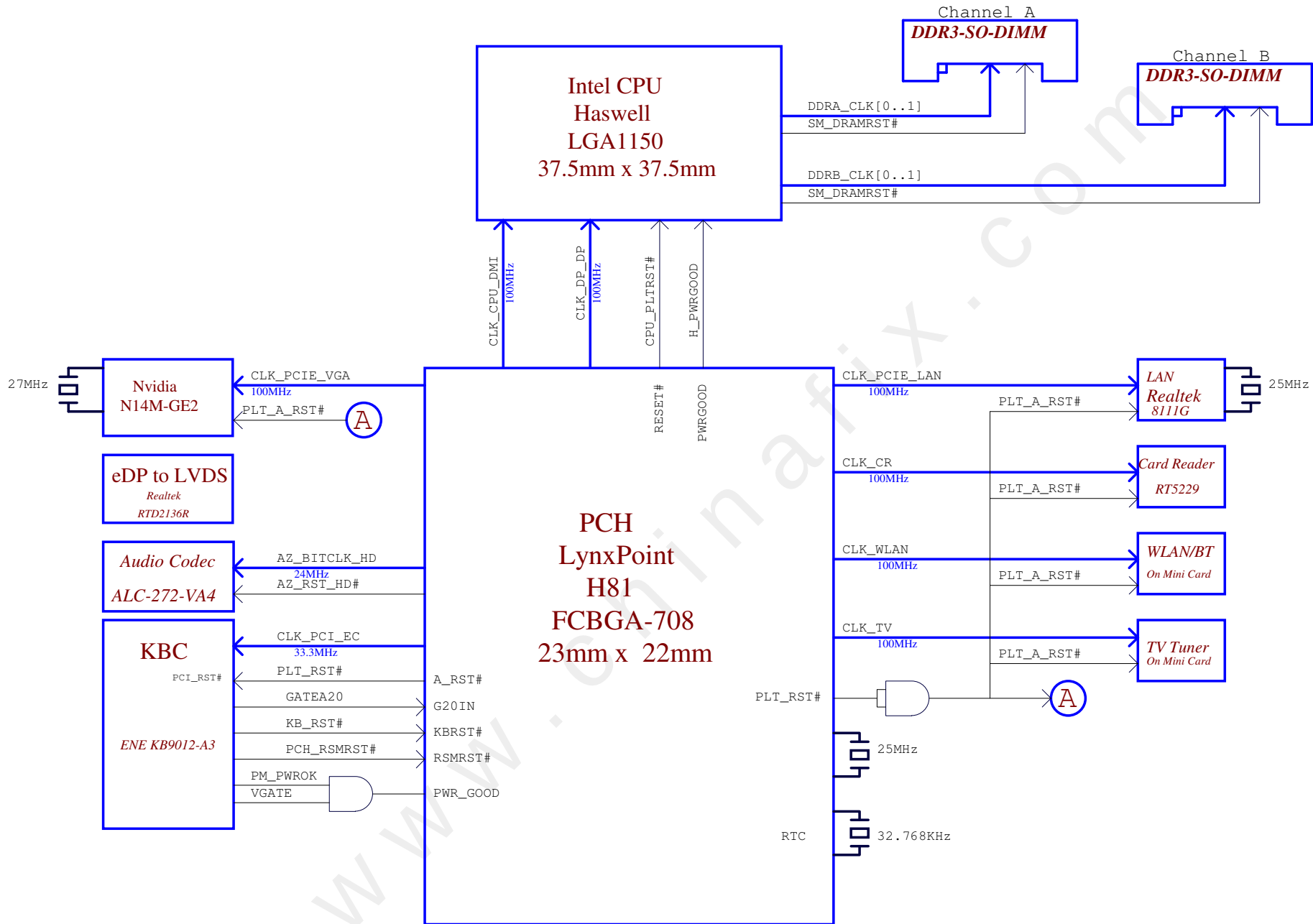
Iocp(set)=5.718A~8.304A

|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            |                          |                             |
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| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | Compal Secret Data |            | Compal Electronics, Inc. |                             |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Date:                    | Tuesday, September 24, 2013 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Sheet                    | 53 of 56                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Rev                      | 0.1                         |

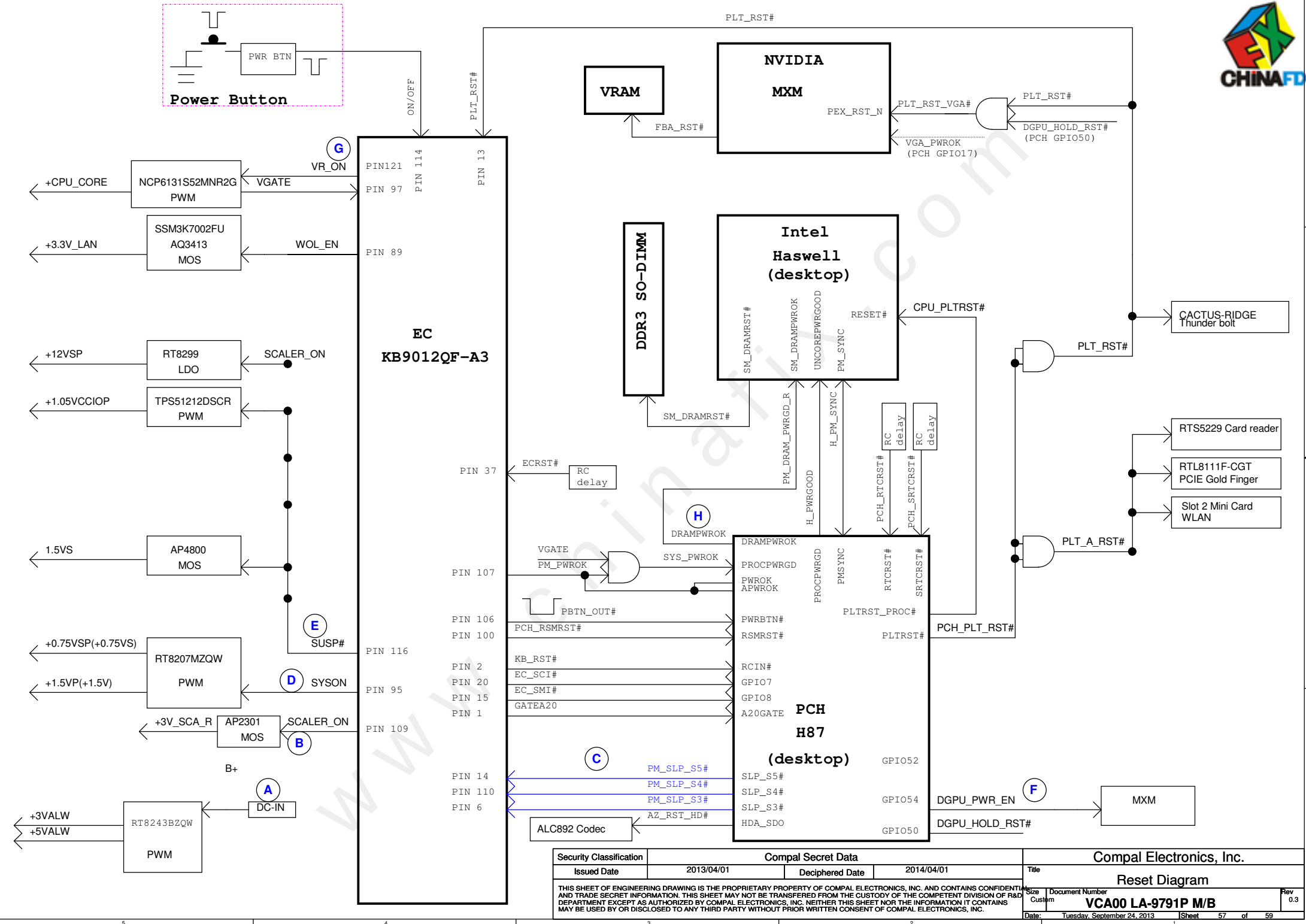


|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            |                                   |                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------|------------|-----------------------------------|----------------|
| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | Compal Secret Data |            | Compal Electronics, Inc.          |                |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2011/09/12 | Deciphered Date    | 2012/09/12 | Title                             | Power Rail     |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Date: Tuesday, September 24, 2013 | Sheet 55 of 59 |

# Clock and Reset Diagram



| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | Compal Secret Data |            | Compal Electronics, Inc. |                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------|------------|--------------------------|-----------------------------|
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2013/04/01 | Deciphered Date    | 2014/04/01 | Title                    | Clock/Reset Diagram         |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Date:                    | Tuesday, September 24, 2013 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Sheet                    | 56 of 59                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Rev                      | 0.3                         |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            |                          |                             |
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| Security Classification                                                                                                                                                                                                                                                                                                                                                                                                                                       |            | Compal Secret Data |            | Compal Electronics, Inc. |                             |
| Issued Date                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2013/04/01 | Deciphered Date    | 2014/04/01 | Title                    |                             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Reset Diagram            |                             |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Date:                    | Tuesday, September 24, 2013 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Sheet                    | 57 of 59                    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                    |            | Rev                      | 0.3                         |

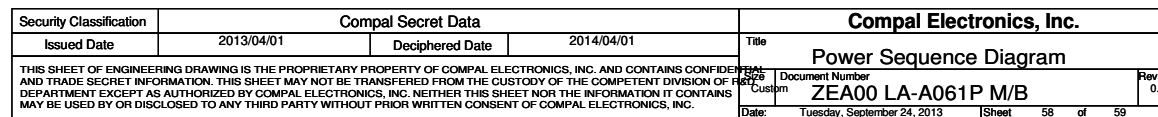


| NO |          | DATE | PAGE | MODIFICATION LIST                             | PURPOSE                 |
|----|----------|------|------|-----------------------------------------------|-------------------------|
| 1. | 20130604 | P47  |      | Add PC154 and Change the pu101 to SA00005A600 | For Pericom issue       |
| 2. | 20130729 | P45  |      | Add PR105 ,PC155 Snaber                       | For EM1 Request         |
| 3. | 20130729 | P45  |      | Add PR1                                       | For EM1 Request         |
| 3. | 20130729 | P45  |      | Add PR39                                      | For S5 power loss issue |

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**DATE:** 2013/04/01



HW PIR (Product Improve Record)

ZEA00 LA-A061P SCHEMATIC CHANGE LIST

REVISION CHANGE: 0.1 --> 0.2

GERBER-OUT DATE: 2013/06/20

| NO  | DATE | PAGE | MODIFICATION LIST                                                                                                                                                                                                    | PURPOSE |
|-----|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1.  |      |      | Change C45 from SF000002V00 to SF000003X00                                                                                                                                                                           |         |
| 2.  |      |      | Change +LCDVDD enable control from EC to LVDS convertor,un-pop R367 and R365 change short pad.                                                                                                                       |         |
| 3.  |      |      | Change LCD_BACKLIGHT control from EC to LVDS convertor,un-pop R364 and R363 change 0 ohm.                                                                                                                            |         |
| 4.  |      |      | Remove un-used components(U18,R335,R336,C357,C359,C360,R338,R339) for eDP to LVDS convertor.                                                                                                                         |         |
| 5.  |      |      | Pop R428 for AZ_SDIN0_HD.                                                                                                                                                                                            |         |
| 6.  |      |      | U2 footprint change from socket to IC.                                                                                                                                                                               |         |
| 7.  |      |      | Add RH11                                                                                                                                                                                                             |         |
| 8.  |      |      | Change Y2 from SJ10000CU00 to SJ10000DE00,change C106 & C107 from 27pF to 4.7pF                                                                                                                                      |         |
| 9.  |      |      | Change R423 location to L45                                                                                                                                                                                          |         |
| 10. |      |      | Change D7 from SC2N202U010 to SC600000B00 for 替代料                                                                                                                                                                    |         |
| 11. |      |      | Change Q29 from SB548000210 to SB000002N00.                                                                                                                                                                          |         |
| 12. |      |      | Change D8&D9 from SCS000002G00 to SCS000002Z00                                                                                                                                                                       |         |
| 13. |      |      | X1 code change:1.Change Q2,Q3,Q4,Q5,Q30,Q31 from SB01000JE00 to SB00000EO00. 2.Change Q9 from SB934130020 to SB934130000. 3.Change Q10 from SB00000FC00 to SB00000F400. 4.Change L1 from SM01000JE00 to SB01000JN00. |         |
| 14. |      |      | Change R551 & R553 pull-high from +3VS to +3VALW_PCH for leakage.                                                                                                                                                    |         |
| 15. |      |      | Add R677 & R678 & R679 for PTC request, Change R473,R490,R679,R677,R678 from 0ohm to PTC(SP040005X00).                                                                                                               |         |
| 16. |      |      | Change Q10 from SB00000FC00 to SB00000L800 for 替代料                                                                                                                                                                   |         |
| 17. |      |      | Remove R469 0ohm for TV.                                                                                                                                                                                             |         |
| 18. |      |      | Add C2134 ,C2135,C2136,C2137,C2138,C2139,C2140,C2141,C2143 for ESD.                                                                                                                                                  |         |
| 19. |      |      | Remove JXDP1,OC1,OC2,RC3,RC4,R125,R126.                                                                                                                                                                              |         |
| 20. |      |      | Pop U7&R231, un-pop R228 for PLT_RST_VGA#.                                                                                                                                                                           |         |
| 21. |      |      | Swap SATA_PRX_DTX_N1 & SATA_PRX_DTX_P1 for m-SATA pin define.                                                                                                                                                        |         |
| 22. |      |      | Un-pop LAN power components Q26,Q27,R573,R574,C562.                                                                                                                                                                  |         |
| 23. |      |      | 0 ohm change to short pad: R347,R585,R507,R674,R644,R645,R646,R647                                                                                                                                                   |         |
| 24. |      |      | Change R453&R457 from 0ohm to 1.1K, R451&R459 from 300ohm to 5.6Kohm.                                                                                                                                                |         |
| 25. |      |      | Pop R438,R439 for ESD request.                                                                                                                                                                                       |         |

PVT change list:

1. Change Q10 from SB00000FC00(EOL soon) to SB00000ZN00(同Q29),SB00000FC00 as 2nd source.Schematic,需驗證

2. Change U23 pin12\_+USB3\_VCCA to +USB3\_VCCB, pop U22, un-pop U24 for USB charger
3. R365 change from short pad to 0ohm.
4. U5 pin5 change from +3VSto +3VALW\_PCH for BCM43142 wake from WLAN issue.
5. Change R473,R490,R677,R678,R679 from SP040005X00\_0603 size to F1,F2,F4,F5,F3 SP040003S00\_1206 size.
6. Change L11 from SM010014520 to SM01000EJ00 for ACL request
7. Change L8 from SM010007W00 to SM010019400 for ACL request
8. Change D7 from SC2N202U010 to SC600000B00(same as D1/D2), SC2N202U010 as 2nd source..
9. Change RP19 from SD309510A80(T88 P/N) to SD309510A10.
11. Change R276 from 10k to 100k for +3VS\_VGA rise time.
12. Change R672 from 10k to 100k for +3VALW\_PCH rise time.
14. Change R438 & R439 from 0\_0603 to short pad.
15. Un-pop C125 & C548 for sequence EA.
16. Change C394, C398,C520 & C514 from 220uF (LELON\_SF0000001F00) to 100uF (Panasonic\_SF0000005100) to meet Inrush EA & ACL request.
17. Change C170 & C171 from 12pF to 10pF for EA.
20. Change C106 & C107 from 4.7pF to 10pF for 25MHz crystal.
21. Add R677 & reserve R678 on U5 AND gate for PLT\_A\_RST#
13. Change JUSB1 & JUSB2 from DC23300AE00 to DC233008R00(VBA11)
24. Change R591 pull-high from +5VSB to VL for power S5 Erp request.
22. Change D20 & D21 from SC3000001Y00 to SC300002F00 for ESD request
23. Change D22 & D23 from SCA00001100 to SCA00000T00 for ESD ACL request
10. Add C2144-C2152 for EMI request.
18. Change R402 from short pad to 22ohm for EMI, R399,R401,R403 & R404 change from short pad to 0 ohm for EMI request.
19. Reserve C2153,C2154,C2155,C2156, add D29 for ESD.
20. Change R282 from 100k to 2k, R277 from 470 to 22 ohm for GPU power sequence.
21. Change Y1 from SJ100001K00 to SJ10000FA00 ,C102 & C107 to 6pF.

- pre-MP change list:
1. Change R399,R401,R402,R403,R404 from 0ohm to short pad.
2. Add C2157 and reserve C2158.
3. Change R8,R470,R669,R670,R416 from 0ohm to short pad.
4. Un-pop JECDB1 & SW1.
5. For R3 P/N, change PCH P/N from SA00006RF00 to SA00006RF20, PCB P/N from DA60011S000 to DA60011S010 and GPU P/N from SA00006ZF00 to SA00006ZF10.
6. Change C520 & C514 from 100uF to 220uF.
7. Pop C2149~C2152 for ESD request.
8. Change C559 & C2128 from 0603 to 0805.
9. Change C2145 from 0.1uF to 470pF, change C2149~C2152 from 330pF to 470pF for EMI.
10. Add C418 for EMI.



|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            |                          |                             |                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------|------------|--------------------------|-----------------------------|----------------|
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            |                          | VCA00 LA-9791P M/B          | 0.3            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    |                 |            | Date                     | Tuesday, September 24, 2013 | Sheet 89 of 99 |